Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

MATHEMATICS

Class – 7

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1. Integers

Solutions

Worksheet – 1

1. Correct answer: C) P

1. Greatest number on the number line

Numbers increase as we move to the right on a number line.
Among A, B, O, and P, P is the farthest to the right.

2. Correct answer: B) −10° to +1°

2. Maximum rise in temperature

Find the increase in each case:

A) $32 - 23 = 9°$

B) $1 - (-10) = 11°$

C) $-11 - (-18) = 7°$

D) $5 - (-5) = 10°$

Maximum rise is 11°.

3. Correct answer: B) Shimla

Lowest temperature

Lower temperature is represented by the point farther left on the number line.
Point P (Shimla) is the leftmost.

4. Correct answer: C) −1

Largest negative integer

Among negative integers, the one closest to zero is the largest.
That is −1.

5. Correct answer: D) Z

Symbol for integers

The set of integers is denoted by Z.

6. Correct answer: C) −998

Successor of −999

The successor of a number is the number one more than it.
$-999 + 1 = -998$

7. Correct answer: B) −101

Predecessor of −100

The predecessor of a number is the number one less than it.
$-100 - 1 = -101$

8. Correct answer: B) Natural numbers

Counting chairs starting from 1

Counting objects beginning from 1 uses natural numbers.

9. Correct answer: C) 0

Number between −2 and 1

Numbers between −2 and 1 are −1 and 0.
Among the options, only 0 lies between them.

10. Correct answer: B) n^2

Finding the number from its successor

Successor of a number = number + 1
Given successor $= n^2 + 1$

So, number $= n^2 + 1 - 1 = n^2$

Worksheet – 2

1. Correct answer: C) 152

Sum of two integers

Sum = 93, one integer = −59
Other integer $= 93 - (-59) = 93 + 59 = 152$

2. Correct answer: B) 5

Moving right means adding a positive number.

$-4 + 9 = 5$

3. Correct answer: B) 8 + (−8)

Which of the following sums is zero?

Check each option:

A) $-9 + (-9) = -18$
B) $8 + (-8) = 0$

C) $-4+(-6)=-10$
D) $5+0=5$

4. Correct answer: C) 216

Product of (−6) × (−6) × (−6)

$(-6)\times(-6)=36, 36\times(-6)=-216$
Negative of −216 is 216.

5. Correct answer: A) −576

Product of (−8) × (−4) × 6 × (−3)

Number of negative signs = 3 → result is negative

$8\times4\times6\times3=576$
So product = −576

6. Correct answer: A) 5,870

$52{,}830\div9$

$52{,}830\div9=5{,}870$

7. Correct answer: C) 0

Number line movement

Start at −2
Move 4 steps left: $-2-4=-6$
Move 6 steps right: $-6+6=0$

8. Correct answer: B) 50

Hot dog packages

$400\div8=50$

9. Correct answer: C) 524

Total bowling score

Jake = 189
Charles = 120
Max = 120
Terry = 95

$189+120+120+95=524$

10. Correct answer: D)

Does not represent an integer

A) $0\div(-7)=0\rightarrow$ integer
B) $20\div(-4)=-5\rightarrow$ integer
C) $(-9)\div3=-3\rightarrow$ integer
D) $4\times(-12)\div5=-48\div5=-9.6\rightarrow$ not an integer

Worksheet – 3

1. Correct answer: C

$(-25)\times[6+4]$ is not same as _____

$(-25)\times(6+4)=(-25)\times10=-250$

Check options:
A) $(-25)\times10=-250$ same
B) $(-25)\times6+(-25)\times4=-150-100=-250$ same
C) $(25)\times6\times4=600$ not same
D) −250 same

2. Correct answer: A) 4300

$(-43)\times(-99)+43$ is equal to

$(-43)\times(-99)=4257$
$4257+43=4300$

3. Correct answer: B) −150

$(-15)\times8+(-15)\times2=$

Using distributive property:

$(-15)(8+2)=(-15)\times10=-150$

4. Correct answer: C)

If $a=-9$ and $b=5$, which statement is correct?

Check options:

A) $a-b=-14, b-a=14$

B) Addition is commutative, so this is false

C) $a\times b=b\times a$ (commutative property)

D) Division is not commutative

5. Statement I:
$a\div(-b)=-\frac{a}{b}$ **and** $(-a)\div b=-\frac{a}{b}\rightarrow$ **Correct**

Statement II:
$(-a)\div(-b)=\frac{a}{b}$, but this is not equal to $b\div a$ in general → Incorrect

Statement III:
Division by zero is not defined and $0\div a=0$ for $a\neq0\rightarrow$ Correct

Statement IV:
$a\div1=a$, but $1\div a\neq a$ (for $a\neq1$) → Incorrect

Incorrect statements:II and IV

6. Correct answer:B

A)$-121\div(-11)=11$

$-11\div(-121)=\frac{1}{11}$

Not same → Statement is correct

B)$-100 \div 25 = -4$
-4 is an integer, but the statement says it is *not* an integer → Incorrect

C)$(-216 \div 36) \div 6 = (-6) \div 6 = -1$
Statement is correct

D)$36 \div (2 + 4) = 36 \div 6 = 6$
$(36 \div 2) + (36 \div 4) = 18 + 9 = 27$
Since $6 \neq 27$, statement is correct

Incorrect alternative:B

7. Correct answer: C) Distributive

Property used in $-4 \times (6 - 8) = (-4 \times 6) - (-4 \times 8)$
This shows multiplication distributed over subtraction.

8. Correct answer: B)

Associative property of integers under addition

Associative property:$a + (b + c) = (a + b) + c$
A) Uses multiplication

B) $\{(-11) + (-3)\} + (-2) = -11 + \{(-3) + (-2)\}$

C) Shows commutative property

D) Involves subtraction

9. Correct answer: A) 17 s

Time taken by the bomb to hit the submarine

Height above sea level = +200 m
Depth below sea level = −650 m

Total distance fallen:$200 + 650 = 850$ m
Speed = 50 m/s

Time taken:$850 \div 50 = 17$ s

10. Correct answer: C) 62 mm

Difference in lengths of the plastic rod

Decrease in temperature:

Drop = 20°C

Every 5°C → decrease 8 mm

Number of intervals = $20 \div 5 = 4$

Decrease in length:$4 \times 8 = 32$ mm
Increase in temperature:

Rise = 15°C

Every 5°C → increase 10 mm

Number of intervals = $15 \div 5 = 3$

Increase in length:$3 \times 10 = 30$ mm
Difference between the two readings:$32 + 30 = 62$ mm

Worksheet – 4

1. Correct answer: D) 1270

What is the value of

$225 - (-145) + 1000 - (50) \times (+2)$?

$225 + 145 + 1000 - 100 = 1270$

2. Correct answer: C) $\frac{6}{5}$

Find the value of

$\{42 \div (-7)\} \div \{(-25) \div 5\}$

$42 \div (-7) = -6$
$(-25) \div 5 = -5$
$(-6) \div (-5) = \frac{6}{5}$

3. Correct answer: C) 110

Simplify:

$(36 \times 25 + 10 \times (-2)) \div 8$

$36 \times 25 = 900$
$10 \times (-2) = -20$
$900 - 20 = 880$
$880 \div 8 = 110$

4. Correct answer: C) 376

Simplify:

$$429 - \left[\frac{1}{2}\{-2 - (5 - (4 - (-1))) + 108\}\right]$$

$4 - (-1) = 5$
$5 - 5 = 0$
$-2 - 0 + 108 = 106$
$\frac{1}{2} \times 106 = 53$
$429 - 53 = 376$

5. Correct Answer: B) $\frac{P}{Q}$

Find values

$P = (21 - 4 + 32 \div 2) - (7 \times 3 + 11)$
$= (17 + 16) - (21 + 11) = 33 - 32$
$= 1$

$Q = (105 + 5) \div 10 + 9 - \{(25 \div 5 \times 7) - 15\}$
$= (110 \div 10 + 9) - (35 - 15)$
$= 20 - 20 = 0$

$R = 42 + [(30 - 15 + 13) \div 7 + 6] - 51$
$= 42 + (28 \div 7 + 6) - 51 = 42 + 10 - 51$
$= 1$

$S = (37 + 3) \times 2 - 49 - (6 \times 5)$
$= 80 - 49 - 30 = 1$

$Q/P = 0/1 \rightarrow$ definite

$P/Q = 1/0 \rightarrow$ undefined

$S/R = 1/1 \rightarrow$ definite

$R/S = 1/1 \rightarrow$ definite

6. Correct Answer: C

Option A

Denominator:
$(7 + 5 \times 3) \div 2 - 10 = (7 + 15) \div 2 - 10$
$= 22 \div 2 - 10 = 11 - 10$
$= 1$

Defined

Option B

Denominator:
$(4 + 5 \times 2) \div 7 + 15 - 3$
$= (4 + 10) \div 7 + 12$
$= 14 \div 7 + 12 = 2 + 12$
$= 14$

Defined

Option C

Denominator:
$(28 \div 4 + 14 - 21)(4 \times 7)$
First bracket: $7 + 14 - 21 = 0$
Whole denominator: $0 \times 28 = 0$
Undefined

Option D

Denominator:
$(9 \times 5) \div 5 + 35 - 31 = 45 \div 5 + 4$
$= 9 + 4 = 13$

Defined

7. Correct Answer: C) 15

$| -15 | = 15, | -18 | = 18$
$| 15 - 18 | = | -3 | = 3$
$3 + 12 = 15$

8. Correct Answer: D) 5°C/min

Cooling Rate

Initial temperature = $6°C$
Freezing temperature = $-34°C$

Temperature drop $= 6 - (-34) = 40°C$

Time = 8 minutes

Rate $= \frac{40}{8} = 5°C/\text{min}$

9. Correct Answer: C) 40 m

Distance from Sea Level

Below sea level = 40 m
Distance is always positive.

10. Correct Answer: C) 46

Simplify

$48 - [18 - \{16 - (5 - 4 - 1)\}]$
$(5 - 4 - 1) = 0$
$\{16 - 0\} = 16$
$[18 - 16] = 2$
$48 - 2 = 46$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: B

Greatest Quotient

(A) $20 \div (-4) = -5$
(B) $20 \div 4 = 5$
(C) $(-20) \div (-5) = 4$
(D) $(-30) \div 5 = -6$

Greatest value = 5

2. Correcte Answer: D

Identify the Wrong Option

For integers a, b:
Addition, subtraction, multiplication → integers
Division → not always an integer

3. Correcte Answer: B

Find the Other Number

Other number $= \frac{-90}{9} = -10$

4. Correcte Answer: D

Evaluate the Expression

$(-45) - (-39) + 42 - 28 - (-152)$
$- 85 + (-9)$
$= -45 + 39 + 42 - 28 + 152 - 85 - 9$
$= 66$

5. Correcte Answer: B) 1

$[65 \div \{5 + (12 - 8 \times 2)\}] - 4$
Vinculum: $12 - 8 = 4$
Parentheses: $4 \times 2 = 8$

Braces:$5 + 8 = 13$

Brackets:$65 \div 13 = 5$

Final Subtraction:$5 - 4 = 1$

6. Correct answer: C

For any whole number a, $a \times 1 = 1 \times a = a$.

So, 1 is the multiplicative identity.

7. Correct answer: B

Sum $= -300$, one number $= 60$

Other number $= -300 - 60 = -360$

8. Correct answer: C

$1 - 1$ forms a pair = 0

102 terms ⇒ 51 pairs

$51 \times 0 = 0$

9. Correct answer: B

9.$[90 \div \{3 \times (8 - 5 + 2)\}] + 5$

Bracket: $8 - 5 + 2 = 5$

$3 \times 5 = 15$, $90 \div 15 = 6$

$6 + 5 = 11$

10. Correcte Answer: D) −58

Determine the integer whose product with $-1$is 58.

$-1 \times x = 58 \Rightarrow x = -58$

11. Answer: A) 360

$(-2) \times 36 \times (-5)$

Negative × Positive = Negative

Negative × Negative = Positive

$$2 \times 36 \times 5 = 360$$

12. Correcte Answer: D) Not Defined

The multiplicative inverse of a number x is $\frac{1}{x}$.

For $x = 0$, $\frac{1}{0}$ is not defined.

13. Correcte Answer: C) 31

Original Expression:$[100 - \{20 + (15 \bar{-} 9 \times 3)\}] \div 2$

Vinculum:$15 - 9 = 6$

Parentheses $6 \times 3 = 18$

Braces $20 + 18 = 38$

Brackets $100 - 38 = 62$

Division $62 \div 2 = 31$

14. Correcte Answer: C) 0

$(-15) \times 0 \times (-18)$

$= 0$(any number $\times 0$

$= 0)$

15. Correcte Answer: A) −490

Sum $= -355$, one integer $= 135$

Other $= -355 - 135 = -490$

16. Correcte Answer: A) −4

"8 more than −12" means

$-12 + 8 = -4$

17. Correcte Answer: B) 15

Subtracting −15 from 0:

$$0 - (-15) = 15$$

18. Correcte Answer: A) 9/8

Multiplicative inverse of $\frac{8}{9}$is its reciprocal: $\frac{9}{8}$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

19. Correct answers: A, B, D

Which of the following statements are true?

A) −7 lies to the left of −3 → True
B) 0 is greater than all negative integers → True
C) −1 is the smallest negative integer → False (there is no smallest negative integer)
D) Every positive integer is greater than every negative integer → True

20. Correct answers: A, B, D

Which of the following operations on integers always give an integer?

A) −8 + 5 = −3 → Integer
B) 9 − (−4) = 13 → Integer
C) 7 ÷ 3 = 7/3 → Not an integer
D) −6 × 4 = −24 → Integer

21. Correct answers: A, C

Which of the following give the same result on both sides?

A) −3 + (−4 + −5) = −12
(−3 + −4) + −5 = −12 → Same
B) −10 − (6 − 2) = −14
(−10 − 6) − 2 = −18 → Not same
C) −2 × (3 × −4) = 24
(−2 × 3) × −4 = 24 → Same
D) 18 ÷ (6 ÷ 3) = 9
(18 ÷ 6) ÷ 3 = 1 → Not same

22. Correct answers: B, C

Which of the following expressions are not defined?

A) 0 ÷ 7 = 0 → Defined
B) 7 ÷ 0 → Not defined
C) −12 ÷ 0 → Not defined
D) 0 ÷ (−5) = 0 → Defined

23. Correct answers: A, C

Which of the following statements are correct?

A) |−8| = 8 → True
B) Absolute value of an integer is always positive → False (can be 0)
C) |0| = 0 → True
D) |−5| = −5 → False

ASSERTION AND REASON TYPE

24. Correct answer: A

Assertion (A): The sum of two negative integers is always a negative integer. (True)
Reason (R): When we add two negative numbers, their absolute values are added and the result is negative. (True)

Both A and R are true, and R correctly explains A.

25. Correct answer: D

Assertion (A): The product of a positive integer and a negative integer is a positive integer. $(e.g., 5\times-2=-10)$. (False)
Reason (R): Multiplying integers with opposite signs gives a negative product. (True)

A is false, but R is true.

26. Correct answer: B

Assertion (A): 0 is an integer but it is neither positive nor negative. (True)
Reason (R): 0 lies between −1 and 1 on the number line. (True)

Both A and R are true, but R does not correctly explain why 0 is neither positive nor negative.

27. Correct answer: A

Assertion (A): Subtracting a negative integer is the same as adding its positive. (True)
Reason (R): −(−5) = +5 (True)

Both A and R are true, and R correctly explains A.

28. Correct answer: D

Assertion (A): Every integer has a reciprocal. (False)
Reason (R): The reciprocal of 1 is 1, and the reciprocal of −2 is −1/2.

A is false (0 has no reciprocal), but R is true.

MATRIX MATCHING TYPE

29. Correct matching:
A–S, B–R, C–Q, D–P

Column-I calculations

A. $-8+(-3)=-11\rightarrow$ S
B. $27\div3\div3=9\div3=3\rightarrow$ R
C. $-4\times-4\times-4+1=-64+1=-63\rightarrow$ Q
D. $(14\div2)\div(49\div7)=7\div7=1\rightarrow$ P

30. Correct matching:
1–A, 2–C, 3–D, 4–B

1. $-6+(-4+-10)=-6-14=-20$
$(-6+-4)+-10=-10-10=-20\rightarrow$ A
2. $-12-(5-7)=-12-(-2)=-10$
$(-12-5)-7=-17-7=-24\rightarrow$ C
3. $-3\times(-2\times6)=-3\times-12=36$
$(-3\times-2)\times6=6\times6=36\rightarrow$ D
4. $-5\times(8-3)=-5\times5=-25$
$(-5\times8)-(-5\times3)=-40+15=-25\rightarrow$ B

COMPREHENSION TYPE

Passage - 1

Given temperatures (°C):Monday −4, Tuesday −6, Wednesday −3, Thursday −7, Friday −2, Saturday 0, Sunday −1

31. Correct answer: D) Saturday

Which day's temperature is the greatest integer?

Greatest temperature = 0°C (highest among all values) ⟶ Saturday

32. Correct answer: C) Saturday has the highest temperature

32. Which statement is true about the temperatures?

A) All are negative ⟶ (Saturday is 0)
B) Zero is the smallest → (−7 is smallest)
C) Saturday has the highest temperature
D) −1 is less than −7 ⟶ (−1 > −7)

33. Correct answer: C) 1°C

If Sunday's temperature increases by 2°C, what is the new temperature?

Sunday = −1°C

$-1 + 2 = 1°C$

Passage 2

Given movement (in floors):
Up 2 ⟶ +2
Down 5 ⟶ −5
Up 3 ⟶ +3
Down 4 ⟶ −4

34. Correct answer: B) −4 floors

Net change in floors

$+2 - 5 + 3 - 4 = -4$

35. Correct answer: C

Correct integer expression

$2 - 5 + 3 - 4$

36. Correct answer: C) It is a negative integer

Statement about the final position

Final position $= -4$(below ground level) ⟶ negative integer

INTEGER TYPE

37. Correcte Answer: 16

$(1 - 3) \times (3 - 5) \times (5 - 1)$
$= (-2) \times (-2) \times 4$
$= 16$

38. Correcte Answer: 5

Additive inverse of −23 is +23

$-18 + 23 = 5$

39. Correcte Answer: 0

Additive identity of −15 = 0
Multiplicative identity of 15 = 1

$0 \times 1 = 0$

40. Correcte Answer: −850

$17 \times (-34) + 16 \times (-17)$
$= -578 - 272 = -850$

41. Find the value of x in:

$(-16) \times 38 + 16 \times 58$
$= (-16) \times (38 - x)$

Let's simplify the left side first:

$(-16 \times 38) + (16 \times 58)$

To match the right side, factor out -16:

$(-16 \times 38) + (-16 \times -58)$
$= (-16) \times (38 + (-58))$
$= (-16) \times (38 - 58)$

Now, compare this with the right side of the equation:

$(-16) \times (38 - 58) = (-16) \times (38 - x)$

By comparing the expressions inside the brackets, we get:

$x = 58$

Previous Year Questions

1. Correct answer:D) Rohan, 40 points

Rohan's total = $45 + (-50) + 75 = 70$
Karan's total = $-45 + 50 + 25 = 30$

Difference = $70 - 30 = 40$

Rohan scored more by 40 points.

2. Correct answer:C) −6

Follow BODMAS:

$(-9) \times 7 = -63$
$(-15) \div (-3) = 5{,}5 \times 6 = 30$
$9 \times 3 = 27$
Now add:$-63 + 30 + 27 = -6$

3. Correct answer:B) (i) F (ii) T (iii) T

(i) False − Whole numbers do not include negative integers.
(ii) True − $140 \div (-35) = -4$.
(iii) True − −1 is the greatest negative integer.

4. Correct answer:D) Team A, 40 points

Team A total $45 + (-50) + 75 = 70$

Team B total $(-45) + 50 + 25 = 30$

Difference $70 - 30 = 40$

Team A scored more by 40 points.

5. Correct answer:A

On the number line:

Start at −4

Subtracting −2 means move 2 steps to the right

You reach −2

The diagram that shows a move from −4 to −2 is Option A.

6. Correct answer:A) 340

First term:$(-17) \times (-4) = 68, 68 \times (-1) = -68, -68 \times (-5) = 340$
Second term:Since the product includes 0,

$(-15) \times (-3) \times 0 \times (-13) \times (-11) = 0$
Now subtract:$340 - 0 = 340$

7. Correct answer:C

Descending order means arranging numbers from greatest to smallest.

Correct descending order of the given integers is:

$$18 > 6 > 0 > -12 > -37 > -43$$

8. Correct answer:C) Q, 45 points

Team P total:$-40 + 10 + 50 - 20 + 15 = 15$
Team Q total:$40 - 20 - 10 + 30 + 20 = 60$
Difference:$60 - 15 = 45$
So, Team Q scored more by 45 points.

9. Correct answer:A

Natural numbers ⊂ Whole numbers ⊂ Integers

Natural numbers (1, 2, 3, …) are part of whole numbers (0, 1, 2, …).

Whole numbers are part of integers (…, −2, −1, 0, 1, 2, …).

So, statement A is true.
All other options are false because integers include negative numbers, which are neither whole nor natural.

10. Correct answer:B) 22

Add profits and losses separately.

Profits:$21 + 11 + 17 + 25 = 74$

Losses:$-19 - 20 - 13 = -52$

Net profit:$74 - 52 = 22$

2. Fractions, Decimals, Rational Numbers

Solutions

Worksheet – 1

1. Correct Answer: A

$4\frac{1}{2} = \frac{9}{2}$

$\frac{9}{2} \times \frac{2}{3} = \frac{18}{6} = 3$

2. Correct Answer: B

$\frac{7}{9} \div \frac{14}{27} = \frac{7}{9} \times \frac{27}{14}$

$= \frac{189}{126} = \frac{3}{2}$

3. Correct Answer: C

4th multiple of 3 $\rightarrow a = 12$

5th multiple of 2 $\rightarrow b = 10$

$\frac{a}{b} = \frac{12}{10} = \frac{6}{5}$

$\frac{6}{5} \times \frac{p}{q} = \frac{2058}{135}$

$\frac{p}{q} = \frac{2058}{135} \div \frac{6}{5} = \frac{2058}{135} \times \frac{5}{6}$

$= \frac{343}{27} = \left(\frac{7}{3}\right)^3$

Result: $\left(\frac{7}{3}\right)^3$

4. Correct Answer: A

If $\frac{3}{5}$ of $\frac{2}{3}$ is added to -1

$\frac{3}{5} \times \frac{2}{3} = \frac{2}{5}$

$-1 + \frac{2}{5} = -\frac{3}{5}$

Result: $-\frac{3}{5}$

5. Correct Answer: C

Weight of 1 bag = $\frac{6}{7}$kg

Number of bags = 3

$3 \times \frac{6}{7} = \frac{18}{7} = 2\frac{4}{7}$

Result: $2\frac{4}{7}$kg

6. Correct Answer: B

$x = \frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$

$y = \frac{3}{4} \times \frac{8}{12} = \frac{1}{2}$

$(x + y) + (x - y) = 2x = 2 \times \frac{1}{2} = 1$

7. Correct Answer: A

Cost of $\frac{8}{15}$ of $1\frac{1}{2}$metres of cloth at Rs. 10 per metre.

$1\frac{1}{2} = \frac{3}{2}$

$\frac{8}{15} \times \frac{3}{2} = \frac{4}{5}$ m

Cost $= \frac{4}{5} \times 10 = Rs.8$

8. Correct Answer: A

I work $\frac{2}{5}$ of 20 hours everyday for 12 days. Wages are $\frac{3}{4}$ of Rs. 12 per hour.

Hours per day $= \frac{2}{5} \times 20 = 8$

Total hours $= 8 \times 12 = 96$

Wage per hour $= \frac{3}{4} \times 12 = 9$

Total earnings $= 96 \times 9 = Rs.864$

9. Correct Answer: A

Cost of $12\frac{1}{2}$kg potatoes at Rs. $40\frac{4}{5}$per kg.

$12\frac{1}{2} = \frac{25}{2}, 40\frac{4}{5} = \frac{204}{5}$

Cost $= \frac{25}{2} \times \frac{204}{5} = 510$

10. Correct Answer: C

In a factory, 1500 are Americans.

Chinese workers = $\frac{3}{5}$of AmericansIndian workers = $\frac{5}{9}$of Chinese workers.

Chinese $= \frac{3}{5} \times \mathbf{1500} = \mathbf{900}$

Indian $= \frac{5}{9} \times \mathbf{900} = \mathbf{500}$

Total workers $= \mathbf{1500} + \mathbf{900} + \mathbf{500} = \mathbf{2900}$

Worksheet – 2

1. Correct Answer: A

$\mathbf{1}$ cm $= \mathbf{10}$ mm

$\mathbf{25} \div \mathbf{10} = \mathbf{2.5}$ cm

2. Correct Answer: A

A rope of length 149.236 m is cut equally into 8 pieces.

$\mathbf{149.236} \div \mathbf{8} = \mathbf{18.6545}$ m

3. Correct Answer: D

If $\mathbf{473} \times \mathbf{2962} \times \mathbf{12} = \mathbf{16812312}$, find $(\mathbf{4.73} \times \mathbf{0.2962} \times \mathbf{1.2}) \times \mathbf{100}$

$\mathbf{4.73} = \frac{\mathbf{473}}{\mathbf{100}}, \mathbf{0.2962} = \frac{\mathbf{2962}}{\mathbf{10000}}, \mathbf{1.2} = \frac{\mathbf{12}}{\mathbf{10}}$

$(\mathbf{4.73} \times \mathbf{0.2962} \times \mathbf{1.2}) \times \mathbf{100}$

$= \frac{\mathbf{473} \times \mathbf{2962} \times \mathbf{12}}{\mathbf{100} \times \mathbf{10000} \times \mathbf{10}} \times \mathbf{100}$

$= \frac{\mathbf{16812312}}{\mathbf{100000}} = \mathbf{168.12312}$

4. Correct Answer: A (17055.7500)

If $\mathbf{72.369} \times \mathbf{200}$ is added to $\mathbf{5.1639} \times \mathbf{500}$

$\mathbf{72.369} \times \mathbf{200} = \mathbf{14473.8}$

$\mathbf{5.1639} \times \mathbf{500} = \mathbf{2581.95}$

Total $= \mathbf{14473.8} + \mathbf{2581.95} = \mathbf{17055.75}$

5. Correct Answer: C

Find the value of $\mathbf{3.25} \times \mathbf{2.32}$

$\mathbf{3.25} \times \mathbf{2.32} = \mathbf{7.54}$

6. Correct Answer: C

$\mathbf{6.05} \times \mathbf{5} = \mathbf{30.25}$
$\mathbf{26.2} + \mathbf{38.05} = \mathbf{64.25}$
$\mathbf{64.25} - \mathbf{30.25} = \mathbf{34}$

7. Correct Answer: C

Convert 75 g into kilograms.

$\mathbf{1}$ kg $= \mathbf{1000}$ g
$\mathbf{75} \div \mathbf{1000} = \mathbf{0.075}$ kg

8. Correct Answer: C

A person travelled 3.4 hrs at 25 kmph, 4.2 hrs at 15 kmph, 6.3 hrs at 35 kmph.

$\mathbf{3.4} \times \mathbf{25} = \mathbf{85}$ km
$\mathbf{4.2} \times \mathbf{15} = \mathbf{63}$ km
$\mathbf{6.3} \times \mathbf{35} = \mathbf{220.5}$ km
Total distance $= \mathbf{85} + \mathbf{63} + \mathbf{220.5}$
$= \mathbf{368.5}$ km

9. Correct Answer: A

Monika walks 1.25 miles in 22.4 minutes.

Time to walk 1 mile?

$\mathbf{22.4} \div \mathbf{1.25} = \mathbf{17.92}$ minutes

10. Correct Answer: A

Thousandths place = 3rd digit after decimal.

3.0095 → 9 is in thousandths

Others do not satisfy.

Worksheet – 3

1. Correct Answer: D) Infinite

Between any two rational numbers, we can always find more rational numbers endlessly.

2. Correct Answer: A

Other number $= \frac{\mathbf{-9}}{\mathbf{16}} \div \frac{\mathbf{-4}}{\mathbf{15}} = \frac{\mathbf{9}}{\mathbf{16}} \times \frac{\mathbf{15}}{\mathbf{4}}$
$= \frac{\mathbf{135}}{\mathbf{64}} = \mathbf{2}\frac{\mathbf{7}}{\mathbf{64}}$

3. Correct Answer: B

Subtract the sum of $-\frac{5}{6}$ and $-\mathbf{1}\frac{\mathbf{3}}{\mathbf{5}}$ from the sum of $\mathbf{2}\frac{\mathbf{2}}{\mathbf{3}}$ and $-\mathbf{6}\frac{\mathbf{2}}{\mathbf{5}}$.

$-\frac{\mathbf{5}}{\mathbf{6}} + \left(-\mathbf{1}\frac{\mathbf{3}}{\mathbf{5}}\right) = -\frac{\mathbf{5}}{\mathbf{6}} - \frac{\mathbf{8}}{\mathbf{5}} = -\frac{\mathbf{73}}{\mathbf{30}}$

$\mathbf{2}\frac{\mathbf{2}}{\mathbf{3}} + \left(-\mathbf{6}\frac{\mathbf{2}}{\mathbf{5}}\right) = \frac{\mathbf{8}}{\mathbf{3}} - \frac{\mathbf{32}}{\mathbf{5}} = -\frac{\mathbf{56}}{\mathbf{15}}$

$$-\frac{56}{15}-\left(-\frac{73}{30}\right)=-\frac{112}{30}+\frac{73}{30}$$
$$=-\frac{39}{30}=-\frac{13}{10}$$

4. Correct Answer: A

Convert to improper fractions:

$-2\frac{15}{17}=-\frac{49}{17}, 3\frac{5}{34}=\frac{107}{34}$

Sum: $-\frac{49}{17}+\frac{107}{34}=-\frac{98}{34}+\frac{107}{34}=\frac{9}{34}$

Difference: $-\frac{49}{17}-\frac{107}{34}=-\frac{98}{34}-\frac{107}{34}=-\frac{205}{34}$

Required value: $\frac{9}{34}\div\left(-\frac{205}{34}\right)=-\frac{9}{205}$

5. Correct Answer: A

If $\frac{2}{3}$ is multiplied by 4 in both numerator and denominator

$\frac{2\times4}{3\times4}=\frac{8}{12}=\frac{2}{3}$

6. Correct Answer: C

Which of the following is a rational number?

A rational number can be written as $\frac{p}{q}$, where $q\neq0$.

$\frac{3}{6}=\frac{1}{2}$

7. Correct Answer: B

Convert the decimal number 1.75 to rational form

$$1.75=\frac{175}{100}=\frac{7}{4}$$

8. Correct Answer: B

Which fraction represents the decimal number 15.75 before simplification?

$$15.75=\frac{1575}{100}$$

9. Correct Answer: C

The standard form of $-\frac{48}{60}$ is

$$-\frac{48}{60}=-\frac{4}{5}$$

10. Correct Answer: C

What is the rational form of 15.75?

$$15.75=\frac{1575}{100}=\frac{63}{4}$$

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: D

$2\frac{1}{4}+3+1\frac{1}{4}$

$2\frac{1}{4}+1\frac{1}{4}=3\frac{1}{2}$

$3\frac{1}{2}+3=6\frac{1}{2}$

2. Correct Answer: C

$20\frac{5}{7}-15\frac{1}{7}$

$$(20-15)+\left(\frac{5}{7}-\frac{1}{7}\right)=5+\frac{4}{7}$$

3. Correct Answer: A

$2\frac{1}{3}\times1\frac{5}{14}\times1\frac{4}{5}$

$2\frac{1}{3}=\frac{7}{3}, 1\frac{5}{14}=\frac{19}{14}, 1\frac{4}{5}=\frac{9}{5}$

$\frac{7}{3}\times\frac{19}{14}\times\frac{9}{5}=\frac{57}{10}=5\frac{7}{10}$

4. Correct Answer: A

$7\div\frac{3}{8}$

$7\times\frac{8}{3}=\frac{56}{3}=18\frac{2}{3}$

5. Correct Answer: A

If $a=\frac{2}{3}$ and $b=\frac{4}{9}$

$\frac{2}{3}=\frac{6}{9}>\frac{4}{9}$

6. Correct Answer: C) Equivalent fractions

The fractions $\frac{1}{2},\frac{2}{4},\frac{3}{6}$ are

$\frac{1}{2}=\frac{2}{4}=\frac{3}{6}$

All have the same value.

7. Correct Answer: C

At Venkat's birthday party, $\frac{5}{7}$ part of the cake was distributed.

Cake left out = ?

$$1-\frac{5}{7}=\frac{2}{7}$$

8. Correct Answer: C

Which represents $\frac{24}{30}$ in its simplest form?

$$\frac{24}{30} = \frac{24 \div 6}{30 \div 6} = \frac{4}{5}$$

9. Correct Answer: D

Which of the following is equivalent to $\frac{13}{25}$?

$$\frac{13}{25} = \frac{52}{100} = 0.52$$

10. Correct Answer: A

Change the mixed fraction $5\frac{1}{2}$ into an improper fraction.

$$5\frac{1}{2} = \frac{(5 \times 2) + 1}{2} = \frac{11}{2}$$

11. Correct Answer: A

Arrange $\frac{3}{-5}, \frac{-7}{10}, \frac{8}{-10}, \frac{-17}{15}$ in ascending order

Convert to comparable values:

$$\frac{-17}{15} \approx -1.13,\ \frac{8}{-10} = -0.8,\ \frac{-7}{10} = -0.7,\ \frac{3}{-5} = -0.6$$

Ascending order (smallest to greatest):

$$\frac{-17}{15},\ \frac{8}{-10},\ \frac{-7}{10},\ \frac{3}{-5}$$

12. Correct Answer: C

If $\frac{2}{3}, \frac{3}{4}, \frac{1}{3}, \frac{5}{6}$ are fractions, their ascending order is

$$\frac{1}{3} < \frac{2}{3} < \frac{3}{4} < \frac{5}{6}$$

13. Correct Answer: C

Three pizzas are cut into 9 equal pieces each.

Eaten: $\frac{8}{9},\ \frac{2}{3} = \frac{6}{9},\ \frac{7}{9}$

Total eaten:

$$\frac{8}{9} + \frac{6}{9} + \frac{7}{9} = \frac{21}{9}$$

Total pizzas $= \frac{27}{9}$

Left:

$$\frac{27}{9} - \frac{21}{9} = \frac{6}{9} = \frac{2}{3}$$

Fraction of 3 pizzas left:

$$\frac{2}{3} \div 3 = \frac{2}{9}$$

14. Correct Answer: C

During an 8-hour workday, Bob spends 2 hours on the phone

$$\frac{2}{8} = \frac{1}{4}$$

15. Correct Answer: C

Find 0.63×0.42

$63 \times 42 = 2646$

Total decimal places $= 4$

$$0.2646$$

16. Correct Answer: A

$$2.9 \div 0.8758$$

$$2.9 \div 0.8758 \approx 3.31$$

17. Correct Answer: D

Compare decimals:

$0.043 > 0.0043$

18. Correct Answer: C

$78.09 + 19.367$

$78.090 + 19.367 = 97.457$

19. Correct Answer: A

6.723, 6.0723, 6.7023, 6.7

Greatest is 6.723.

20. Correct Answer: A

Cost of $6\frac{1}{4}$ kg apples is Rs. 400. Find rate per kg.

$6\frac{1}{4} = 6.25$ kg

$400 \div 6.25 = 64$

21. Correct Answer: D

Total juice in two bottles:

Apple = 1.42 L, Grape = 1.89 L

$1.42 + 1.89 = 3.31$ litres

22. Correct Answer: B) $223.62

Six friends split the cost of gasoline evenly.Each friend paid $37.27.

$37.27 \times 6 = 223.62$

23. Correct Answer: A) $28.99

Cost of washing machine = $428.98

Sale price = $399.99

$428.98 - 399.99 = 28.99$

24. Correct Answer: D) 98.44 square inches

Area of rectangular opening

Length = 11.25 inches, Width = 8.75 inches

$11.25 \times 8.75 = 98.4375$

Rounded to nearest hundredth:

98.44 square inches

25. Correct Answer: A) 14.5 L

A tank contains 72.5 litres of water shared equally among 5 containers.

$72.5 \div 5 = 14.5$ L

26. Correct Answer: C

A non-terminating recurring decimal goes on forever and repeats.

0.25 → terminating

1.375 → terminating

0.333... → repeating forever

0.101001 → terminating

27. Correct Answer: A

Which pair of rational numbers has the same value?

$\frac{6}{8} = \frac{3}{4}, \frac{9}{12} = \frac{3}{4}$

Both are equal.

28. Correct Answer: B

Which of the following comparisons is correct?

Solution:

A: $-\frac{7}{11} < -\frac{5}{11}$(False)

B: $-\frac{3}{8} < -\frac{1}{4}$(True)

C: $\frac{6}{7} > \frac{6}{9}$(False)

D: $\frac{4}{5} > \frac{3}{5}$(False)

Only option B is correct.

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

29. Correct Answer: B, D

Which of the following fractions are already in simplest form?

$\frac{6}{8} = \frac{3}{4}$

$\frac{7}{11}$

$\frac{9}{15} = \frac{3}{5}$

$\frac{5}{13}$

30. Correct Answer: C

Which cannot be written as rational numbers?

0.25 → terminating → rational

1.375 → terminating → rational

0.101001000... → non-terminating, non-recurring → irrational

0.666... → recurring → rational

31. Correct Answer: A, C

Which decimals are like decimals?

(Like decimals have the same number of decimal places)

2.35 and 7.48 → both have 2 decimal places

4.6 and 3.245 → 1 and 3 decimal places

9.125 and 1.125 → both have 3 decimal places

0.7 and 0.70 → 1 and 2 decimal places

32. Correct Answer: A, B, C

Which statements about decimal place value are correct?

In 4.58, 5 is in tenths place

In 6.203, 3 is in thousandths place

In 9.04, 4 is in hundredths place

In 7.6, 6 is in tenths place (not hundredths)

33. Correct Answer: A, B, D

Which statements are correct?

Positive rational numbers lie to the right of 0

Negative rational numbers lie to the left of 0

Only integers can be shown on number line

Zero is neither positive nor negative

ASSERTION AND REASON TYPE

34. Correct Answer: C(Assertion is correct, Reason is incorrect)

Assertion: Every fraction is a rational number.
Reason: Rational numbers are only whole numbers.

Explanation:

Assertion is true (every fraction can be written as $\frac{p}{q}$).

Reason is false (rational numbers include fractions, integers, and decimals, not only whole numbers).

35. Correct Answer: A(Both Assertion and Reason are correct and Reason is the correct explanation)

Assertion: 0.25 is a terminating decimal.
Reason: It can be expressed as the fraction $\frac{1}{4}$.

Explanation:

0.25 ends after two decimal places → terminating.

$0.25 = \frac{1}{4}$, which correctly explains why it is terminating.

36. Correct Answer: C*(Assertion is correct, Reason is incorrect)*

Assertion: Rational numbers can be positive or negative.
Reason: Rational numbers are always greater than zero.

Explanation:

Assertion is correct (rational numbers can be positive, negative, or zero).

Reason is incorrect (negative rational numbers exist).

37. Correct Answer: A
(Both Assertion and Reason are correct and Reason is the correct explanation.)

Assertion: Every terminating or recurring decimal is a rational number.
Reason: Every terminating or recurring decimal can be written in the form $\frac{p}{q}$, where p and q are integers and $q \neq 0$.

Explanation:

The Assertion is correct.

The Reason is correct and properly explains the assertion.

38. Correct Answer: A
(Both Assertion and Reason are correct and Reason is the correct explanation.)

Assertion: $-\frac{3}{7}$ is a rational number.
Reason: Rational numbers include negative fractions.

Explanation:

The Assertion is correct.

The Reason is correct and explains why the assertion is true.

MATRIX MATCHING TYPE

39. 1. $\frac{2}{3} + \frac{5}{6} - \frac{2}{9}$

LCM of 3, 6, 9 = 18

$\frac{12}{18} + \frac{15}{18} - \frac{4}{18} = \frac{23}{18}$

Matches with (a)

2. $\frac{5}{6} \times 6\frac{3}{5}$

$6\frac{3}{5} = \frac{33}{5}$

$\frac{5}{6} \times \frac{33}{5} = \frac{33}{6} = \frac{11}{2}$

Matches with (b)

3. $12 \div \frac{36}{37}$

$12 \times \frac{37}{36} = \frac{37}{3}$

Matches with (c)

4. 6 of $\frac{3}{12}$

$6 \times \frac{3}{12} = \frac{18}{12} = \frac{3}{2}$

Matches with (d)

40. Match Column-I with Column-II

1. $2.3 \times 4.2 = 9.66$
→ Matches with (b)

2. $2.16 + 4.92 = 7.08$
→ Matches with (a)

3. $60.009 \times 100 = 6000.9$
→ Matches with (d)

4. $99.623 - 1.63 = 97.993$
→ Matches with (c)

COMPREHENSION TYPE

Passage - 1

Given:

Total slices = 8

Riya ate = 2 slices

Aarav ate = 1.5 slices

Meera ate = 2.5 slices

Kabir ate = remaining slices

41. Correct Answer: B

Aarav ate 1.5 slices. What fraction of the pizza is this?

$$\frac{1.5}{8} = \frac{3/2}{8} = \frac{3}{16}$$

42. Correct Answer: B

What decimal represents Meera's share of the pizza?

$\frac{2.5}{8} = \frac{5/2}{8} = \frac{5}{16}$

$\frac{5}{16} = 0.3125$

43. Correct Answer: C

Kabir's share of the pizza as a rational number is

Slices eaten by others:

$2 + 1.5 + 2.5 = 6$

Remaining slices:

$8 - 6 = 2$

Fraction: $\frac{2}{8} = \frac{1}{4}$

Passage – 2

Given data

Day 1 = 127.5 L

Day 2 = 132.25 L

Day 3 = 128 L

44. Correct Answer: C (387.75 liters)

What is the total amount of water used over the three days?

$127.5 + 132.25 + 128 = 387.75$ liters

45. Correct Answer: C(It lies between the minimum and maximum daily usage)

Which statement is TRUE about the average water usage?

Daily usages range from 127.5 L (minimum) to 132.25 L (maximum). The average of numbers always lies between the minimum and maximum values.

46. Correct Answer: C (129.25 liters)

$\frac{387.75}{3} = 129.25$ liters

INTEGER TYPE

47. Answer: 40 pieces

Shikha cuts 50 m of cloth into pieces of 1.25 m each. $50 \div 1.25 = 40$

48. Correct Answer: $-\frac{1}{24}$

Subtract $-\frac{5}{6}$ from $-\frac{7}{8}$.

$-\frac{7}{8} - \left(-\frac{5}{6}\right) = -\frac{7}{8} + \frac{5}{6}$

LCM of 8 and 6 = 24

$$-\frac{21}{24} + \frac{20}{24} = -\frac{1}{24}$$

49. Correct Answer: $-\frac{135}{4}$

Product of $6\frac{2}{3} \times \left(-5\frac{1}{16}\right)$

$6\frac{2}{3} = \frac{20}{3}, -5\frac{1}{16} = -\frac{81}{16}$

$\frac{20}{3} \times \left(-\frac{81}{16}\right) = -\frac{1620}{48} = -\frac{135}{4}$

50. Correct Answer: 69

Two rational numbers $\frac{23}{29} = \frac{x}{87}$ are equal. Find x.

$$x = \frac{23 \times 87}{29} = 23 \times 3 = 69$$

Previous Year Questions

1. Correct Answer: A) ₽14.56

Total cost of wheat: $145.6 \times 11 =$ ₽1601.6

Cost per person: $1601.6 \div 110 =$ ₽14.56

2. Correct Answer: B) 238.172

First find the sum:

$208.056 + 48.685 = 256.741$

Now subtract:

$256.741 - 18.569 = 238.172$

Correct Answer: B) 238.172

3. Correct Answer: B) $\frac{45}{33}$

Let the other fraction be x.

$\frac{3}{5} \times x = \frac{9}{11}$

$x = \frac{9}{11} \div \frac{3}{5} = \frac{9}{11} \times \frac{5}{3} = \frac{45}{33}$

4. Correct Answer: A) 41.40

p=6.09÷0.7=8.7

$q = 9.81 \div 0.3 = 32.7$

$p + q = 8.7 + 32.7 = 41.4$

5. Correct Answer: A) ₽878.70

Total amount spent:

$736.45 + 384.85 = 1121.30$

Amount returned:

$$2000 - 1121.30 = 878.70$$

6. Correct Answer: D) 56.25

Let the number be x.

$\frac{3}{7}x = 33.75$

$x = 33.75 \times \frac{7}{3} = 78.75$

Now find $\frac{5}{7}$ of the number:

$$\frac{5}{7} \times 78.75 = 56.25$$

7. Correct Answer: A) 7612.43 kg

Add the quantities sold on three days:

$3152.50 + 2905.75 + 1554.18 = 7612.43$ kg

8. Correct Answer: C) 2004

Convert the mixed numbers into improper fractions:

$15\frac{3}{14} = \frac{213}{14}$

$11\frac{5}{21} = \frac{236}{21}$

Find the difference:

LCM of 14 and 21 = 42

$\frac{213}{14} = \frac{639}{42}, \frac{236}{21} = \frac{472}{42}$

$\frac{639}{42} - \frac{472}{42} = \frac{167}{42}$

Multiply by 504:

$$\frac{167}{42} \times 504 = 167 \times 12 = 2004$$

9. Correct Answer: B) 125 minutes

Convert all hours to minutes:

$\frac{5}{6} \times 60 = 50$ minutes

$\frac{3}{4} \times 60 = 45$ minutes

$\frac{1}{2} \times 60 = 30$ minutes

Total time: $50 + 45 + 30 = 125$ minutes

10. Correct Answer: C) ₽1233.75

Cost of shoes:

$5 \times 215.25 = ₽1076.25$

Cost of socks:

$3 \times 52.50 = ₽157.50$

Total cost:

$1076.25 + 157.50 = ₽1233.75$

3. Data Handling

Solutions

Worksheet – 1

1. Correct Answer: B) 3/11

Lace-up shoes = 3 × 10 = 30

Rest = (4 + 3 + 4) × 10 = 110

Required fraction = 30 / 110 = **3/11**

2. Correct Answer: C) 5

Tallies are usually marked in a bunch of **5**.

3. Correct Answer:

🍎 = 3 oranges

Venkat = 4 apples = 4 × 3 = 12 oranges
Durga Sai = 6 apples = 6 × 3 = 18 oranges

Difference = 18 − 12 = 6 oranges ✅

4. Correct option B) 81

Sunday = 16
Monday = 9
Tuesday = 10
Wednesday = 13
Thursday = 16
Friday = 8
Saturday = 9

Now add them:

$$16 + 9 + 10 + 13 + 16 + 8 + 9 = 81$$

5. Correct Answer: B) Observation

Each numerical value collected in data is called an **observation**.

Example:
If marks are 45, 60, 75 — each number (45, 60, 75) is an **observation**.

6. Correct Answer: B) Fantasy

6. Fashion magazines = 4

Drama = 3

Fantasy = 6

Sports fiction = 5

Biographies = 2

Historical fiction = 4

The maximum number of votes is 6 for Fantasy.

7. Correct Answer: B) 48

Count the monkey symbols:

Mon = 4

Tue = 2

Wed = 5

Thurs = 1

Total bananas =
4 + 2 + 5 + 1 = **12**

(The question also confirms total = 12.)

Now multiply by 4: $12 \times 4 = 48$

8. Correct Answer= C) 189

Count the symbols:

Rose = 7

Tulip = 4

Orchid = 6

Lily = 4

Total symbols =

7 + 4 + 6 + 4 = 21

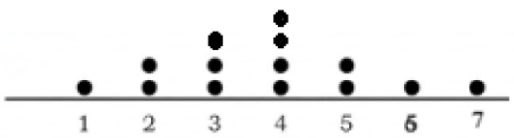
Now multiply by 9:

21×9=189

9. Correct Answer: A) 4

Arrange in order

1, 2, 2, 3, 3, 3, 4, 4, 4, 4, 5, 5, 6, 7



10. Correct Answer: B) 1,200

8–9 h = 50
9–10 h = 400
10–11 h = 250
11–12 h = 250

12–13 h = 150
13–14 h = 100

Now add them:

$$\mathbf{50 + 400 + 250 + 250 + 150 + 100 = 1,200}$$

Worksheet-II

1. Correct Answer: A) 2 : 3

1. Mode : Median = 9 : 7
Using relation (empirical):

$$\text{Mode} = 3\text{Median} - 2\text{Mean}$$

Let Mode = 9x, Median = 7x

$$9x = 3(7x) - 2\text{Mean}$$
$$9x = 21x - 2\text{Mean}$$
$$2\text{Mean} = 12x$$
$$\text{Mean} = 6x$$

Mean : Mode = 6x : 9x = **2 : 3**

2. Correct Answer: B) 34

Mean of 17 observations = 30
Total = 17 × 30 = 510

Deleted: 20 + 30 + 40 = 90
New total = 510 − 90 = 420

Add 90 ⟶ New total = 420 + 90 = 510

New number of observations = 15

New mean = 510 ÷ 15 = **34**

3. Correct Answer: A) 19

Mean of 19 observations = 20
Total = 19 × 20 = 380

Deleted: 30 + 20 = 50
New total = 380 − 50 = 330

Added: 15 + 25 + 10 = 50
New total = 330 + 50 = 380

New observations = 20

Mean = 380 ÷ 20 = **19**

4. Correct Answer: A) 7

Data: 13, 16, 12, 14, 19, 12, 14, 13, 14

14 appears 3 times ⟶ Mode = 14

½ of mode = 14 ÷ 2 = **7**

5. Correct Answer: B) 1

Mean = 12
Sum = 132

$$n = \frac{132}{12} = 11$$

Since n = 10 + ___

Blank = **1**

6. Correct Answer: D) 8

Numbers: 5, 7, x, 10, 7, 5
Mean = 7

Total required = 6 × 7 = 42

Sum of known numbers = 5 + 7 + 10 + 7 + 5 = 34

$$x = 42 - 34 = 8$$

7. Correct Answer: C) 9

Median of 9, 12, 11, 10, 8, 9, 7

Arrange in ascending order:
7, 8, 9, 9, 10, 11, 12

Middle value = 9

8. Correct Answer: B) 8

Mode of 7, 8, 9, 8, 9, 10, 6, 11, 11, 8

Count frequency:
8 appears 3 times (highest)

9. Correct Answer: B) 3

There are 10 numbers.
Median = average of 5th and 6th terms.

Given arranged data:
11, 12, 14, 18, **x+2, x+4**, 30, 32, 35, 41

Median = 24

$$\frac{(x+2)+(x+4)}{2} = 24$$
$$\frac{2x+6}{2} = 24$$
$$2x + 6 = 48$$
$$2x = 42$$
$$x = 21$$
$$\frac{x}{7} = \frac{21}{7} = 3$$

10. Correct Answer: A) 4

Mean of first 7 natural numbers

Numbers: 1, 2, 3, 4, 5, 6, 7

Mean = (1+2+3+4+5+6+7) ÷ 7
= 28 ÷ 7 = 4

COMPITITIVE CORNER

SINGLE ANSWER CORRECT TYPE

1. Correct Answer: C) 9

Mean of 5 observations = 11

Observations:
x, x+2, x+4, x+6, x+8

Sum = 5x + 20

$$\frac{5x+20}{5} = 11$$
$$5x + 20 = 55$$
$$5x = 35$$
$$x = 7$$

First 3 observations:
7, 9, 11

Mean =

$$\frac{7+9+11}{3} = \frac{27}{3} = 9$$

2. Answer: C) $0.48

Prices:
1.79, 1.61, 1.96, 1.84, 1.75, 2.09

Highest = 2.09
Lowest = 1.61

Range =

$$2.09 - 1.61 = 0.48$$

3. Correct Answer: A) 24

1. Arrange in Order

Since $x$is positive, the larger the denominator, the smaller the value.

$$\frac{x}{5} < \frac{x}{4} < \frac{x}{3} < \frac{x}{2} < x$$

2. Identify the Median

There are 5 numbers, so the median is the middle (3rd) value.

The middle value is:

$$\frac{x}{3}$$

3. Solve for x

$$\frac{x}{3} = 8$$
$$x = 24$$

4. Correct Answer: D) 80

Key: 🔵 = 20 videos

October has **4 circles**.

So, $4 \times 20 = 80$

5. Correct Answer: B) 52 units

Scale: 1 cm = 10 units
Length of bar = 5.2 cm

$$5.2 \times 10 = 52$$

6. Correct Answer: D) 26

From the bar graph:

Sports = 7

Dance = 5

Painting = 6

Drama = 8

Total students =

$$7 + 5 + 6 + 8 = 26$$

7.

8. Correct Answer: B) 60

From the pie chart:

Magazines = 35%

Novels = 25%

Journals = 20%

Total so far = 35 + 25 + 20 = 80%

So, **Newsletters = 100% – 80% = 20%**

Total books = 300

$$20\% \text{ of } 300 = \frac{20}{100} \times 300 = 60$$

9. Answer: A) x + 4

Observations:
x, x+2, x+4, x+6, x+8

Sum =
5x + 20

Mean =

$$\frac{5x+20}{5} = x + 4$$

10.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct Answer: B, C

Answer: A) x + 4
Arrange in order:
25, 31, 32, 37, 39, 42, 43, 45, 46

Median = 5th observation = **39**

✓ **B) 39**
✓ **C) 5th observation is median**

12. **Correct Answer: D**

Mean of 5 numbers = 27
Total = 5 × 27 = 135
Mean of remaining 4 numbers = 25
Total = 4 × 25 = 100
Excluded number = 135 − 100 = **35**
✓ **D) 35**

13. **Correct Answer: A, B**

Data: 7, 21, 11, 5, 9, 20, 32, 28
Highest = 32
Lowest = 5
Range = 32 − 5 = **27**
✓A) 32-5 **B) 27**

14. **Correct Answer: A, B, D**

(Fair coin)
Probability of Head = 1/2
Probability of Tail = 1/2
Head and Tail equally likely
✓ **A, B, and D**

15. **Correcct Answer: A, C**

(Fair die)
Getting 3 → 1/6 ✓
Getting even number → 3/6 = 1/2 ✗
Getting 6 → 1/6 ✓
Getting number less than 7 → 6/6 = 1 ✗
✓ **A and C**

ASSERTION AND REASON TYPE

16. **Correct Answer: C) A is true, but R is false**

Assertion (A): True ✓ (Mode is the most frequent value.)
Reason (R): False ✗ (Middle value is called median, not mode.)

17. **Correct Answer: B) Both A and R are true, but R is not the correct explanation of A**

Assertion (A): True ✓ (Bar graphs compare categories.)
Reason (R): True ✓ (Bars have equal width.)
But equal width is not the explanation for comparing categories.

18. **Correct Answer: C) A is true, but R is false**

Assertion (A): True ✓ (Mean = sum ÷ number of items.)
Reason (R): False ✗ (Most frequent value is mode, not mean.)

19. **Correct Answer: A) Both A and R are true, and R is the correct explanation of A**

Assertion (A): True ✓ (Pictographs use symbols to represent data.)
Reason (R): True ✓ (Each symbol represents a fixed number of items.)
The reason correctly explains the assertion.

20. **Correct Answer: C) A is true, but R is false**

Assertion (A): True ✓ (Data handling helps in making decisions.)
Reason (R): False ✗ (Data handling uses actual data, not random guessing.)

MATCH MATRIX TYPE

21. **Correct Answer: A–2, B–1, C–3, D**

A. **Mode** → **2** (Value that appears most often)
B. **Mean** → **1** (Sum of values ÷ number of values)
C. **Median** → **3** (Middle value when data is arranged)
D. **Pictograph** → **4** (Representation using symbols)

22. **Correct Answer: A–2, B–1, C–3, D–4**

A. Tossing a coin → 2 (1/2)
B. Rolling a die → 1 (1/6)
C. Getting an even number on a die → 3 (3/6)
D. Drawing a red card from 52 cards → 4 (26/52)

COMPREHENSION TYPE

Passage-1

23. **Correct Answer: A) 2/5**

Probability of picking a red ball

$$\frac{4}{10} = \frac{2}{5}$$

24. Answer: A) 1/5

Probability of picking a green ball

$$\frac{2}{10} = \frac{1}{5}$$

25. Answer: A) 7/10

Probability of picking a ball that is NOT blue

Not blue = Red + Green + Yellow
= 4 + 2 + 1 = 7

$$\frac{7}{10}$$

26. Correct Answer: A) 1 July–8 July

Greatest rise

1–8 July → 135 to 170 (increase of 35) ✓
15–22 July → 142 to 158 (increase of 16)
22–29 July → 158 to 160 (increase of 2)
5–12 Aug → 150 to 165 (increase of 15)

Greatest rise = 1 July–8 July

Passage-2

27. Correct Answer: D) 5

Weeks with more than 145 deaths

8 July (170)
22 July (158)
29 July (160)
5 Aug (150)
12 Aug (165)

Total = 5 weeks

28. Correct Answer: C) 5

Weeks with less than 165 deaths

1 July (135)
15 July (142)
22 July (158)
29 July (160)
5 Aug (150)

Total = 5 weeks

INTEGER TYPE

29. Answer: 13

Mean of 6 numbers = 18

Total sum =

$$6 \times 18 = 108$$

Sum of five numbers = 95

Sixth number =

$$108 - 95 = 13$$

30. Correct Answer: 18

Data: 12, 18, 25, 10, 30, 22, 15

Arrange in order:
10, 12, 15, 18, 22, 25, 30

Middle value (4th term) = **18**

31. Correct Answer: 90°

25% of total circle

Total angle in a circle = 360°

$$\frac{25}{100} \times 360 = 90°$$

32. Correct Answer: 1/3

Multiples of 3 on a die: 3, 6

Total outcomes = 6

$$\frac{2}{6} = \frac{1}{3}$$

Tickets Sold

Day	PVR X	PVR Y
Mon	400	650
Tue	600	800
Wed	300	500
Thu	800	250
Fri	850	750

Previous Year Questions

1. Correct Answer: A) 14 : 31

Required Ratio

PVR X (Mon + Wed)
= 400 + 300 = 700

PVR Y (Tue + Fri)
= 800 + 750 = 1550

Ratio = 700 : 1550

Divide by 50:

2. Correct Answer: None of these

Money earned difference

Total tickets sold:

PVR X total
= 400 + 600 + 300 + 800 + 850
= 2950

PVR Y total
= 650 + 800 + 500 + 250 + 750
= 2950

Both totals are equal.

So difference in earning = ₱0

3.

4. Correct Answer: D) None of these

Total Males

120 (P) + 90 (Q) + 120 (R) + 190 (S) + 150 (T) = 670

2. Males who like R and T

120 + 150 = 270

3. Calculation

Percentage
= (270 / 670) × 100

= 27000 / 670

= 2700 / 67 ≈ 40.29%

Looking at the options, 40 5/8 % = 40.625%.
Since 40.29% does not exactly match the given options (A, B, or C),

5. Correct Answer: C) 10 : 17

Ratio of males (P and Q together) to females (S and T together)

Males (P + Q):
110 + 90 = 200

Females (S + T):
200 + 150 = 350

Ratio:
200 : 350

Divide by 50:
4 : 7

Equivalent option:
10 : 17

6. Correct Answer: C) Mod

She wants to know which ice cream is most liked by the children.

The measure that tells the most frequently occurring value is Mode.

7. Answer: D) Statement–I is false but Statement–II is true

Statement–I

Given data (already arranged):

15, 17, 20, 20, 24, a+5, 29, 31, 31, 33, 34

There are 11 numbers.

Median = 6th term = a + 5

Given median = 27

So,a + 5 = 27=>a = 22

Now find mean:Replace a = 22

Data becomes: 15, 17, 20, 20, 24, 27, 29, 31, 31, 33, 34

Sum = 281

Mean = 281 ÷ 11 = 25.55 (approx)

Mean ≠ 20So, Statement–I is false.

Statement–II

Data: 70, 52, 47, 64, 47, 71, 58

Arrange:47, 47, 52, 58, 64, 70, 71

Median = 58 ✓Mode = 47 ✓So, Statement–II is true.

8. Correct Answer: A) 88 kg

Average Weight Calculation

Total weight of 10 students:
10 × 36 = 360 kg

Total weight of 8 students:
8 × 34 = 272 kg

Sum of weights of remaining 2 students:
360 − 272 = 88 kg

9. Correct Answer: D) 52

Data: 37, 57, 48, 33, 39, 64, 86

Sum = 364

Mean = 364 ÷ 7 = 52

10. Correct Answer: A) 9.625

First 8 prime numbers:

2, 3, 5, 7, 11, 13, 17, 19

Sum = 77

Mean = 77 ÷ 8 = 9.625

4. Simple Equations

Solutions

Worksheet – 1

1. Correct Answer: C) 12

$$7x - 81 + 81 = 3 + 81$$
$$7x = 84$$
$$\frac{7x}{7} = \frac{84}{7}$$
$$x = 12$$

2. Correct Answer: A) −2

$$7.5x + 4 = 6.5x + 2$$
$$7.5x - 6.5x = 2 - 4$$
$$x = -2$$

3. Correct Answer: C) $3 - 2\sqrt{2}$

$$1 + x = \sqrt{2}(1 - x)$$
$$1 + x = \sqrt{2} - \sqrt{2}x$$
$$x + \sqrt{2}x = \sqrt{2} - 1$$
$$x(1 + \sqrt{2}) = \sqrt{2} - 1$$
$$x = \frac{\sqrt{2} - 1}{1 + \sqrt{2}}$$

After rationalizing:

$$x = 3 - 2\sqrt{2}$$

4. Correct Answer: D) 7

$$P - (2P + 5) - 5(1 - 2P) = 2(3 + 4P) - 3(P - 4)$$

Simplifying both sides:

LHS = $9P - 10$
RHS = $5P + 18$

$$9P - 10 = 5P + 18$$
$$4P = 28$$
$$P = 7$$

5. Correct Answer: A) $\frac{5}{4}$

Given $x = \frac{1}{2}$

$$x + \frac{1}{1 + \frac{1}{1 + \frac{1}{x}}}$$

Stepwise:

$$\frac{1}{x} = 2$$
$$1 + 2 = 3$$
$$\frac{1}{3}$$
$$1 + \frac{1}{3} = \frac{4}{3}$$
$$\frac{1}{4/3} = \frac{3}{4}$$

Final: $\frac{1}{2} + \frac{3}{4} = \frac{5}{4}$

6. Correct Answer: D) $\frac{88}{27}$

$$0.5x + 0.3 = 0.23x + 1.18$$
$$0.5x - 0.23x = 1.18 - 0.3$$
$$0.27x = 0.88$$
$$x = \frac{0.88}{0.27}$$

Multiply numerator & denominator by 100:

$$x = \frac{88}{27}$$

7.

1. Multiply both sides by 6:

$$6(m) - 6\left(\frac{m-1}{2}\right) = 6(1) - 6\left(\frac{m-1}{3}\right)$$

2. Simplify:

$$6m - 3(m - 1) = 6 - 2(m - 1)$$

3. Distribute the negative signs:

$$6m - 3m + 3 = 6 - 2m + 2$$

4. Combine like terms:

$$3m + 3 = 8 - 2m$$

5. Add 2m to both sides and subtract 3 from both sides:

$$5m = 5$$
$$m = 1$$

8. Correct Answer: D) 40

Let the number be x.

Given:

$$\frac{x}{8} = 5$$
$$x = 5 \times 8 = 40$$

9. Correct Answer: A) $\frac{58}{21}$

$$\frac{3p}{10} + \frac{2p}{5} = \frac{7p}{25} + \frac{29}{25}$$

Convert all to denominator 50:

$$\frac{3p}{10} = \frac{15p}{50}$$
$$\frac{2p}{5} = \frac{20p}{50}$$

LHS:

$$\frac{35p}{50}$$

RHS:

$$\frac{14p}{50} + \frac{58}{50}$$
$$\frac{35p}{50} = \frac{14p + 58}{50}$$

Multiply by 50:

$$35p = 14p + 58$$
$$21p = 58$$
$$p = \frac{58}{21}$$

10. Correct Answer: A) 1

$$3(m + 6) = 21$$
$$m + 6 = 7$$
$$m = 1$$

Worksheet-2

1. Correct Answer: (C) 5

$$2x + 16 = 26$$
$$2x = 10$$
$$x = 5$$

2. Correct Answer: (A) 2

$$x + 2 = 4$$
$$x = 2$$

3. Correct Answer: (B) 3

$$5p + 2 = 17$$
$$5p = 15$$
$$p = 3$$

4. Correct Answer: (D) 6

$$3m - 14 = 4$$
$$3m = 18$$
$$m = 6$$

5. Correct Answer: (B) 3

$$5z + 5 = 20$$
$$5z = 15$$
$$z = 3$$

6.

7. Correct Answer: (B) 20

If 8 is subtracted from a number, the result is 12.

Let the number be x.

$$x - 8 = 12$$
$$x = 20$$

8. Correct Answer: (A) 5

If 7 is added to a number, the result is 12.

Let the number be x.

$$x + 7 = 12$$
$$x = 5$$

9. Correct Answer: (D) 4

The product of a number and 8 is 32.

Let the number be x.

$$8x = 32$$
$$x = 4$$

10. Correct Answer: B) 11

Given equation:
3x + 2 = 11

The **RHS (Right-Hand Side)** of an equation is the part written **to the right of the equality sign (=)**.

Here,
LHS = 3x + 2
RHS = 11

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: B) 5

$$\frac{2}{3}(3x - 3) = \frac{4}{3}(2x - 3) - \frac{4}{3}$$

LHS: $\frac{2}{3}(3x - 3) = 2x - 2$

RHS: $\frac{4}{3}(2x - 3) = \frac{8x}{3} - 4$

$$\frac{8x}{3} - 4 - \frac{4}{3} = \frac{8x}{3} - \frac{16}{3}$$

So, $2x - 2 = \frac{8x}{3} - \frac{16}{3}$

Multiply both sides by 3: $6x - 6 = 8x - 16$

$$10 = 2x$$
$$x = 5$$

2. Correct Answer: A) 11

Given: $q = 2$

$$pq - 3 = 19$$
$$2p - 3 = 19$$
$$2p = 22$$
$$p = 11$$

3. Correct Answer: B) 1/2

$$\frac{5p - 1}{3} - \frac{2p - 2}{2} = 1$$

Multiply by 6 (LCM):

$$2(5p - 1) - 3(2p - 2) = 6$$
$$10p - 2 - 6p + 6 = 6$$
$$4p + 4 = 6$$
$$4p = 2$$
$$p = \frac{1}{2}$$

4. Correct Answer: A) 1

$$0.5x + 3.5 = 3.5x + 0.5$$
$$3.5 - 0.5 = 3.5x - 0.5x$$
$$3 = 3x$$
$$x = 1$$

5. Correct Answer: B) 3

$$\frac{x}{3} - \frac{1}{4} = \frac{3}{2} - \frac{x}{4}$$

Multiply by 12: $4x - 3 = 18 - 3x$

$$7x = 21$$
$$x = 3$$

6. Correct Answer: D) x = -2

$$\frac{2x - 1}{2x - 3} = \frac{3x + 1}{3x - 1}$$

Cross multiply: $(2x - 1)(3x - 1) = (3x + 1)(2x - 3)$

Expand:

LHS:

$$6x^2 - 5x + 1$$

RHS:

$$6x^2 - 7x - 3$$

Equate:

$$6x^2 - 5x + 1 = 6x^2 - 7x - 3$$
$$2x = -4$$
$$x = -2$$

(Valid since denominator ≠ 0)

7. Correct Answer: C) Solution

The value that makes an equation true is called its **solution**.

For example:
If 2x = 10, then x = 5.
Here, 5 satisfies the equation, so it is the **solution**.

8. Correct Answer: B) $\frac{20}{3}$

Given:

$(5x - 1)(9x + 1)$ and $3a(3x - 1)$ are equal when $x = 1$

Substitute $x = 1$

LHS:

$$(5(1) - 1)(9(1) + 1) = (5 - 1)(9 + 1) = 4 \times 10 = 40$$

RHS:

$$3a(3(1) - 1) = 3a(3 - 1) = 3a \times 2 = 6a$$

Since equal:

$$6a = 40$$
$$a = \frac{40}{6} = \frac{20}{3}$$

9. Correct Answer: C) 3

$$\frac{3x + 5}{3x - 5} = \frac{7}{2}$$

Cross multiply:

$$2(3x + 5) = 7(3x - 5)$$
$$6x + 10 = 21x - 35$$
$$45 = 15x$$
$$x = 3$$

(Valid since $3x - 5 \neq 0$)

10. Correct Answer: B) 7

Given equation:
7x + 3 = 10

The **coefficient** of a variable is the numerical value that multiplies the variable.

Here, **7** is multiplying **x**.

11. Correct Answer: C) 2

$$\frac{10z + 8}{2z} = 7$$

Multiply both sides by $2z$:

$$10z + 8 = 14z$$

$$8 = 4z$$
$$z = 2$$

12. Correct Answer: A) $\frac{228}{11}$

$$\frac{5x-4}{8} - \frac{x-3}{5} = \frac{x+6}{3}$$

LCM of 8, 5, 3 = 120

Multiply throughout by 120:

$$15(5x-4) - 24(x-3) = 40(x+6)$$

Expand:

$$75x - 60 - 24x + 72 = 40x + 240$$
$$51x + 12 = 40x + 240$$
$$11x = 228$$
$$x = \frac{228}{11}$$

13. Correct Answer: A) 4

$$3y - 3 = 4y - 7$$
$$-3 + 7 = 4y - 3y$$
$$4 = y$$
$$y = 4$$

14. Correct Answer: A) 1

$$2x - 1 = \frac{3x-1}{2}$$

Multiply both sides by 2:

$$4x - 2 = 3x - 1$$
$$4x - 3x = -1 + 2$$
$$x = 1$$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

15. Correct Answer: A (The value of x is a multiple of 3) B) $x^2 = 81$ C) x = 9

$$x - \left(2x - \frac{3x-4}{7}\right) = \frac{4x-27}{3} - 3$$

Simplifying,

$$-\frac{4(x+1)}{7} = \frac{4(x-9)}{3}$$

Solving:

$$x = 6$$

16. Correct Answer: A, B, C

$$8(2p-5) - 6(3p-7) = 1$$

Expanding:

$$16p - 40 - 18p + 42 = 1$$
$$-2p + 2 = 1$$
$$p = \frac{1}{2}$$

17. Correct Answer: A (Even number), B (32), C (Equals 4 × 8)

Given $r = 6$

$$\frac{9r}{2} + 5 = \frac{9 \times 6}{2} + 5$$
$$= 27 + 5 = 32$$

18. Correct Answer: A, B

$$3(n-4) + 2(4n-5) = 5(n+2) + 16$$

Expanding:

$$11n - 22 = 5n + 26$$
$$6n = 48$$
$$n = 8$$

19. Correct Answer: A) x= $\frac{-5}{6}$ B) 6x+5=0

$$5(x+2) - 2(3-4x) = 3(x+5) - 4(4-x)$$

Expanding:

$$13x + 4 = 7x - 1$$
$$6x = -5$$
$$x = -\frac{5}{6}$$

20. Correct Answer: A, B, C

Four times a number reduced by 5 equals 19.

$$4x - 5 = 19$$
$$4x = 24$$
$$x = 6$$

ASSERTION AND REASON TYPE

21. Correct Answer: A

Assertion: If x + 5 = 9, then x = 4.

✓ Subtract 5 from both sides:

x + 5 – 5 = 9 – 5

x = 4 ✓ (True)

Reason: Subtracting the same number from both sides keeps equality unchanged.

✓ This is the Subtraction Property of Equality (True)

- The reason correctly explains the assertion.

22. Correct Answer: A

Assertion: In 7x = 21, x = 3.
✓ Divide both sides by 7:
x = 21 ÷ 7 = 3 ✓ (True)

Reason: Dividing both sides by the same non-zero number does not change equality.
✓ This is the Division Property of Equality (True)

- The reason correctly explains the assertion.

23. Correct Answer: A

Assertion: If 5 is transposed from LHS to RHS, its sign changes.
✓ True (Transposition changes the sign)

Reason: Transposition means changing a term from one side to the other with sign changed.
✓ This is exactly the definition (True)

- The reason correctly explains the assertion.

24. Correct Answer: A

Assertion: (2,1) is a solution of 4x + 2y = 10.

Substitute x = 2 and y = 1:
4(2) + 2(1) = 8 + 2 = 10 ✓
So, the Assertion is true.

Reason: Substituting x = 2 and y = 1 gives 4(2) + 2(1) = 10.
✓ This correctly verifies the assertion.
So, the Reason is true and correctly explains the Assertion.

25. Correct Answer: A

Assertion: In the equation 2x + 3 = 11, the RHS is 11.
✓ RHS means the part on the right side of the equality sign.
So, RHS = 11 → Assertion is true.

Reason: The part to the right of the equality sign is called RHS.
✓ This is the definition of RHS → Reason is true.

The reason correctly explains the assertion.

MATRIX MATCHING TYPE

26. A. $x + y = 4$
Check (0,4):
0 + 4 = 4 ✓
➡ A → 1

B. $2x - y = 1$
Check (1,1):
2(1) – 1 = 1 ✓
➡ B → 2

C. $3x + y = 7$
Check (1,4):
3(1) + 4 = 7 ✓
➡ C → 3

D. $x - y = 0$
Check (2,2):
2 – 2 = 0 ✓
➡ D → 4

27. A. Linear equation → 4 (A mathematical statement with '=')

B. Variables → 2 (Symbols representing unknowns)

C. Solution of equation → 3 (A pair that satisfies the equation)

D. Graph of equation → 1 (Has a graph that is a straight line)

COMPREHENSION TYPE

Passage–1

Given equation:

$$5x + 3 = 28$$

28. Correct Answer: B) Subtract 3 from both sides

The correct first step is:

To isolate $5x$, subtract 3 from both sides.

$$5x + 3 - 3 = 28 - 3$$
$$5x = 25$$

29. Correct Answer: A) 5x = 25

After subtracting 3, the equation becomes:

$$5x = 25$$

30. Correct Answer: C) 5

The value of x is:

$$5x = 25$$
$$x = 5$$

Passage–2

Let:
x= cost of one Math book
y= cost of one Science book

Given:

Aman: $3x + 2y = 390$
Riya: $2x + 3y = 370$
Ravi buys some notebooks for ₹120.
If he buys 3 more notebooks, he must pay ₹180.
Each notebook costs the same amount.

31. Correct Answer: A. $3x + 120 = 180$

Let the cost of one notebook be ₱x. Which equation represents the situation?

Let the original number of notebooks be n.

Cost of notebooks = ₱120
So,

$$nx = 120$$

If he buys 3 more notebooks, total notebooks = $n + 3$
Total cost = ₱180

$$(n + 3)x = 180$$

From $nx = 120$, substitute:

$$120 + 3x = 180$$

32. Correct answer: C. ₱20

Solve the equation

$$3x + 120 = 180$$
$$3x = 180 - 120$$
$$3x = 60$$
$$x = 20$$

Cost of one notebook = ₱20

33. Correct answer: C. 6

How many notebooks does Ravi buy originally?

$$nx = 120$$
$$n \times 20 = 120$$
$$n = 6$$

INTEGER TYPE

34. Correct Answer: $x = \frac{9}{2}$

$$\frac{x}{2} - \frac{1}{4} = \frac{x}{3} + \frac{1}{2}$$

Multiply both sides by 12 (LCM):

$$6x - 3 = 4x + 6$$
$$6x - 4x = 6 + 3$$
$$2x = 9$$
$$x = \frac{9}{2}$$

35. Correct Answer: –3

$$\frac{x + 7}{3x + 16} = \frac{4}{7}$$

Cross multiply:

$$7(x + 7) = 4(3x + 16)$$
$$7x + 49 = 12x + 64$$
$$49 - 64 = 12x - 7x$$
$$-15 = 5x$$
$$x = -3$$

36. Correct Answer: $\frac{145}{21}$

$$\frac{n}{5} - \frac{5}{7} = \frac{2}{3}$$

Multiply by LCM 105:

$$21n - 75 = 70$$
$$21n = 145$$
$$n = \frac{145}{21}$$

37. Correct Answer: x = 5

Let present age = x

After 15 years:

$$x + 15 = 4x$$
$$15 = 3x$$
$$x = 5$$

Previous Year Questions

1. Correct Answer: B (13)

$$\frac{x - 3}{5} - \frac{2x + 1}{3} = -7$$

Multiply by 15:

$$3(x - 3) - 5(2x + 1) = -105$$
$$3x - 9 - 10x - 5 = -105$$
$$-7x - 14 = -105$$
$$-7x = -91$$
$$x = 13$$

2. Correct Answer: B (40)

Let males = m
Females = $50 - m$

$$5m + 8(50 - m) = 280$$
$$5m + 400 - 8m = 280$$
$$-3m = -120$$
$$m = 40$$

3. Correct Answer: C (10 years)

Let Megha's age = x
Mother's age = $x + 20$

After 10 years:

$$x + 20 + 10 = 2(x + 10)$$
$$x + 30 = 2x + 20$$
$$x = 10$$

4. Correct Answer: A (13)

$$(-91) \div X + 68 \div (-17) = -11$$
$$\frac{-91}{X} - 4 = -11$$
$$\frac{-91}{X} = -7$$
$$X = 13$$

5. Correct Answer: B (73)

Let unit digit = x
Tens digit = $2x + 1$

$$x + (2x + 1) = 10$$
$$3x + 1 = 10$$
$$x = 3$$

Tens digit = 7

Number = **73**

6. Correct Answer: B (Both statements are false)

Statement I:

If $3m - 2 = \frac{6m+1}{5}$

Multiply by 5:

$$15m - 10 = 6m + 1$$
$$9m = 11$$
$$m = \frac{11}{9}$$

So 3m ≠ 5 ❌ (False)

Statement II:

$$m = 3n - 5$$

If $m = 49$:

$$49 = 3n - 5$$
$$3n = 54$$
$$n = 18$$

Statement says n = 16 ❌ (False)

7. Correct Answer: B (3)

$$7z - 5 = \frac{4}{5}(11 + 3z)$$

Multiply by 5:

$$35z - 25 = 4(11 + 3z)$$
$$35z - 25 = 44 + 12z$$
$$35z - 12z = 44 + 25$$
$$23z = 69$$
$$z = 3$$

8. Correct Answer: B (60 m)

Let height = x

Painted red = $\frac{1}{4}x$

Painted blue = $\frac{2}{5}x$

Remaining = 21 m

$$x - \left(\frac{1}{4}x + \frac{2}{5}x\right) = 21$$

LCM = 20

$$\frac{5x + 8x}{20} = \frac{13x}{20}$$

Remaining:

$$x - \frac{13x}{20} = 21$$
$$\frac{7x}{20} = 21$$
$$7x = 420$$
$$x = 60$$

9. Correct Answer: C (20)

Let number = x

$$\frac{x}{4} + 15 = x$$
$$15 = x - \frac{x}{4}$$
$$15 = \frac{3x}{4}$$
$$x = 20$$

10. Correct Answer: C (40 years)

Let Varun's age = x

Karan's age = $4x$

Five years ago:

$$4x - 5 = 7(x - 5)$$
$$4x - 5 = 7x - 35$$
$$30 = 3x$$
$$x = 10$$

Karan's age = $4 \times 10 = 40$

5. Lines and Angles

Solutions

Worksheet – 1

1. Correct Answer: A) Opening a book fully (A fully opened book makes a straight angle of 180°, i.e., supplementary angles.)

Which of the following situations represents supplementary angles?

A) Opening a book fully
B) Turning a fan once
C) Clock hands at 3:00
D) Overlapping clock hands

2. Correct Answer: D) 45°

The angles $135^\circ$and x° form a linear pair on a straight line.
Linear pair angles are supplementary.

$x + 135^\circ = 180^\circ$
$x = 180^\circ - 135^\circ = 45^\circ$

3. Correct Answer: C) Reflex angle

A rotating fan blade turns through 270°.

An angle between 180° and 360° is called a reflex angle.
Since $270^\circ > 180^\circ$, it is a reflex angle.

4. Correct Answer B: 15°

On a straight line, the sum of angles is 180°.
The adjacent angles are $5y^\circ$, y°, and $6y^\circ$.

$5y + y + 6y = 180$
$12y = 180$
$y = 15$

5. Correct Answer: A) $360^\circ - 122^\circ$

The given angle $122^\circ$and angle Btogether make a complete angle around the point.

A complete angle = 360°

$\angle B = 360^\circ - 122^\circ$
$\angle B = 238^\circ$

6. Correct Answer: A) $52^\circ, 47^\circ$

Let the angles be x and y.

$x + y = 99$

$x - y = 5$
Add both equations: $2x = 104 \Rightarrow x = 52$
Substitute in $x + y = 99$:

$52 + y = 99 \Rightarrow y = 47$

7. Correct Answer: B) ∠AGC

An angle must have its vertex at point O.

∠AOF → vertex at O

∠COB → vertex at O

∠BOA → vertex at O

∠AGC → vertex at G (not at O)

8. Correct Answer: (A) 26°

Find $\angle EOF$
Angles on one side of straight line ABadd up to 180°:

$79^\circ + \angle EOF + 37^\circ = 180^\circ$
$\angle EOF = 64^\circ$
Find the complement of $\angle EOF$

$90^\circ - 64^\circ = 26^\circ$

9. Correct Answer: (C) ∠EBF and ∠FBC

Adjacent angles must:

Have the same vertex

Share a common arm

Have no overlap

∠EBF and ∠FBC:

Both have vertex B

Share the common arm BF

Lie next to each other

So, they form a pair of adjacent angles.

10. Correct Answer: (D)

(A) ∠AOB and ∠DOC are vertically opposite angles, so they are equal, not necessarily supplementary → True

(B) ∠FOA and ∠DOC are vertically opposite angles → True

(C) ∠EOF and ∠EOD share vertex O and a common arm OE, so they are adjacent angles → True

(D) Complementary angles add up to 90°.∠DOC and ∠COB are not complementary. → False

Worksheet-2

1. Correct Answer: (D) 135°

$\angle 1$ and $\angle 2$ form a linear pair, so their sum is $180°$.

Given ratio:$\angle 1 : \angle 2 = 3:1$
Let $\angle 1 = 3x$, $\angle 2 = x$

$3x + x = 180°$
$4x = 180° \Rightarrow x = 45°$
So, $\angle 2 = 45°$
$\angle 3$ is corresponding to $\angle 2$ (since $l \parallel m$), hence:

$\angle 3 = 180° - 45° = 135°$

2. Correct Answer A: $\boxed{80°}$

Find $\angle AOM$
Since $AB \parallel PQ$, $\angle AOM = \angle OMQ = 25°$(alternate interior angles).

Find $\angle AON$

$\angle AON = \angle AOM + \angle MON = 25° + 75° = 100°$

Find $x = \angle ONX$
Since $AB \parallel XY$, $\angle AON$and x are supplementary(Consecutive interior angles are supplementary,)

$x = 180° - 100° = 80°$

3. Correct Answer B: $\boxed{93°}$

Given $AB \parallel CD$.

The angle between the upper slanted line and line CD is $55°$.
By alternate interior angles,

$\angle 1 = 55°$
The angle between the lower slanted line and line $AB$is $38°$.
By alternate interior angles, $\angle 2 = 38°$
At point O, angle $a$is the sum of $\angle 1$and $\angle 2$:

$a = 55° + 38° = 93°$

4. Correct Answer C: $\boxed{90°}$

Given $AB \parallel CD$.

$\angle ABO = 50°$.
By alternate interior angles, the angle between $OB$and the horizontal at $O$is also $50°$.

$\angle CDO = 40°$.
By alternate interior angles, the angle between $OD$and the horizontal at $O$is $40°$.

At point O, $\angle BOD$is the sum of these two angles:

$\angle BOD = 50° + 40° = 90°$

5. Correct Answer: (D) Alternate angles

$BC \parallel AD$ (shown by arrows).

Line AC acts as a transversal.

$\angle ACB$ and $\angle CAD$ lie on opposite sides of the transversal and between the parallel lines.

Therefore, they are alternate angles.

6. Correct Answer:B $\boxed{\angle PRQ = 50°}$

Given:

$QP \parallel RA$

$\angle QPR = 70°$

$\angle ARB = 60°$

To find: $\angle PRQ$

Use parallel lines property

Since $QP \parallel RA$and $PR$is a transversal,

$\angle QPR = \angle PRA = 70°$
(corresponding angles)

Find $\angle PRB$

At point R, $\angle ARB = 60°$.
So,$\angle PRA + \angle ARB = \angle PRB$
$70° + 60° = 130°$
Find $\angle PRQ$

$\angle PRB$ and $\angle PRQ$ form a linear pair:

$\angle PRQ = 180° - 130° = 50°$

7. Correct Answer B: $\boxed{50°}$

Identify the relationship
$\angle ARS$ and $\angle CSR$ are co-interior (consecutive interior) angles because $AB \parallel CD$and PQ is a transversal.
So, they are supplementary:

$\angle ARS + \angle CSR = 180°$

$(2y + 12) + (7y - 3) = 180$
$9y + 9 = 180$
$9y = 171 \Rightarrow y = 19$
Find $\angle PRB$
$\angle PRB$and $\angle ARS$ are vertically opposite angles, so they are equal:

$\angle PRB = 2y + 12 = 2(19) + 12 = 50°$

8. Correct Answer: (A) $107°$and $73°$

Let the angles be $x°$and $y°$.

Supplementary angles:

$x + y = 180°$
Difference given:

$x - y = 34°$
Adding both equations:$2x = 214° \Rightarrow x = 107°$
Then,$y = 180° - 107° = 73°$

9. Correct option: (B)

Since $l \parallel m$and $p \parallel q$, the given angle $110°$and angle $x$are corresponding angles formed by parallel lines.

Corresponding angles are equal.

But the shown $110°$is an exterior angle, so the corresponding interior angle is:

$180° - 110° = 70°$
Answer: $x = 70°$

10. Correct Answer:(B) $180°$

∠a and ∠b are adjacent angles on a straight line.

Adjacent angles forming a straight line are a linear pair.

The sum of angles in a linear pair is: $180°$

Compititive Corner

SINGLE ANSWER CORRECT TYPE

1. Correct Answer: A) $100°$

Two supplementary angles are in the ratio 4: 5. Find the measure of the greater angle.

Let the angles be $4x$ and $5x$.

Supplementary angles sum to $180°$:

$4x + 5x = 180°$
$9x = 180° \Rightarrow x = 20°$
Greater angle $= 5x = 5 \times 20° = 100°$

2. Correct Answer: A) Alternate interior angles

In the given figure, a and $b$are

Angles a and b lie between two parallel lines.

They are on alternate sides of the transversal.

So, a and b are alternate interior angles.

3. Correct Answer A: $35°$

The given angles are consecutive (same-side) interior angles.

Consecutive interior angles between parallel lines are supplementary.

Set up the equation:$3x + (x + 40) = 180$
$4x + 40 = 180$
$4x = 140$
$x = 35$

4. Correct Answer B: $\boxed{45°}$

In the following figure, ∠x = ?

The angles $65°$, x, and $70°$lie on a straight line.

Angles on a straight line add up to $180°$.

$65° + x + 70° = 180°$
$x + 135° = 180°$
$x = 45°$

5. Correct Answer C: $\boxed{32}$

The angles $(2x+60)°$and $(3x-40)°$ form a linear pair.

Linear pair angles are supplementary.

$(2x + 60) + (3x - 40) = 180$
$5x + 20 = 180$
$5x = 160$
$x = 32$

6. Correct Answer B: $y = 20°$

The vertical line and the horizontal line form a right angle of $90°$.

The slanted line makes an angle of $30°$with the horizontal line on the left.

Therefore, the angle between the slanted line and the vertical line on the right is:

$90° - 30° = 60°$
This angle is labeled $3y$.

$3y = 60°$
$y = 20°$

7. Correct Answer C: 8.25

The vertical line is perpendicular to the horizontal line, so the angle between them is $90°$.

The two angles inside this right angle are: $(5x + 9)°$ and $(3x + 15)°$
These two angles together form $90°$.

Set up the equation: $(5x + 9) + (3x + 15) = 90$
$8x + 24 = 90$
$8x = 66$
$x = \frac{66}{8} = 8.25$

8. Correct Answer B:46

A straight line forms an angle of $180°$.

The two angles at point $O$are $(3x-5)°$and $(x+1)°$.

Since they form a straight line, their sum is $180°$.

$(3x - 5) + (x + 1) = 180$
$4x - 4 = 180$
$4x = 184$
$x = 46$

9. Correct Answer C: $x = 35$

The three angles on the straight line AB at point $O$are:

$(x - 10), (4x - 25), (x + 5)$

Angles on a straight line add up to $180°$.

Set up the equation:$(x - 10) + (4x - 25) + (x + 5) = 180$
$6x - 30 = 180$
$6x = 210$
$x = 35$

10. Correct Answer:(C) $38°$

OA is perpendicular to OC, so

$\angle AOC = 90°$
The given angle is: $\angle BOC = 128°$
Since $\angle BOC = \angle BOA + \angle AOC$,

$\angle AOB = 128° - 90° = 38°$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct answers: A, B, C

Which of the following pairs are complementary angles?

(Complementary angles sum to $90°$)

A) $30° + 60° = 90°$

B) $45° + 45° = 90°$

C) $50° + 40° = 90°$

D) $100° + 80° = 180°$

12. Correct answer: A) $27°$

If $(3x+20)°$and $(2x+25)°$are supplementary angles, find x.

Supplementary ⇒ sum $= 180°$

$(3x + 20) + (2x + 25) = 180$
$5x + 45 = 180 \Rightarrow x = 27$

13. Correct answers: A, B, D

If two angles form a linear pair, which statements are always true?

A) They are adjacent (TRUE)

B) Their sum is $180°$(TRUE)

C) They are equal (False)

D) Their non-common arms form a straight line (TRUE)

14. Correct answers: A, B

Two angles are complementary and one of them is $35°$. Which are correct?

Other angle $= 90° - 35° = 55°$

A) The other angle is $55°$

B) Both angles are acute

15. Correct answers: A, B

At 9:00 PM, the hands of a clock form angles. Which statements are correct?

At 9:00 → smaller angle $= 90°$, larger angle $= 270°$

A) The smaller angle is $90°$

B) The larger angle is a reflex angle

ASSERTION AND REASON TYPE

16. Correct Answer: A) Both A and R are true, and R is the correct explanation of A

Assertion (A) is true because adjacent angles on a straight line together form a straight angle.

Reason (R) is true since the measure of a straight angle is $180°$.

Reason (R) correctly explains why the sum of adjacent angles on a straight line is 180°.

17. Correct Answer: D) A is false but R is true

Assertion (A) is false (acute angles are less than 90°).

Reason (R) is true.

18. Correct Answer: A) Both A and R are true, and R is the correct explanation of A

Assertion (A): If two angles form a linear pair, they are supplementary.
Reason (R): Supplementary angles add up to 180°.

Explanation:

Assertion (A) is true (linear pair angles always sum to 180°).

Reason (R) is true and correctly explains Assertion (A).

19. Correct Answer: A) Both A and R are true, and R is the correct explanation of A

Assertion (A): All right angles are equal.
Reason (R): A right angle measures 90°.

Explanation:

Assertion (A) is true.

Reason (R) is true.

Since every right angle measures 90°, they are all equal.

Thus, R correctly explains A.

20. Correct Answer: A) Both A and R are true, and R is the correct explanation of A

Assertion (A): Alternate exterior angles are non-adjacent angles.
Reason (R): They are formed at different points of intersection on the transversal.

Explanation:

Assertion (A) is true (alternate exterior angles lie on opposite sides of the transversal and are not adjacent).

Reason (R) is true and explains why they are non-adjacent.

MATRIX MATCHING TYPE

21. **1.** Two angles share a common arm and vertex, and their measures are 35°and **55°.**

They share a common arm and vertex ⟶ Adjacent angles
Match: 1 – D

2.Two angles formed at the intersection of two straight lines are opposite each other and each measures 70°.

Opposite angles at an intersection are vertically opposite angles
Match: 2 – C

3.Two angles are adjacent and together form a straight line.

Adjacent angles forming a straight line ⟶ Linear pair of angles
Match: 3 – A

4.The angles inside a right angle are divided into 25°and 65°.

$25° + 65° = 90°$⟶ Complementary angles
Match: 4 – B

22. **1.** A straight road without any turn

Represents 180°⟶ Straight angle
Match: 1 – B

2. The angle formed by clock hands at 2:00

Angle between hands at 2:00 is 60°⟶ Acute angle
Match: 2 – D

3. One full turn of a ceiling fan blade

One full rotation = 360°⟶ Complete angle
Match: 3 – C

4. A person turning around more than half a circle

More than 180°⟶ Reflex angle
Match: 4 – A

COMPREHENSION TYPE

Passage-1

23. Correct Answer: D) $360°$

Angles around a point make one complete turn.

24. Correct Answer: C) $80°$

Sum of angles at a point $= 360°$

Remaining angle

$$= 360° - (70° + 90° + 120°)$$

$$= 360° - 280° = 80°$$

25. Correct Answer: B) $90°$

Total angle around a point $= 360°$

Divided equally into 4 angles:

$$360° \div 4 = 90°$$

Passage – II:

26. Correct Answer: C) Obtuse

At 5:00, the angle between the hands is $150°$.

This is greater than $90°$and less than $180°$.

27. Correct Answer: A) $90°$

At 3:00, the hands form a right angle.

28. Correct Answer: B) Straight angle

At 6:00, the hands are opposite each other, forming $180°$.

INTEGER TYPE

29. x and y are complementary angles.

$x + y = 90°$

Given $x = 70°$,

$y = 90° - 70° = \boxed{20°}$

30. In the figure, POR is a straight line, so angles form a linear pair.

$(3a + 5) + (2a - 25) = 180$

$5a - 20 = 180$

$5a = 200 \Rightarrow a = \boxed{40}$

31. x, y, z are complementary angles.

$x + y + z = 90°$

Given $x + y = 60°$,

$z = 90° - 60° = \boxed{30°}$

32. Supplementary angles mean the sum of the angles is $\boxed{180°}$

33. Two angles are equal and supplementary.

Each angle $= \frac{180°}{2} = \boxed{90°}$

Previous Year Questions

1. Correct Option: A

(i)Supplementary angles sum to $180°$.

$(6x + 20) + 10x = 180$

$16x + 20 = 180 \Rightarrow 16x = 160 \Rightarrow x = 10$

Answer: $10°$

(ii)$\frac{3}{5} \times 45° = 27°$

Complement of an angle $= 90° -$given angle

$90° - 27° = 63°$

Answer: $63°$

(iii)Let the angles be x and $5x$.

$x + 5x = 90° \Rightarrow 6x = 90° \Rightarrow x = 15°$

Smallest angle: $15°$

2. Correct Answer: B

Angles around a point always add up to $360°$, not $270°$.

Hence, Option B is incorrect.

All other statements are correct properties of lines and angles.

3. Correct Answer: A ($22.5°$)

Since ABC is a straight line,

$\angle ABC = 180°$

$\angle DBE = 90°$(right angle)

The remaining angle on the straight line is:

$180° - 90° = 90°$

This $90°$is divided equally into four angles:

$\angle a = \angle b = \angle c = \angle d$

So,$4a = 90° \Rightarrow a = \frac{90°}{4} = 22.5°$

4. Correct Answer: C ($36°$)

Let the three angles be $2x$, $3x$, $5x$.

Greatest angle $= 5x$

Smallest angle $= 2x$

Given difference:

$5x - 2x = 54°$

$3x = 54° \Rightarrow x = 18°$
Smallest angle:

$2x = 2 \times 18° = 36°$

5. Correct Answer: A (90°)

Point O lies on a straight-line AB, so the angle at point O is:180°
The angles around point O are:

$$\frac{a}{2} + 2°$$
$$\frac{a}{2} - 2°$$
$$a$$
$$2a$$

Sum of angles on a straight line:$\left(\frac{a}{2}+2\right) + \left(\frac{a}{2}-2\right) + a + 2a = 360°$
Simplify:
$\frac{a}{2} + \frac{a}{2} + a + 2a = 360°$
$4a = 360°$
$a = 90°$

6. Correct Answer: A
$\boxed{y = 65°, z = 115°, w = 65°}$

Finding y (Vertically Opposite Angles)

Vertically opposite angles are equal.

$y = x = 65°$

Finding z(Linear Pair)

Angles in a linear pair sum to $180°$.

$z = 180° - 65° = 115°$

Finding w(Alternate Interior Angles)

Alternate interior angles are equal for parallel lines.

$w = y = 65°$

7. Correct Answer: B $\boxed{x = 110°, y = 145°}$

Given:DAE ∥ CBH ∥ FG

The slanted line AGcuts the three parallel lines.

At G, the angle between $AG$and $FG$is $70°$.

By alternate interior angles, the angle between $AG$and AE(right direction) at $A$is also $70°$.

But x is measured with respect to AD(left direction), which is a straight line with AE.

$x = 180° - 70° = \boxed{110°}$
Find y

At A, the angle between $AB$and $AE$is $35°$.

Since $AE \parallel CBH$, the corresponding angle between $AB$and CBHat $B$is also $35°$.

The angle $y$is the linear pair with this angle at B.

$y = 180° - 35° = \boxed{145°}$

8. Correct Answer: D. 20°

Let the angles be $4x$ and $5x$.
Since they are supplementary:

$4x + 5x = 180°$
$9x = 180° \Rightarrow x = 20°$
Angles are:

$4x = 80°, 5x = 100°$
Difference:$100° - 80° = 20°$

9. Correct Answer C: The value of $a + b$ is $124°$.

Find the value of a
Since $l \parallel m$, the angle $a$and the given angle are corresponding angles.
Corresponding angles are equal, so $a = 73°$
Find the value of b
Since $l \parallel m$, the angle $b$and the given angle are alternate interior angles.
Alternate interior angles are equal, so $b = 51°$
Find $a + b$

$a + b = 73° + 51° = 124°$

10. Correct Answer: B. 82°

Since $AB \parallel CD \parallel EF$, all three lines are parallel.

At point B, the given angle is $158°$.
So the acute angle between BD and the parallel line is:

$180° - 158° = 22°$
At point F, the given angle is $120°$.
So the acute angle between DF and the parallel line is:

$$180° - 120° = 60°$$

The required angle $\angle BDF$ is the sum of these two acute angles:

$$\angle BDF = 22° + 60° = 82°$$

6. Plane Geometry

Solutions

Worksheet – 1

1. Correct answer: (C) 60°

In an equilateral triangle, all angles are equal and their sum is $180°$.
Each angle $= 180° \div 3 = 60°$.

2. Correct answer: (B) an equilateral

A triangle with all three sides equal is called an equilateral triangle.

3. Correct answer: (A) an isosceles

A triangle with two equal sides is called an isosceles triangle.

4. Correct answer: (C) 45°, 60°, 75°

Ratio of angles = 3: 4: 5.
Sum of ratios $= 3 + 4 + 5 = 12$.
Each part $= 180° \div 12 = 15°$.
Angles $= 45°, 60°, 75°$.

5. Correct answer: (A) 72°, 18°, 90°

Given: $\angle A = 4\angle B$, $\angle C = 5\angle B$.
Let $\angle B = x$.
Then $4x + x + 5x = 10x = 180° \Rightarrow x = 18°$.
So, $\angle A = 72°$, $\angle B = 18°$, $\angle C = 90°$.

6. Correct answer: (C) 45:40:36

Given:
$\frac{2}{3}\angle A = \frac{3}{4}\angle B = \frac{5}{6}\angle C = x$
$\angle A = \frac{3x}{2}, \angle B = \frac{4x}{3}, \angle C = \frac{6x}{5}$
Sum of angles of a triangle:
$\frac{3x}{2} + \frac{4x}{3} + \frac{6x}{5} = 180°$
Taking LCM $= 30$:
$45x + 40x + 36x = 180 \times 30 \Rightarrow 121x = 5400$
So, ratio:
$$\angle A : \angle B : \angle C = 45 : 40 : 36$$

7. Correct answer: (A) 20°

Given: $AB = AC \Rightarrow$ base angles $\angle B = \angle C$.
$\angle ACD = 100°$ is an exterior angle.

Exterior angle =sum of interior opposite angles:
$$\angle A + \angle B = 100°$$
Also, $\angle A + 2\angle B = 180°$
Subtracting: $\angle B = 80° \Rightarrow \angle A = 20°$

8. Correct answer: (C) 55°

Given:

$\angle A + \angle B = 145°$
$\angle B + \angle C = 90°$
Adding: $\angle A + 2\angle B + \angle C = 235°$
But $\angle A + \angle B + \angle C = 180°$
$$180° + \angle B = 235° \Rightarrow \angle B = 55°$$

9. Correct Answer: B) 12 cm

For a triangle to exist, the sum of any two sides must be greater than the third side.
$$6 + 8 > x \Rightarrow x < 14$$
$$6 + x > 8 \Rightarrow x > 2$$
So, $2 < x < 14$.

Among the options, 12 cm satisfies this condition.

10. Correct Answer: A) Right angled triangle

Sum of angles of a triangle = 180°
Let one angle = sum of the other two = x
So, $x + x = 180° \Rightarrow x = 90°$

Hence, the triangle is a right-angled triangle.

Worksheet – 2

1. Correct Answer: (C) 75°

Exterior angle = 105°, opposite interior angles in ratio 3 : 4

Let opposite interior angles be $3x$ and $4x$.

$3x + 4x = 105° \Rightarrow 7x = 105° \Rightarrow x = 15°$

Angles = 45°and 60°

Other interior angle:

$$180° - 105° = 75°$$

Greatest interior angle = 75°

2. Correct Answer: (C) 3

How many altitudes does a triangle have?

Each vertex has one altitude.

3. Correct Answer: (D) (10, 27, 26)

Which is NOT a Pythagorean triplet?

Check $a^2 + b^2 = c^2$:

(3, 4, 5)

(7, 24, 25)

(9, 12, 15)

(10, 27, 26)

4. Correct Answer: (C) 65°

Angles reduced by 15°, 10°, 5° give ratio 4 : 5 : 6

Let reduced angles be $4x, 5x, 6x$

Original angles:$P = 4x + 15, Q = 5x + 10, R = 6x + 5$
Sum of angles:

$(4x + 15) + (5x + 10) + (6x + 5) = 180$
$15x + 30 = 180 \Rightarrow x = 10$
$R = 6(10) + 5 = 65°$

5. Answer: (D) An acute-angled triangle

Interior angles in ratio 5 : 7 : 8

Sum of ratios = 20

Each part:

$\frac{180°}{20} = 9°$

Angles: $45^\circ, 63^\circ, 72^\circ$

All angles < 90°

6. Correct Answer: (A) 45

Exterior angles:
$(2x+11)^\circ, (3x-6)^\circ, (2x+40)^\circ$

Sum of one exterior angle at each vertex of a triangle = 360°

$(2x + 11) + (3x - 6) + (2x + 40) = 360$
$7x + 45 = 360 \Rightarrow 7x = 315 \Rightarrow x = 45$

7. Correct Answer: (B) 85°

Find the value of $x°$

At point C, given exterior angle = 135°

Interior angle at C:$180° - 135° = 45°$
Sum of interior angles of triangle:

$x + 50° + 45° = 180°$
$x + 95° = 180° \Rightarrow x = 85°$

8. Correct Answer: (A) 115°

Given: ΔABC is isosceles with AB = BC = 50 cm

Since AB = BC, the base angles at A and C are equal.

Given interior angle at B = 50°.

Sum of angles in a triangle:

$$A + B + C = 180°$$
$$A + C = 180° - 50° = 130°$$

Since $A = C$:

$$A = C = \frac{130°}{2} = 65°$$

Angle x is the exterior angle at C:

$$x = 180° - 65° = 115°$$

9. Correct Answer: (A) 60°, 75°

Exterior angle = 135°, opposite interior angles in ratio 4 : 5

Let angles be $4x$and $5x$.

$4x + 5x = 135° \Rightarrow 9x = 135° \Rightarrow x = 15°$
Angles:

$$4x = 60°, 5x = 75°$$

10. Correct Answer: (D) 70°

Sum of angles of a triangle = 180°

Let each of the equal angles be $x°$.

$$x + x + 40° = 180°$$
$$2x = 140°$$
$$x = 70°$$

∴ Measure of each equal angle = 70°

Worksheet-3

1. Correct Answer: (C) Every line segment joining two interior points lies completely inside

Which statement is true for a convex quadrilateral?

In a convex figure:

All interior angles are less than 180°

All diagonals lie inside

The line joining any two interior points lies completely inside

2. Answer: (C) T

Which point lies on the boundary of quadrilateral ABCD?

U, V → inside the quadrilateral

S → outside the quadrilateral

T → lies on side DC (boundary)

3. Correct Answer: (B) 120°

A quadrilateral has one interior angle measuring 210°

In a convex quadrilateral, all interior angles are less than 180°.
An angle of 210° (> 180°) is a reflex angle, so the quadrilateral is concave.

4. Correct Answer: (D) Concave

Three angles are in the ratio 2: 3: 4and the fourth angle is 90°

Sum of interior angles of a quadrilateral:

$$360°$$

Let the three angles be $2x, 3x, 4x$.

$$2x + 3x + 4x + 90° = 360°$$
$$9x = 270°$$
$$x = 30°$$

Angles:

$$2x = 60°, 3x = 90°, 4x = 120°$$

Largest angle = 120°

5. Correct Answer: (B) 72

Interior angles of a quadrilateral

$(x+20)^\circ, (2x-10)^\circ, (x+30)^\circ, (x-40)^\circ$

Sum of interior angles of a quadrilateral = 360°

$$(x + 20) + (2x - 10) + (x + 30) + (x - 40) = 360$$
$$5x = 360$$
$$x = 72$$

6. Correct Answer: (B) 26 cm

Parallelogram PQRS

$$PQ = 8 \text{ cm}, QR = 5 \text{ cm}$$

Perimeter (total distance around the figure):

$$2(PQ + QR) = 2(8 + 5) = 26 \text{ cm}$$

7. Correct Answer: (C) 90°

Opposite sides equal and parallel, all angles equal

This describes a rectangle.

Each angle in a rectangle: 90°

8. Correct Answer: (C) Square

Which quadrilateral satisfies all the properties of a parallelogram, kite, and rectangle?

Parallelogram → opposite sides equal & parallel

Rectangle → all angles 90°

Kite → two pairs of adjacent equal sides

Only a square satisfies all these properties.

9. Correct Answer: (C) 30

In parallelogram ABCD, $\angle A = 3x°$ and $\angle B = (2x + 30)°$

Adjacent angles in a parallelogram are supplementary:

$$\angle A + \angle B = 180°$$
$$3x + (2x + 30) = 180$$
$$5x = 150$$
$$x = 30$$

10. Correct Answer: (A) 1

BASE is a rectangle. Diagonals intersect at O

In a rectangle, diagonals bisect each other, so:

$$OB = OE$$

Given:

$$OB = 5x + 1, OE = 2x + 4$$
$$5x + 1 = 2x + 4$$
$$3x = 3$$
$$x = 1$$

Competitive Corner

SINGLE OPTOIN CORRECT TYPE

1. Correct Answer A: 137°

Find $y°$

The two slanted sides have equal marks, so the triangle formed is isosceles.

Angle at the top intersection $C = 94°$.

Interior angle at $C$of the triangle:

$$180° - 94° = 86°$$

Base angles of the isosceles triangle are equal:

$$\frac{86°}{2} = 43°$$

So, interior angle at $B = 43°$.

Angle $y°$is the exterior angle at B:

$$y = 180° - 43° = 137°$$

2. Correct Answer: (C) 60°

Quadrilateral angle problem

Let the smaller angle be $x°$.
Then the larger angle = $2x°$.

Given the other two angles are 90°each.

Sum of interior angles of a quadrilateral:

$$360°$$
$$x + 2x + 90 + 90 = 360$$
$$3x + 180 = 360$$
$$3x = 180$$
$$x = 60°$$

3. Correct Answer: (A) 45°

Find $\frac{x+y}{2}$

Use straight line at point C
Angles $\angle ACB$ and $\angle ACD$ form a straight line, so their sum is 180°.

Triangle ABC:

$x + 60° + \angle ACB = 180°$
$\angle ACB = 120° - x$
Triangle ACD:

$30° + y + \angle ACD = 180°$
$\angle ACD = 150° - y$
Straight line at C:

$(120 - x) + (150 - y) = 180$
$270 - (x + y) = 180$
$x + y = 90$
$\frac{x+y}{2} = \frac{90}{2} = 45°$

4. Correct Answer: (C) 61°, 59°

Angles of a triangle

$(3x-5)°, (2x+15)°, (5x-50)°$

Sum of angles of a triangle = 180°

$(3x - 5) + (2x + 15) + (5x - 50) = 180$
$10x - 40 = 180$
$x = 22$

Angles:

$$3x - 5 = 61°$$
$$2x + 15 = 59°$$
$$5x - 50 = 60°$$

Greatest angle = 61°
Least angle = 59°

5. Correct Answer: (D) 90°

ABCD is a rhombus and a linear pair of angles are equal

A linear pair sums to 180°.
If both angles are equal:

$$\frac{180°}{2} = 90°$$

6. Correct Answer: (B) A square

In a rhombus, if diagonals are equal

A rhombus with equal diagonals becomes a square.

7. Correct Answer: (D) 12

ABCD is a parallelogram

Given:
$AC = 14$cm, $BD = 10$cm
Diagonals intersect at point P.

In a parallelogram, diagonals bisect each other.

$$AP = \frac{AC}{2} = \frac{14}{2} = 7 \text{ cm}$$
$$BP = \frac{BD}{2} = \frac{10}{2} = 5 \text{ cm}$$
$$AP + BP = 7 + 5 = 12 \text{ cm}$$

8. Correct Answer: (B) 80°

Three interior angles of a quadrilateral are equal, fourth angle = 120°

Sum of interior angles of a quadrilateral:

$$360°$$

Sum of the three equal angles:

$$360° - 120° = 240°$$

Each equal angle:

$$\frac{240°}{3} = 80°$$

9. Correct Answer: (B) 20°

Kite $ABCD$

Given:

$AB = AD$and $CB = CD$(kite)

$$\angle ABD = 25°$$
$$\angle BDC = 35°$$

Use properties of a kite

In a kite, the diagonal joining the equal sides bisects the vertex angles.

So, diagonal BD bisects $\angle A$ and $\angle C$.

Find $\angle A$

$\angle A = 2 \times \angle ABD = 2 \times 25° = 50°$

Find $\angle C$

$\angle C = 2 \times \angle BDC = 2 \times 35° = 70°$

Find $\angle A - \angle C$

$| 50° - 70° | = 20°$

10. Correct Answer: (A) 4

Find x if the perimeter of the triangle is 19 cm

The three sides of the triangle are:$x + 1, x + 2, x + 4$

Perimeter of a triangle = sum of its sides:

$$(x + 1) + (x + 2) + (x + 4) = 19$$
$$3x + 7 = 19$$
$$3x = 12$$
$$x = 4$$

11. Correct Answer: (C) 48

Find the perimeter of ΔABC

From the figure:

$$AB = 17$$
$$AC = 10$$

$AD = 8$ is perpendicular to BC at D

In right triangle ABD:

$$BD = \sqrt{AB^2 - AD^2} = \sqrt{17^2 - 8^2}$$
$$= \sqrt{289 - 64} = \sqrt{225} = 15$$

In right triangle ACD:

$$DC = \sqrt{AC^2 - AD^2} = \sqrt{10^2 - 8^2}$$
$$= \sqrt{100 - 64} = \sqrt{36} = 6$$

$$BC = BD + DC = 15 + 6 = 21$$

Perimeter of ΔABC $AB + BC + AC = 17 + 21 + 10 = 48$

12. Correct Answer: (B) 13

Find QV

Given:

QV and WS are medians of $\triangle UVW$

$$QV = 7x - 1$$
$$QW = 4x + 5$$

Since QV is a median, Q is the midpoint of VW, so:

$$QV = QW$$
$$7x - 1 = 4x + 5$$
$$3x = 6$$
$$x = 2$$
$$QV = 7x - 1 = 7(2) - 1 = 13$$

13. Correct Answer: (B) 30

Find AB

Given:KM is a median, so M is the midpoint of AB

$$AM = 2x + 7$$
$$MB = 5x - 5$$

Use midpoint property

$$AM = MB$$
$$2x + 7 = 5x - 5$$
$$3x = 12$$
$$x = 4$$

Find AB

$$AB = AM + MB$$
$$= (2x + 7) + (5x - 5)$$
$$= (2 \cdot 4 + 7) + (5 \cdot 4 - 5)$$
$$= 15 + 15 = 30$$

14. Correct Answer: (B) 4.8 cm

Find the length of altitude AD

Triangle $ABC$is right-angled at A with:

$$AB = 6 \text{ cm}, AC = 8 \text{ cm}$$

First find hypotenuse BC:

$$BC = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100}$$
$$= 10 \text{ cm}$$

Altitude from the right angle to the hypotenuse:

$$AD = \frac{AB \times AC}{BC} = \frac{6 \times 8}{10} = \frac{48}{10} = 4.8 \text{ cm}$$

15. Correct Answer: (A) 12, 35, 37

Which of the following is a Pythagorean triplet?

Check $a^2 + b^2 = c^2$:

$12^2 + 35^2 = 144 + 1225 = 1369 = 37^2$

Others do not satisfy the condition

16. Correct Answer: (D) $5\sqrt{3}$cm

Length of the altitude of an equilateral triangle

Side of equilateral triangle $a = 10$cm.

Formula: Altitude $= \frac{\sqrt{3}}{2} \times a$

$$= \frac{\sqrt{3}}{2} \times 10 = 5\sqrt{3} \text{ cm}$$

17. Correct Answer: (D) 5 cm, 7 cm, 11 cm

Which of the following is the set of measures of the sides of a triangle?

Use the triangle inequality rule:
Sum of any two sides must be greater than the third side.

A) $8 + 4 = 12 < 20$ ❌

B) $9 + 17 = 26 < 30$ ❌

C) $11 + 16 = 27 < 28$ ❌

D) $5 + 7 = 12 > 11, 7 + 11 > 5, 5 + 11 > 7$

18. Correct Answer: (A) 60°

Given $AB \parallel CD$, find $\angle CED$

Angle between BE and $AB = 75°$

Angle between ED and $CD = 45°$

These two angles form a straight angle at point E:

$$\angle CED = 180° - (75° + 45°)$$
$$\angle CED = 60°$$

19. Correct Answer: (D) Kite

In quadrilateral ABCD, $AC \perp BD$and $AB = AD$

$AB = AD \rightarrow$ two adjacent sides equal

Diagonals are perpendicular

These are properties of a kite.

20. Correct Answer: (B) 75

In rectangle ABCD, each angle = $(x+15)°$

Each angle of a rectangle is $90°$.

$$x + 15 = 90$$
$$x = 75$$

ONE OR MORE THAN ONE ANSWER CORRECT TYPE

21. Correct answers: A, B, C

Angles of a triangle are in the ratio $2:3:4$

Sum of ratios $= 9$

Each part $= \frac{180°}{9} = 20°$

Angles: $2 \times 20 = 40°, 3 \times 20 = 60°, 4 \times 20 = 80°$
Smallest angle = $40°$

Largest angle = $80°$

All angles < $90° \rightarrow$ acute-angled

22. Correct answers: A, B, C

Exterior angle = $120°$, opposite interior angles in ratio $1:2$

Let angles be x and $2x$

$x + 2x = 120° \Rightarrow 3x = 120° \Rightarrow x = 40°$
Interior opposite angles: $40°$ and $80°$
Third interior angle: $180° - 120° = 60°$
Smaller opposite angle = $40°$

Larger opposite angle = $80°$

Third interior angle = $60°$

23. Correct answers: A, C, D

Which sets can form a triangle?

Use triangle inequality rule
(sum of any two sides > third side)

A) $5 + 7 = 12 > 9$

B) $6 + 8 = 14 = 14$

C) $7 + 10 = 17 > 12$

D) $9 + 9 = 18 > 9$

24. Correct answers: A, B, C

Three angles of a quadrilateral are $85°, 95°, 110°$

Sum of interior angles of a quadrilateral: $360°$
Fourth angle: $360 - (85 + 95 + 110) = 360 - 290 = 70°$
Now check statements:

(A) Fourth angle is $70°$

(B) All angles are less than $180° \rightarrow$ convex

(C) Sum of all angles is $360°$

25. Correct answers: A, B, D

Parallelogram $PQRS$, $PQ = 9$cm, $QR = 6$cm

Properties of a parallelogram:

Opposite sides are equal

Diagonals bisect each other

Perimeter $= 2(PQ + QR)$

Perimeter:$2(9 + 6) = 30$ cm

Check statements:

(A) Opposite sides are equal

(B) Perimeter is 30cm

(D) Diagonals bisect each other

ASSERTION AND REASON TYPE

26. Correct Answer: (B)Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

Assertion: The sum of the angles of a triangle is always $180°$.
Reason: A triangle is a polygon with three sides.

Assertion is true

Reason is true

But the reason does not explain why the sum is $180°$

27. Correct Answer: (C)Assertion is true, but Reason is false.

Assertion: An equilateral triangle has all sides equal.
Reason: All angles of an equilateral triangle measure $90°$.

Assertion is true

Reason is false (each angle is $60°$, not $90°$)

28. Correct Answer: (B)Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.

Assertion: A right triangle has one angle equal to $90°$.(True)
Reason: The other two angles in a right triangle are always acute.(True)

But the reason does not explain why one angle is $90°$

29. Correct Answer: (A)Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Assertion (A): In a convex quadrilateral, every line segment joining two interior points lies completely inside it.(True)
Reason (R): No interior angle of a convex quadrilateral is greater than $180°$.(True)

Reason correctly explains the property of convexity used in the assertion

30. Correct Answer: (A)Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.

Assertion (A): If three angles of a quadrilateral are $70°, 95°$,and $85°$, then the fourth angle is $110°$.
Reason (R): The sum of interior angles of a quadrilateral is $360°$.

Check the assertion:

$70° + 95° + 85° = 250°$
Fourth angle $= 360° - 250° = 110°$

Assertion is true

Check the reason:

The sum of interior angles of a quadrilateral is $360°\rightarrow$ true

This reason is exactly why the fourth angle is $110°$.

MATRIX MATCHING TYPE

31. Answer:1–B, 2–A, 3–D, 4–C

1. Angle sum property → B. Sum of angles is $180°$
2. Exterior angle property → A. Exterior angle = sum of opposite interior angles
3. Triangle inequality → D. Sum of any two sides > third side
4. Hypotenuse → C. Longest side in a right triangle

32. Answer :1–B, 2–D, 3–A, 4–C

COLUMN–I (Quadrilateral) ↔ COLUMN–II (Property)

1. Trapezium → B. One pair of parallel sides
2. Rectangle → D. All angles $90°$
3. Parallelogram → A. Two pairs of opposite sides parallel

4. Rhombus → C. All sides equal, diagonals perpendicular

COMPREHENSION TYPE

Passage-I

33. Correct Answer: (C) Isosceles triangle

Triangle with sides 6 cm, 6 cm, and 10 cm

Two sides are equal → isosceles triangle

$6 + 6 > 10$→ valid triangle

No right angle given

34. Correct Answer: (D) 100°, 40°, 50°

Which set of angles can NOT form a triangle?

Sum of angles of a triangle must be $180°$.

A) $30 + 60 + 90 = 180$

B) $45 + 45 + 90 = 180$

C) $70 + 50 + 60 = 180$

D) $100 + 40 + 50 = 190$

35. Correct Answer: (C) Obtuse isosceles triangle

Triangle with two equal sides and one angle > 90°

Two equal sides → isosceles

One angle > 90°→ obtuse

Passage-II

36. Correct Answer: (C) Parallelogram

Identify the quadrilateral PQRS

Given properties:

Opposite sides are equal and parallel

Diagonals intersect at their midpoint

These are defining properties of a parallelogram.

37. Correct Answer: (A) 70°

One interior angle of parallelogram PQRS is $110°$

In a parallelogram, adjacent angles are supplementary.

$$\text{Adjacent angle} = 180° - 110° = 70°$$

38. Correct Answer: (B) Rhombus

All sides equal and diagonals intersect at right angles

All sides equal

Diagonals perpendicular

This describes a rhombus (angles need not be 90°, so not necessarily a square).

INTEGER TYPE

39. Correct Answer: $\boxed{175°}$

In the given figure, find $x° + y°$

At point C, the interior angle is $70°$.
So the exterior angle:

$x = 180° - 70° = 110°$
In $\triangle ABC$, sum of interior angles:

$45° + y + 70° = 180°$
$y = 180° - 115° = 65°$
Now,$x + y = 110° + 65° = 175°$

40. Correct Answer: $\boxed{36° \text{ and } 54°}$

In a right-angled triangle, the two acute angles are in the ratio $2:3$

Sum of the two acute angles in a right triangle:$90°$
Let the angles be $2x$ and $3x$:

$$2x + 3x = 90°$$
$$5x = 90° \Rightarrow x = 18°$$

Angles:$2x = 36°, 3x = 54°$

41. Correct Answer: $50°$

Find the value of $y°$

$$PQ \parallel BC$$

At point C, exterior angle $= 110°$

Interior angle at C:$180° - 110° = 70°$
At point A, the straight line gives:$60° + y° + z° = 180°$
Also, $z°$and the interior angle at $C$are corresponding (parallel lines):

$$z = 70°$$

So,$60° + y + 70° = 180°$

$$y = 50°$$

42. Correct Answer: $70°$

In parallelogram PQRS, $\angle P = 70°$

Opposite angles of a parallelogram are equal.

$$\angle R = \angle P = 70°$$

43. Correct Answer: 65°

One angle of a parallelogram is 115°

Adjacent angles in a parallelogram are supplementary:

$$180° - 115° = 65°$$

PREVIOUS YEAR QUESTIONS

1. Correct Answer: A. 14°

Since ABC is an equilateral triangle,

$$\angle A = \angle B = \angle C = 60°$$

In triangle ABD, $DA = DB$, so it is isosceles.
Given angle at D:

$$\angle ADB = 88°$$

Sum of angles in triangle ABD:$\angle DAB + \angle DBA = 180° - 88° = 92°$
Since it is isosceles:$\angle DAB = \angle DBA = 46°$
At point B, angle of equilateral triangle is 60°.
Therefore,$x = 60° - 46° = \boxed{14°}$

2. Correct Answer: C. 49 m

The broken part of the tree forms the hypotenuse of a right-angled triangle.

Vertical height = 24 m, horizontal distance = 7 m.

Using Pythagoras theorem:

$$\text{Length of broken part} = \sqrt{24^2 + 7^2} = \sqrt{576 + 49} = \sqrt{625} = 25 \text{ m}$$

Actual height of the tree:$24 + 25 = \boxed{49 \text{ m}}$

3. Correct Answer: D. 26 cm, 24 cm, 10 cm

For a right-angled triangle, sides must satisfy Pythagoras theorem:

$$a^2 + b^2 = c^2$$

Check option C (largest side = 25 cm):

$$23^2 + 24^2 = 529 + 576 = 1105 \neq 25^2(625)$$

(Incorrect)

Check option D (largest side = 26 cm):

$$24^2 + 10^2 = 576 + 100 = 676 = 26^2$$

(Correct)

4. Correct Answer: C. (i) 70°, (ii) 2.5 cm

(i)Sum of angles of a triangle = 180°

$$(x - 20) + (x - 20) + (x + 10) = 180$$
$$3x - 30 = 180$$
$$x = 70°$$

(ii)From the left right-angled triangle:

$$\text{Height}^2 = 10^2 - 8^2 = 100 - 64 = 36$$
$$\text{Height} = 6 \text{ cm}$$

From the right right-angled triangle:

$$x^2 = 6.5^2 - 6^2 = 42.25 - 36 = 6.25$$
$$x = 2.5 \text{ cm}$$

5. Correct Answer: B 42°

In △ABD,

$$\angle ADC = \angle ABD + \angle BAD \text{[Exterior angle property]}$$
$$\Rightarrow \angle ABD = 95° - 25° = 70° \ldots (i)$$

Now, in △BEC,

$$\angle ABC = \angle BEC + \angle BCE \text{[Exterior angle property]}$$
$$\Rightarrow \angle BCE = \angle DCE = 70° - 28° = 42° \text{[From (i)]}$$

6. Correct Answer: A 2°

As given, $\angle EFA = 90°$and △FGB is an equilateral triangle.

$$\therefore \angle BFG = \angle FGB = \angle GBF = 60°$$
$$\therefore \angle BFC = \angle BFG - \angle CFG = 60° - x \ldots (i)$$

Also,$\angle BFC = \angle AFE - \angle BFE = 90° - y$ $(\because \angle EFA = 90°) \ldots (ii)$
And,$\angle BCF = 180° - 92° =$ 88°(Linear pair)
In △BFC,

$$\angle BFC = 180° - (\angle BCF + \angle CBF) \text{[By angle sum property]}$$
$$\Rightarrow \angle BFC = 180° - (88° + 60°) = 32°$$
$$\Rightarrow 90° - y = 32° \text{[From (ii)]}$$
$$\Rightarrow y = 58°$$

From (i),

$$\angle BFC = 60° - x$$
$$\Rightarrow 32° = 60° - x \Rightarrow x = 60° - 32° = 28°$$
$$\therefore \ y - 2x = 58° - 56° = \boxed{2°}$$

7. Correct Answer A: 90°

In △CED, we have

$$CE = ED$$
$$\therefore \angle EDC = \angle ECD$$

[Angles opposite to equal sides are equal]

$$\Rightarrow \angle ECD = 28^\circ$$

Also,

$\angle ECD = \angle BCA$(Vertically opposite angles)
$\Rightarrow \angle BCA = 28^\circ$
In △BCA,

$y = 62^\circ + 28^\circ$[Exterior angle property]
$\Rightarrow y = 90^\circ$

8. Correct Answer:
C. $\boxed{x = 20^\circ, y = 20^\circ, z = 80^\circ}$

In triangle ASR, angles at A and R are given as:

$$60^\circ + 100^\circ = 160^\circ$$

Hence,$z = 180^\circ - 160^\circ = 20^\circ$
At point R, angle z and 80° form a linear pair:

$$z + 80^\circ = 180^\circ \Rightarrow z = 100^\circ$$

The interior marked angle is:

$$z = 80^\circ$$

In triangle PRQ:$y + 80^\circ + z = 180^\circ \Rightarrow y + 80^\circ + 80^\circ = 180^\circ \Rightarrow y = 20^\circ$
At point P, the straight angle is split into x and y:

$$x + y = 40^\circ \Rightarrow x = 40^\circ - 20^\circ = 20^\circ$$

9. Correct Answer: C. 5, 7, 12

A Pythagorean triplet satisfies:

$$a^2 + b^2 = c^2$$

A: $7^2 + 24^2 = 49 + 576 = 625 = 25^2$

B: $15^2 + 20^2 = 225 + 400 = 625 = 25^2$

C: $5^2 + 7^2 = 25 + 49 = 74 \neq 12^2(144)$

D: $20^2 + 21^2 = 400 + 441 = 841 = 29^2$

10. Correct Answer: C. $\boxed{62^\circ}$

In △PQU,

$\angle PQU + \angle QPU$
$= \angle SUQ$(Exterior angle property)
$\Rightarrow m + 30^\circ = 85^\circ \Rightarrow m = 85^\circ - 30^\circ = 55^\circ$
In △QTR,

$\angle QTR + \angle QRT$
$= \angle PQU$(Exterior angle property)
$\Rightarrow n + 48^\circ = 55^\circ \Rightarrow n = 55^\circ - 48^\circ = 7^\circ$
$\therefore\ m + n = 55^\circ + 7^\circ = \boxed{62^\circ}$

7. Congruency of Triangles

Solutions

Worksheet – 1

1. Correct Answer: A) ΔPLO ≅ ΔPMO

Given:
PL ⊥ OB and PM ⊥ OA such that PL = PM.

Since PL ⊥ OB, ∠PLO = 90°.

Since PM ⊥ OA, ∠PMO = 90°.

PL = PM (Given).

PO = PO (Common side).

Thus, in triangles ΔPLO and ΔPMO:

∠PLO = ∠PMO (Each is 90°)

PL = PM

PO = PO

Therefore, by the RHS (Right angle–Hypotenuse–Side) congruence rule,

2. Correct Answer: B) 5 cm

Using Pythagoras theorem:

$$BC^2 = CA^2 + AB^2$$
$$13^2 = 12^2 + AB^2$$
$$169 = 144 + AB^2$$
$$AB^2 = 25$$

$AB = 5$cm

Since AB corresponds to EF,

EF = 5 cm

3. Correct Answer: B) 15°

Given: ΔABC ≅ ΔPQR
∠A corresponds to ∠P
∠A = (5x + 4)°
∠P = 79°

Since corresponding angles of congruent triangles are equal:

$$5x + 4 = 79$$
$$5x = 75$$
$$x = 15$$

4. Correct Answer: C) 3

Given: ΔABC ≅ ΔPQR
BC = 3x + 4
QR = 2x + 7

Since corresponding sides of congruent triangles are equal:

$$3x + 4 = 2x + 7$$
$$3x - 2x = 7 - 4$$
$$x = 3$$

5. Correct Answer: A) 90°

Given: ΔPQR ≅ ΔP'Q'R'
∠P = 2∠Q'
∠Q' = 30°

∠P = 2 × 30° = 60°

Sum of angles of triangle:
∠P + ∠Q + ∠R = 180°

60° + 30° + ∠R = 180°
∠R = 180° – 90°
∠R = 90°

6. Correct Answer: D) 80°

Given: ΔABC ≅ ΔPQR
∠A = 2x
∠Q = (3/2)x
∠C = 40°

Since triangles are congruent,
∠B = ∠Q

Using angle sum in ΔABC:

2x + (3/2)x + 40 = 180

Multiply by 2:

4x + 3x + 80 = 360
7x = 280
x = 40

∠P = ∠A = 2x = 80°

7. Correct Answer: A) 25 – 5x

Given: ΔABC ≅ ΔA'B'C'
AC = 3x + 4
B'C' = 2x + 1
Perimeter of ΔA'B'C' = 30 cm

Since triangles are congruent, perimeters are equal.

Perimeter = AB + BC + CA

= (3x + 4) + (2x + 1) + AB

= 5x + 5 + AB

Given perimeter = 30

AB = 30 − (5x + 5)
AB = 25 − 5x

8. Correct Answer: C) 15 cm

Given:
QR = x + 8
P'Q' = 2x − 1
PR = 2x + 3
Perimeter of ΔP'Q'R' = 45

Perimeter = (x + 8) + (2x − 1) + (2x + 3)

= 5x + 10

5x + 10 = 45
5x = 35
x = 7

Q'R' = x + 8 = 15 cm

9. Correct Answer: D) 64°

Given:
∠A + ∠B = 116°

Sum of angles of triangle:

∠A + ∠B + ∠C = 180°

116° + ∠C = 180°

∠C = 64°

10. Correct Answer: B) A.A.A.

AAA proves similarity, not congruence.

11. Correct Answer: C.

Explanation:

For three numbers **a, b, and c** to be a **Pythagorean triplet**, they must satisfy the condition:

$$a^2 + b^2 = c^2$$

Here, **c** is the largest number (hypotenuse).

This condition comes from the **Pythagoras Theorem**, which states that in a right-angled triangle:

$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$

COMPITITIVE CORNER

SINGLE ANSWER CORRECT TYPE

1. Correct Answer : B) ΔAED ≅ ΔAEB, ΔBEC ≅ ΔCED

From the given figure:

- Opposite sides are marked parallel.
- Diagonals intersect at E.
- Markings show equal segments on diagonals.

The two pairs of congruent triangles are:

2. Correct Answer: B) S.A.S

To prove ΔABC ≅ ΔADC:

- AB = DC (opposite sides of rectangle)
- AC is common
- ∠BAC = ∠ACD (right angles)

So the correct postulate is:

3. Correct Answer: C) ΔABC ≅ ΔRPQ

From the figure:

- ∠A = ∠Q (top angles marked equal)
- ∠B = ∠R (base angles marked equal)
- One corresponding side is marked equal

So triangles are congruent by **A.A.S.**

Correct matching order:

4. Correct Answer: C) RHS

From the figure:

- Right angles at B and E
- Hypotenuse AC = DF (marked equal)
- One side AB = DE (marked equal)

So triangles are congruent by RHS (Right angle–Hypotenuse–Side)

5. Correct Option: A) ΔADC ≅ ΔCBA

We are given that **AB = CD** and **AD = BC**. We need to find which congruence statement is true for the larger overlapping triangles.

In ΔADC and ΔCBA:

AD = CB (Given)

DC = BA (Given)

AC = AC (Common side)

By the SSS (Side–Side–Side) postulate, ΔADC ≅ ΔCBA.

Note: Order matters! A in the first triangle corresponds to C in the second because of the side lengths.

6. Correct Answer: A) S.A.S.

Given:

- $PQ = SR$
- $\angle PQR = \angle QRS$
- $QR$is common

Thus, two sides and the included angle are equal.

Postulate Used: S.A.S.

7. Correct Answer: D) ΔBQC ≅ ΔCPB

Given:

- $BP \perp AC$and $CQ \perp AB \rightarrow$ Both triangles are right-angled.
- $BQ = CP$(Given)
- $BC = CB$(Common side)

So the triangles are congruent by **RHS (Right angle–Hypotenuse–Side)**.

8. Correct Answer: B) ΔOAB ≅ ΔODC

Given:

- $BC \parallel AD$
- AD and BC intersect at O
- Vertical angles at $O$are equal
- Alternate interior angles are equal
- $AO = OD$(marked equal)

Thus triangles are congruent by **ASA**.

9. Correct Answer: A) BC = CD

Since $AB = AC$, base angles of isosceles triangle ΔABC are equal:

$$\angle ABC = \angle BCA$$

Because $AB \parallel CD$,

$$\angle ACD = \angle CAB$$

(Alternate interior angles)

1. Given $\angle ACD = 18°$, so

$$\angle CAB = 18°$$

2. Using angle properties and symmetry in the isosceles triangle, we get two triangles with equal angles and a common side.
3. By ASA congruence, the required sides become equal.

$$\boxed{BC = CD}$$

10. Correct Answer: D) 40°

Angle at A of big triangle:

$$\angle A = \angle BAD + \angle DAC$$

From symmetry of equal sides,

$$\angle BAD = 40°$$

So,

$$\angle A = 40° + 50° = 90°$$

Now using angle sum property:

$$\angle ABC = 180° - (90° + 50°)$$
$$\angle ABC = 40°$$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct Answer: C) x = 64°

D) x = 180° – (29° + 87°) (since 29° × 3 = 87°, result = 64°)

Find the value of x

Given:

- $\angle B = 29°$
- $AD = BD$
- $CD = AC$

Step 1: In ΔABD

AD = BD ⇒ Isosceles triangle

Base angles are equal:

$$\angle BAD = \angle ABD = 29°$$

So,

$$\angle ADB = 180° - 2(29°)$$
$$\angle ADB = 180° - 58° = 122°$$

Step 2: Straight line at D

Since C, D, B are collinear:

$$\angle CDA + \angle ADB = 180°$$
$$\angle CDA = 180° - 122° = 58°$$

Step 3: In ΔACD

AC = CD ⇒ Isosceles triangle

Base angles are equal:

$$\angle CAD = \angle CDA = 58°$$

So,

$$x = \angle ACD$$
$$x = 180° - 2(58°)$$
$$x = 180° - 116° = 64°$$

12. Correct Answer:

A) AD = AE

B) ΔABD ≅ ΔACE

C) ∠ADE = ∠AED

D) BD = CE

Given AB = AC and ∠BAD = ∠CAE

Step 1: Since AB = AC

ΔABC is isosceles

$$\angle ABC = \angle ACB$$

Step 2: Compare ΔABD and ΔACE

AB = AC

∠BAD = ∠CAE (given)

∠ABD = ∠ACE (base angles of isosceles triangle)

Therefore,

$$\triangle ABD \cong \triangle ACE (\text{SAS})$$

From congruence:

AD = AE

BD = CE

Since AD = AE, ΔADE is isosceles:

$$\angle ADE = \angle AED$$

13. Correct Answer:

A) ∠ABD = ∠ACD ✓
B) ∠BAD = ∠CAD ✓
C) ∠BDA = ∠CDA ✓
D) ΔABD ≅ ΔACD ✓

Given: AB = AC and BD = CD

We need to find which statements are **true**.

Step 1: Consider ΔABD and ΔACD

Given:

- $\text{AB} = \text{AC}$
- $\text{BD} = \text{CD}$
- $\text{AD} = \text{AD}$ (common side)

So, in triangles **ABD** and **ACD**:

- $AB = AC$
  $BD = CD$
  $AD = AD$

Therefore,

$$\triangle ABD \cong \triangle ACD (SSS)$$

Step 2: Corresponding parts are equal (CPCT)

From congruency:

- $\angle \text{ABD} = \angle \text{ACD}$
- $\angle \text{BAD} = \angle \text{CAD}$
- $\angle \text{BDA} = \angle \text{CDA}$

14. Correct Answer:

A) x − p = 0 ✓

B) y = q ✓

C) x + y = p + q ✓

D) The value of x − p is y − q ✓ (both are 0)

Given: AB = AC and DB = DC

We have two isosceles triangles:

- ΔABC with $\text{AB} = \text{AC}$
- ΔDBC with $\text{DB} = \text{DC}$

Let us analyze.

Step 1: In ΔABC

Since **AB = AC**, base angles are equal:

$$\angle ABC = \angle BCA$$

From the figure:

$$\angle ABC = x$$
$$\angle BCA = p$$

So,

$$x = p$$

Step 2: In ΔDBC

Since **DB = DC**, base angles are equal:

$$\angle DBC = \angle BCD$$

From the figure:

$$\angle DBC = y$$
$$\angle BCD = q$$

So,

$$y = q$$

Step 3: Whole angle at B and C

Whole angle at **B**:

$$\angle ABC + \angle DBC = x + y$$

Whole angle at **C**:

$$\angle BCA + \angle BCD = p + q$$

Since $x = p$and $y = q$,

$$x + y = p + q$$

15. Correct Answer:

B) The given information follows the S.A.S. pattern ✓

D) ΔABC ≅ ΔDFE ✓

Compare the given information

From markings:

$$AB = DF$$
$$BC = DE$$
$$\angle B = \angle E$$

The equal angle is included between the two equal sides.

So this follows:

Side – Angle – Side (SAS)

Step 2: Conclusion

By SAS congruency rule:

$\triangle ABC \cong \triangle DFE$

ASSERTION AND REASON TYPE

16. Correct Answer: Both A and R are true, and R is the correct explanation of A.

Assertion (A): Two triangles are congruent if their three sides are equal.
Reason (R): SSS (Side-Side-Side) is a valid criterion for triangle congruence.

The assertion is **true** (this is exactly the SSS rule).
The reason is **true**.
The reason correctly explains the assertion.

17. Correct Answer: Assertion is false, but Reason is true.

Assertion (A): Two triangles having equal areas are always congruent.
Reason (R): Congruent triangles must have equal areas.

✗ The assertion is **false** (triangles can have equal areas but different shapes and sizes).
✓ The reason is **true** (congruent triangles always have equal areas).

18. Correct Answer: Both A and R are true, and R is the correct explanation of A.

Assertion (A): The ASA rule can be used to prove triangle congruence.
Reason (R): ASA stands for Angle-Side-Angle.

The assertion is **true**.
The reason is **true**.
The reason correctly explains the assertion (it defines the rule used).

19. Correct Answer: Both A and R are true, and R is the correct explanation of A.

Assertion (A): A triangle can be congruent to another even if their orientations are different.
Reason (R): Congruent triangles are identical in shape and size.

Solution:
Congruent triangles have the same shape and size. Their position or orientation (rotation or reflection) does not affect congruency. Therefore, triangles can be congruent even if they are placed differently.

20. Correct Answer: Both A and R are true, but R is not the correct explanation of A.

Assertion (A): In two congruent triangles, corresponding parts are equal.
Reason (R): CPCT is the short form for "Corresponding Parts of Congruent Triangles".

Solution:
When two triangles are congruent, all their corresponding sides and angles are equal. This property is known as CPCT (Corresponding Parts of Congruent Triangles). The reason correctly gives the

meaning of CPCT, but it does not explain why corresponding parts are equal.

MATCH MATRIX TYPE

21. Match the Congruence Rule with Its Name

Column A (Abbreviation) → Column B (Full Form)

A. SSS → **4. Side–Side–Side**
B. SAS → **3. Side–Angle–Side**
C. ASA → **2. Angle–Side–Angle**
D. RHS → **1. Right angle–Hypotenuse–Side**

22. Match the Conditions with the Triangle Pairing

Column A (Condition) → Column B (Type of Congruence)

A. All sides equal → **4. SSS**
B. Two angles and the included side equal → **1. ASA**
C. One right angle, hypotenuse, and one side equal → **3. RHS**
D. Two sides and the included angle equal → **2. SAS**

COMPREHENSION TYPE

Passage-1

23. Correct Answer: B) S.A.S.

Which congruence rule is applicable here?

Given in the passage:
Two sides and the included angle of one triangle match exactly with the other.

This follows the **Side–Angle–Side (SAS)** rule.

24. Correct Answer: C) Yes, because they satisfy the S.A.S. criterion.

Can the two triangles be considered congruent? Why?

If two sides and the included angle are equal, then by the **SAS congruence criterion**, the triangles are congruent. Their position (on different sides of the flag) does not affect congruency.

25. Correct answers: A

Name any two corresponding parts of the triangles that must be equal.

If the triangles are congruent, then by CPCT (Corresponding Parts of Congruent Triangles):

- Corresponding sides are equal
- Corresponding angles are equal
- From the options:
- PR = XZ ✓

(These are valid corresponding parts.)

Passage - II :

26. Correct Answer: B) Congruent

The two support triangles are:

Given:
Both triangles have exactly equal three sides (4 m, 5 m, 6 m).

If all three sides are equal in both triangles, they are **congruent**.

27. Correct Answer: C) SSS

Which congruence rule applies to them?

Since all three corresponding sides are equal, the rule used is:

SSS (Side–Side–Side)

28. Correct Answer: B) It ensures symmetry and equal load distributio

Why is congruence important in bridge construction?

Congruent triangles on both sides ensure:

- Proper symmetry
- Equal load distribution
- Structural balance

INTEGER TYPE

29. Correct Answer: y = 5

(Top Question – ASA Congruence)

Given:
$\Delta ABC \cong \Delta PQR$

So corresponding angles are equal:

$\angle B = \angle Q$
$\angle C = \angle R$

Given:
∠B = (x + 15)°
∠Q = 70°

So,
x + 15 = 70
x = 55

Now,

∠C = (5y + 10)°
∠R = 35°

So,
5y + 10 = 35
5y = 25
y = 5

30. Correct Answer: 2

(Isosceles Triangle Question)

Given:
AB = AC

Let each base angle = B

Vertex angle = A

Sum of angles:
A + 2B = 180

Given:
Vertex angle = twice the sum of base angles

A = 2(2B)
A = 4B

Substitute:

4B + 2B = 180
6B = 180
B = 30°

Now,

1/15 of ∠B = 30/15 = 2

31. Correct Answer: (Second Question – Angle Relation)

Given:
∠B = 2∠C

Let ∠C = x
Then ∠B = 2x

AC = BC ⇒ base angles equal

So, ∠A = ∠B = 2x

Sum of angles:

2x + 2x + x = 180
5x = 180
x = 36

∠BAC = ∠A = 72°

Now,

1/5 of ∠BAC = 72/5 = 14.4

But since integer type:

72 ÷ 5 = **14.4**

(If integer required, answer = 14.4)

32. Correct Answer: 1

If ΔABD ≅ ΔACD, then corresponding sides are equal.

So,

x / 4 = 1

Previous Year Questions

1. Correct Answer: B

In the given figure, choose the correct order of congruency.

From the figure:

AB = AD (single tick marks)
BC = CD (double tick marks)
∠ABC = ∠ADC (marked angles)
AC is common to both triangles

Now compare triangles ABC and ADC:

AB = AD
BC = CD
Included angles ∠ABC = ∠ADC

Therefore, by the **SAS congruence rule**, triangle ABC is congruent to triangle ADC.

Correct order of corresponding vertices:

A corresponds to A
B corresponds to D
C corresponds to C

Hence,

ΔABC ≅ ΔADC

2. Correct Answer: A) Faces = 5, Edges = 8

The solid given below is a pyramid. How many faces and edges does it have?

From the figure, the base of the pyramid is quadrilateral (4 sides).

So it is a square (or quadrilateral) pyramid.

For a pyramid with a 4-sided base:

Faces:

- 1 base face
- 4 triangular side faces

Total faces = 5

Edges:

- 4 edges of the base
- 4 edges from the apex to each base vertex

Total edges = 8

3. Correct Answer: B) Kartik and Sid

Kartik:

"Two squares are congruent, if they have same area."
✓ Correct.
If two squares have the same area, their sides are equal. Hence, they are congruent.

Sid:
"Two circles are congruent, if they have same diameter."
✓ Correct.
Circles with equal diameter (or radius) are congruent.

4. Correct Answer: C) SAS

By which congruence criteria is ΔQSP ≅ ΔRSP?

From the figure:

- QS = SR (tick marks on the vertical line)
- ∠QTS and ∠RTS are right angles (right angle mark at T)
- SP is common to both triangles

So in triangles QSP and RSP:

QS = SR
∠QSP = ∠RSP (marked equal angles at S)
SP = SP (common side)

Thus, two sides and the included angle are equal.

Therefore, the triangles are congruent by **SAS** rule.

5.

6. Correct Answer: C) ΔPQR ≅ ΔSRQ

Compare triangles PQR and SRQ

In ΔPQR and ΔSRQ:

PQ = SR (given)
PR = SQ (given)
QR = RQ (common side)

So, all three corresponding sides are equal.

Therefore, triangles are congruent by **SSS** rule.

Correct order of correspondence:

P corresponds to S
Q corresponds to R
R corresponds to Q

Hence,

ΔPQR ≅ ΔSRQ

8. Ratio and Its Applications

Solutions

Worksheet – 1

1. Correct Answer: (A) 2

Number of terms in a ratio

A ratio compares two quantities.

2. Correct Answer: (B) 10 : 7

Ratio between 1.2 kg and 840 grams

Convert both to the same unit:
1.2 kg = 1200 g

Ratio = 1200 : 840
Divide by 120 → 10 : 7

3. Correct Answer: (C) 16 : 21

If A : B = 6 : 7 and B : C = 8 : 9, find A : C

Make B common:
A : B = 6 : 7 → multiply by 8 → 48 : 56
B : C = 8 : 9 → multiply by 7 → 56 : 63

So, A : C = 48 : 63 = 16 : 21

4. Correct Answer: (A) 3 : 4 : 6

If A : B = 3 : 4 and B : C = 2 : 3, find A : B : C

Make B same:
A : B = 3 : 4 → ×2 → 6 : 8
B : C = 2 : 3 → ×4 → 8 : 12

So, A : B : C = 6 : 8 : 12 = 3 : 4 : 6

5. Correct Answer: (A) 4 : 6 : 9

Find the continued ratio of 2 : 3 and 6 : 9

Write as 2 : 3 : 9
Simplify by dividing by 1 → 2 : 3 : 9
Convert to simplest form → 4 : 6 : 9

6. Correct Answer: (B) 3 : 28

If 0.7 of one number = 0.075 of another number

Let numbers be x and y:
0.7x = 0.075y

x : y = 0.075 : 0.7
= 75 : 700= 3 : 28

7. Correct Answer: (A) 16 : 25

A class has 32 boys and 18 girls

Total students = 32 + 18 = 50

Required ratio = boys : total
= 32 : 50 = 16 : 25

8. Correct Answer: (A) 15 : 28

Which of the following represents the compound ratio of 5 : 7 and 3 : 4?

Compound ratio
= (5 × 3) : (7 × 4)= 15 : 28

9. Correct Answer: (A) 12

If 6, x, and 24 are in proportion, then the value of x is

6 : x = x : 24

$\Rightarrow x^2 = 6 \times 24$
$\Rightarrow x^2 = 144$
$\Rightarrow x = 12$

10. Correct Answer: (B) 49 : 81

The duplicate ratio of 7 : 9 is

Duplicate ratio = square of each term
$= 7^2 : 9^2 = 49 : 81$

Worksheet – 2

1. Correct Answer: C. ₹225

Given: 12 kg rice = ₹540
Cost of 1 kg = 540 ÷ 12 = ₹45
Cost of 5 kg = 45 × 5 = **₹225**

2. Correct Answer: B. ₹3,000

Given: Wages of 8 workers = ₹1,600
Wages of 1 worker = 1600 ÷ 8 = ₹200
Wages of 15 workers = 200 × 15 = **₹3,000**

3. Correct Answer: B. ₹300

Given: Earnings for 8 days = ₹480
Earnings for 1 day = 480 ÷ 8 = ₹60
Earnings for 5 days = 60 × 5 = **₹300**

4. Correct Answer: B. 120 litres

Given: 300 litres in 5 hours
Water filled in 1 hour = 300 ÷ 5 = 60 litres
Water filled in 2 hours = 60 × 2 = **120 litres**

5. Correct Answer: C. 8

Given: Cost of 15 chocolates = ₹75
Cost of 1 chocolate = 75 ÷ 15 = ₹5
Chocolates for ₹40 = 40 ÷ 5 = **8 chocolates**

6. Correct Answer: C. 300 km

Given: 180 km in 3 hours
Speed = 180 ÷ 3 = 60 km/h
Distance in 5 hours = 60 × 5 = **300 km**

7. Correct Answer: C. 10 kg

Given: Cost of 15 kg sugar = ₹675
Cost of 1 kg = 675 ÷ 15 = ₹45
Quantity for ₹450 = 450 ÷ 45 = **10 kg**

8. Correct Answer: B. ₹64

Given: Cost of 1 notebook = ₹40
Cost of 8 notebooks = 8 × 40 = ₹320
Cost of 5 books = ₹320
Cost of 1 book = 320 ÷ 5 = **₹64**

9. Correct Answer: C. 480

Given: 7 machines make 280 toys
1 machine makes = 280 ÷ 7 = 40 toys
12 machines make = 40 × 12 = **480 toys**

10. Correct Answer: B. 900

Given: 360 bulbs in 6 days using 8 machines
Bulbs per day per machine = 360 ÷ (6 × 8) = 7.5
Bulbs in 10 days using 12 machines = 7.5 × 10 × 12 = **900 bulbs**

Worksheet – 3

1. Correct Answer: D) 175%

The fraction $\frac{7}{4}$ in percentage

$$\frac{7}{4} = 1.75$$
$$1.75 \times 100 = 175\%$$

2. Correct Answer: D) 102.6

Value of the expression

$$\frac{25}{17} \times 68\% + 8.7 \times (3 + 9) - 2.8$$

Convert percentage:

$$68\% = \frac{68}{100}$$
$$\frac{25}{17} \times \frac{68}{100} = \frac{1700}{1700} = 1$$
$$8.7 \times 12 = 104.4$$
$$1 + 104.4 - 2.8 = 102.6$$

3. Correct Answer: C) $\frac{1}{8}$

$$0.125 = \frac{125}{1000} = \frac{1}{8}$$

4. Correct Answer: B) $62\frac{1}{2}$%

The circle is divided into 8 equal parts.

5 parts are shaded.

$$\text{Shaded fraction} = \frac{5}{8}$$
$$\frac{5}{8} \times 100 = 62.5\% = 62\frac{1}{2}\%$$

5. Correct Answer: C) $\frac{7}{10}$

$$70\% = \frac{70}{100} = \frac{7}{10}$$

6. Correct Answer: B) 15%

Profit percentage

Cost Price (C.P.) = ₹800
Selling Price (S.P.) = ₹920

Profit = 920 – 800 = ₹120

$$\text{Profit \%} = \frac{120}{800} \times 100 = 15\%$$

7. Correct Answer: B) 10%

Loss percentage

Selling Price (S.P.) = ₹720
Loss = ₹80

Cost Price (C.P.) = 720 + 80 = ₹800

$$\text{Loss \%} = \frac{80}{800} \times 100 = 10\%$$

8. Correct Answer: C) 20%

Profit percentage

Cost Price (C.P.) = ₹1,000
Selling Price (S.P.) = ₹1,200

Profit = 1,200 – 1,000 = ₹200

$$\text{Profit \%} = \frac{200}{1000} \times 100 = 20\%$$

9. Correct Answer: C) ₹5,400

Simple Interest – Amount to be repaid

Principal (P) = ₹4,500
Rate (R) = 8% p.a.
Time (T) = 2½ years = 2.5 years

$$\text{S.I.} = \frac{P \times R \times T}{100}$$

$$\text{S.I.} = \frac{4500 \times 8 \times 2.5}{100} = ₹900$$

Amount = P + S.I. = 4500 + 900 = ₹5,400

10. Correct Answer: B) ₹1,020

Selling price after discount

Marked Price (M.P.) = ₹1,200
Discount = 15% of 1,200

$$\text{Discount} = \frac{15}{100} \times 1200 = ₹180$$

Selling Price = 1,200 − 180 = ₹1,020

Competitive Corner

SINGLE ANSWER CORRECT TYPE

1. Correct Answer: A) 20

Age of the elder brother

Ratio (elder : younger) = 8 : 5
Younger's age = 12 yrs 6 months = 12.5 years

1 part = 12.5 ÷ 5 = 2.5 years
Elder's age = 8 × 2.5 = 20 years

2. Correct Answer: A) 456

Divide ₹1216 in the ratio $\frac{1}{3}:\frac{1}{5}$; find B's share

Convert ratio:

$\frac{1}{3}:\frac{1}{5} = 5:3$
Total parts = 5 + 3 = 8
Value of 1 part = 1216 ÷ 8 = 152

B's share = 3 × 152 = ₹456

3. Correct Answer: D) 20

Ratio of two quantities is 3 : 4

Antecedent (3 parts) = 15
1 part = 15 ÷ 3 = 5

Consequent = 4 × 5 = 20

4. Correct Answer: B) 12 : 13

If $x:y = 3:4$, find $x^2y + xy^2 : x^3 + y^3$

Let $x = 3k, y = 4k$

Numerator: $x^2y + xy^2 = (3k)^2(4k) + (3k)(4k)^2 = 36k^3 + 48k^3 = 84k^3$
Denominator: $x^3 + y^3 = (3k)^3 + (4k)^3 = 27k^3 + 64k^3 = 91k^3$
Ratio = 84 : 91 = 12 : 13

5. Correct Answer: D) $\frac{4}{5}$

$$80\% = \frac{80}{100} = \frac{4}{5}$$

6. Correct Answer: C) 70%

Percentage equivalent of $\frac{7}{10}$

$$\frac{7}{10} \times 100 = 70\%$$

7. Correct Answer: C) 27 : 29

$\frac{5x - 3y}{5y - 3x} = \frac{3}{4}$
$4(5x - 3y) = 3(5y - 3x)$
$20x - 12y = 15y - 9x$
$29x = 27y \Rightarrow x:y = 27:29$

8. Correct Answer: B) 15 : 20 : 8

Make B common (LCM of 4 and 5 = 20)

$A:B = 3:4 \Rightarrow 15:20$
$B:C = 5:2 \Rightarrow 20:8$
$A:B:C = 15:20:8$

9. Correct Answer: A) 144

If 81, x, x, 256 are in proportion

$81:x = x:256 \Rightarrow x^2 = 81 \times 256$
$x^2 = 20736 \Rightarrow x = 144$

10. Correct Answer: B) 24, 32

Age problem (Krishna : Balaram)

Present ratio = 3 : 4
Let ages = 3x and 4x

Four years earlier:

$\frac{3x - 4}{4x - 4} = \frac{5}{7}$
$7(3x - 4) = 5(4x - 4)$
$21x - 28 = 20x - 20 \Rightarrow x = 8$
Present ages = $3x = 24,\ 4x = 32$

11. Correct Answer: C) ₹81

Cost of 18 articles

Cost of 40 articles = ₹180
Cost of 1 article = 180 ÷ 40 = ₹4.5

Cost of 18 articles = 18 × 4.5 = ₹81

12. Correct Answer: C) 270

Number of shawls woven

5 persons weave 180 shawls in 12 days

Shawls per person per day

$$= \frac{180}{5 \times 12} = 3$$

For 6 persons in 15 days:

$$6 \times 15 \times 3 = 270$$

13. Correct Answer: B) ₹2,000

Find the principal

Simple Interest (S.I.) = ₹360
Rate (R) = 9% p.a.
Time (T) = 2 years

$$\text{S.I.} = \frac{P \times R \times T}{100}$$

$$360 = \frac{P \times 9 \times 2}{100}$$

$$P = \frac{360 \times 100}{18} = 2000$$

14. Correct Answer: C) ₽765

Selling price at 15% loss

Cost Price (C.P.) = ₹900

Loss = 15% of 900 = ₹135

Selling Price = 900 – 135 = ₹765

15. Correct Answer: A) 32

If 15 : 8 = 60 : x

$$\frac{15}{8} = \frac{60}{x}$$

$$15x = 480 \Rightarrow x = 32$$

16. Correct Answer: A) $\frac{26}{9}$

Fourth proportional to $\frac{1}{2}, \frac{1}{3}, 4\frac{1}{3}$

Fourth proportional $x$satisfies:

$$\frac{1}{2} : \frac{1}{3} = 4\frac{1}{3} : x$$

Convert mixed fraction:

$$4\frac{1}{3} = \frac{13}{3}$$

$$x = \frac{\frac{1}{3} \times \frac{13}{3}}{\frac{1}{2}} = \frac{13}{9} \times 2 = \frac{26}{9}$$

17. Correct Answer: C) 25%

Profit percentage

Cost Price (C.P.) = ₹500
Profit = ₹125

$$\text{Profit \%} = \frac{125}{500} \times 100 = 25\%$$

18. Correct Answer: A) 22500

Value NOT equivalent to 225%

$$225\% = \frac{225}{100} = 2.25 = 2\frac{1}{4} = \frac{9}{4}$$

Option (A) 22500 is not equivalent.

19. Correct Answer: C) ₽1,200

Find the cost price

Selling Price (S.P.) = ₹1,260
Gain = 5%

$$\text{C.P.} = \frac{100}{105} \times 1260 = ₹1200$$

20. Correct Answer: A) 1

Number to be added

Let the number be x.

$$\frac{5+x}{7+x} = \frac{8+x}{11+x}$$

$$(5+x)(11+x) = (7+x)(8+x)$$

$$x^2 + 16x + 55 = x^2 + 15x + 56$$

$$x = 1$$

21. Correct Answer: C) 20%

Profit percentage

Selling Price (S.P.) = ₹1,440
Profit = ₹240

Cost Price (C.P.) = 1440 – 240 = ₹1,200

$$\text{Profit \%} = \frac{240}{1200} \times 100 = 20\%$$

22. Correct Answer: B) ₹450

Simple Interest for 9 months

Principal (P) = ₹10,000
Rate (R) = 6% p.a.
Time (T) = 9 months = $\frac{9}{12} = \frac{3}{4}$years

$$\text{S.I.} = \frac{P \times R \times T}{100}$$

$$= \frac{10000 \times 6 \times \frac{3}{4}}{100} = ₹450$$

ONE OR MORE THAN ONE CORRECT ASWER TYPE

23. Correct Answers: A, B, C and D

Compound ratio = (product of antecedents) : (product of consequents)

A. 3:4 and 5:6 $\rightarrow 3 \times 5 : 4 \times 6 = 15 : 24$

B. a:b and c:d $\rightarrow ac : bd$

C. 4:5 and 8:10 $\rightarrow 4 \times 8 : 5 \times 10 = 32 : 50$

D. m:n compounded with itself $\rightarrow m^2 : n^2$

24. Correct Answers: A, B and C

Factors affecting simple interest

Simple Interest depends on:

Principal

Rate of interest

Time

Selling price ✗

25. Correct Answer: B) 120

Marks needed to secure 90%

Total marks = 600

Marks obtained = 70% of 600

$$= \frac{70}{100} \times 600 = 420$$

Marks for 90%

$$= \frac{90}{100} \times 600 = 540$$

Extra marks needed

$$= 540 - 420 = 120$$

26. Correct Answers: A, B and D

If the marked price is ₹1,000 and discount is ₹150

Marked Price (M.P.) = ₹1,000
Discount = ₹150

Selling Price (S.P.) = 1000 – 150 = ₹850

Discount %= $\frac{150}{1000} \times 100 = 15\%$

Check statements:

A. Selling price is ₹850

B. Discount percentage is 15%

C. Discount percentage is calculated on S.P. (It is calculated on M.P.)

D. Discount is less than marked price

27. Correct Answers: A, C and D

Correct statements about discount

A. Discount is the reduction on marked price

B. Discount = Selling Price – Marked Price
(Correct formula: Discount = Marked Price – Selling Price)

C. Discount reduces the selling price

D. Discount can be expressed as a percentage

ASSERTION AND REASON TYPE

28. Correct Answer: A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.

Assertion (A): The percentage increase from 100 to 125 is 25%.
Increase = 125 – 100 = 25

$$\frac{25}{100} \times 100 = 25\%$$

Assertion is true.

Reason (R): Percentage increase = (Change / Original) × 100.
Reason is true and correctly explains A.

29. Correct Answer: D)A is false, but R is true.

Assertion (A): A discount of 15% on ₹2,000 is equal to ₹250.

$$15\% \text{ of } 2000 = \frac{15}{100} \times 2000 = ₹300$$

Assertion is false.

Reason (R): Discount = (Discount % × Marked Price) ÷ 100.

Reason is true.

30. Correct Answer: D) A is false, but R is true.

Assertion–Reason

Assertion (A): If $a : b = c : d$, then $a \times b = c \times d$.
This is incorrect.
Correct property is: $a \times d = b \times c$.

Reason (R): In a proportion, the product of extremes equals the product of means.
This statement is true.

31. Correct Answer: A) Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.

Assertion (A): A loss occurs when the cost price is more than the selling price.
True.

Reason (R): Loss = Cost Price – Selling Price.
True and correctly explains the assertion.

32. Correct Answer: A)Both Assertion (A) and Reason (R) are true, and R is the correct explanation of A.

Assertion (A): Profit = ₱100,

Cost Price = ₹500

$$\text{Profit \%} = \frac{100}{500} \times 100 = 20\%$$

Assertion is true.

Reason (R):
Profit % = (Profit / Cost Price) × 100

Reason is true and it correctly explains the assertion.

MATRIX MATCHING TYPE

33. Correct Matching: A–1, B–3, C–4, D–2

A. Simple Interest → 3. (P × R × T) ÷ 100

B. Loss → 4. Cost Price – Selling Price

C. Profit → 1. Selling Price – Cost Price

D. Discount → 2. Marked Price – Selling Price

34. Correct Matching: A–3, B–4, C–1, D–2

A. Profit → 1. Bought at ₹50, sold at ₹70

B. Loss → 3. Bought at ₹90, sold at ₹60

C. Percentage Increase → 4. Price raised from ₹100 to ₹125

D. Percentage Decrease → 2. Marks dropped from 80 to 60

COMPREHENSION TYPE

Passage I:

35. Correct Answer: c) ₹300

Amount of discount

Marked Price (M.P.) = ₹2,000
Discount = 15%

$$\text{Discount} = \frac{15}{100} \times 2000 = ₹300$$

36. Correct Answer: b) ₹1,700

Selling price after discount

Selling Price (S.P.) = M.P. – Discount

$$= 2000 - 300 = ₹1{,}700$$

37. Correct Answer: a) Profit of ₹100

Profit or loss for the shopkeeper

Cost Price (C.P.) = ₹1,600
Selling Price (S.P.) = ₹1,700

$$\text{Profit} = 1700 - 1600 = ₹100$$

Passage II:

38. Correct Answer: c) ₹2,400

Simple interest earned in 2½ years

Principal (P) = ₹12,000
Rate (R) = 8% p.a.
Time (T) = 2½ years = 2.5 years

$$\text{S.I.} = \frac{P \times R \times T}{100} = \frac{12000 \times 8 \times 2.5}{100} = ₹2{,}400$$

39. Correct Answer: c) ₹14,400

Total amount received at the end

Amount = Principal + Simple Interest

$$= 12000 + 2400 = ₹14{,}400$$

40. Correct Answer: c) ₱960

Interest earned per year

Annual interest

$$= 8\% \text{ of } 12000 = \frac{8}{100} \times 12000 = ₹960$$

INTEGER TYPE

41. Correct Answer: 2

If (4x + 5) : (3x + 11) = 13 : 17

$$\frac{4x+5}{3x+11} = \frac{13}{17} \Rightarrow 17(4x+5) = 13(3x+11)$$

$$68x + 85 = 39x + 143 \Rightarrow 29x = 58 \Rightarrow x = 2$$

42. Correct Answer: 1

If $x : y = 4 : 3$, find $x^3 - 7y^2$

Let $x = 4$, $y = 3$

$$x^3 - 7y^2 = 64 - 63 = 1$$

43. Correct Answer: 6

$$a:b = 12:60 = 1:5$$
$$a + b = 1 + 5 = 6$$

44. Correct Answer: 6

Mean proportional of 4 and 9

$$\sqrt{4 \times 9} = \sqrt{36} = 6$$

45. Correct Answer: 8

If the compound ratio of 7:5 and 8:x is 84:60

$$7 \times 8 : 5x = 56 : 5x$$
$$\frac{56}{5x} = \frac{84}{60} \Rightarrow 56 \times 60 = 84 \times 5x$$
$$3360 = 420x \Rightarrow x = 8$$

PREVIOUS YEAR QUESTIONS

1. Correct Answer: D) 104

For a proportion: $\frac{35}{65} = \frac{56}{x}$

Cross-multiply:

$35x = 65 \times 56$

$35x = 3640$

$x = \frac{3640}{35} = 104$

2. Correct Answer: B) 5%

Simple Interest formula:

$$\text{S.I.} = \frac{P \times R \times T}{100}$$

Given:

$$P = ₹10{,}800$$
$$\text{S.I.} = ₹1{,}350$$

$T = 2\frac{1}{2} = 2.5$years

$$1350 = \frac{10800 \times R \times 2.5}{100}$$
$$1350 = 270R \Rightarrow R = 5$$

3. Correct Answer: A) 0.0001%

Numerator: $25\% \times 50\% \times 100\% = 0.25 \times 0.5 \times 1 = 0.125$

Denominator:

"25 of 100" = $25 \times 100 = 2500$

$$50\% \text{ of } 100 = 50$$
$$2500 \times 50 = 125000$$

Whole expression:

$$\frac{0.125}{125000} = 0.000001$$

Convert to percentage:

$$0.000001 \times 100 = 0.0001\%$$

4. Correct Answer: C) 25%

Total ratio parts = $5 + 4 + 3 = 12$

English part = 3

Number of English questions:

$\frac{3}{12} \times 180 = 45$

Percentage of English questions:

$$\frac{45}{180} \times 100 = 25\%$$

5. Correct Answer: B) ₹62,900

Monthly income = ₹85,000

House rent (7½%)

7.5% of 85,000 = ₹6,375

Remaining amount:

$85{,}000 - 6{,}375 = ₹78{,}625$

Education expense (20% of remaining):

20% of 78,625 = ₹15,725

Money left:

$$78{,}625 - 15{,}725 = ₹62{,}900$$

6. Correct Answer: A) $7\frac{1}{5}\%$

Let the principal = P

Given:

Interest = $\frac{9}{25}P$

Amount:

$P + \frac{9}{25}P = \frac{34}{25}P = 306$

$P = \frac{306 \times 25}{34} = 225$

Interest: $I = 306 - 225 = 81$

Now use simple interest formula:

$I = \frac{P \times R \times T}{100}$

$81 = \frac{225 \times R \times 5}{100}$

$81 = \frac{1125R}{100} \Rightarrow R = \frac{8100}{1125} = 7.2\%$

$7.2\% = 7\frac{1}{5}\%$

7. Correct Answer: C) $6\frac{2}{3}\%$, 6 years 8 months

Simple Interest formula:

S.I. $= \frac{P \times R \times T}{100}$

Given:

S.I. $= \frac{4}{9}P$

$\frac{P \times R \times T}{100} = \frac{4}{9}P$

Cancel P:

$\frac{R \times T}{100} = \frac{4}{9} \Rightarrow R \times T = \frac{400}{9}$

Given R = T (numerically)

$R^2 = \frac{400}{9} \Rightarrow R = \frac{20}{3} = 6\frac{2}{3}$

Time:

$T = 6\frac{2}{3}$ years $= 6$ years 8 months

8. Correct Answer: D) ₹3600

Let the principal be ₹P.

SI for 8 months at 4% p.a.:

$SI_1 = \frac{P \times 4 \times \frac{8}{12}}{100} = \frac{2P}{75}$

SI for 15 months at 5% p.a.:

$SI_2 = \frac{P \times 5 \times \frac{15}{12}}{100} = \frac{P}{16}$

Given:

$SI_2 - SI_1 = 129$

$\frac{P}{16} - \frac{2P}{75} = 129$

LCM of 16 and 75 = 1200

$\frac{75P - 32P}{1200} = 129 \Rightarrow \frac{43P}{1200} = 129$

$43P = 154800 \Rightarrow P = 3600$

9. Correct Answer: C) ₹8.58

Total cost price:

$20 \times 6 + 30 \times 7 = 120 + 210 = ₹330$

Total quantity:$20 + 30 = 50$ kg

Cost price per kg:

$\frac{330}{50} = ₹6.60$

Selling price per kg for 30% gain:

$$6.60 \times \frac{130}{100} = ₹8.58$$

10. Correct Answer: C) 123750

Total population = 2,75,000

Females = 40% of 2,75,000

$= 0.40 \times 275000 = 110000$

Males = Total − Females$= 275000 - 110000 = 165000$

Males who moved = 25% of 165000

$= 0.25 \times 165000 = 41250$

Males left in the town

$= 165000 - 41250 = 123750$

9. Perimeter and Area

Solutions

Worksheet – 1

1. Correct Answer: C) 32 cm

The side of a square field is 8 cm, its perimeter is
Perimeter of square = 4 × side
= 4 × 8 = 32 cm

2. Correct Answer: D) 22 cm

The perimeter of a square field is 88 cm, then its side is
Side = Perimeter ÷ 4
= 88 ÷ 4 = 22 cm

3. Correct Answer: C) 16 cm^2

The side of a square is 4 cm, then its area is
Area = $side^2$
= 4^2 = 16 cm^2

4. Correct Answer: C) 2x units

The area of a square is $4x^2$ sq. units, then its side is
Side = $\sqrt{(4x^2)}$ = 2x

5. Correct Answer: B) 280 sq.cm

Length = 20 cm, Breadth = 14 cm. Find area.
Area = l × b= 20 × 14 = 280 sq.cm

6. Correct Answer: A) 50 cm

Length = 15 cm, Breadth = 10 cm. Find perimeter.
Perimeter = 2(l + b)
= 2(15 + 10)= 2 × 25= 50 cm

7. Correct Answer: B) 40 units

Length and breadth are in ratio 4:1, area = 400 sq. units. Find length.

Let length = 4x, breadth = x
Area = 4x × x = $4x^2$

$4x^2 = 400$
$x^2 = 100$
x = 10

Length = 4x = 4 × 10 = 40 units

8. Correct Answer: 11 cm

Area = 286 sq.cm, Length = 26 cm. Find breadth.

Breadth = Area ÷ Length
= 286 ÷ 26= 11 cm

9. Correct Answer: A) 10 cm

The perimeter of a rectangle is 24 cm and breadth is 2 cm, then its length is ___

Perimeter = 2(l + b)
24 = 2(l + 2)

Divide both sides by 2:
12 = l + 2

l = 12 – 2 = 10 cm

10. Correct Answer: B) 96 cm

The area of a rectangle is 512 sq.cm and the length is twice its breadth, then the perimeter is ____

Let breadth = x
Length = 2x

Area = l × b
512 = 2x × x
$512 = 2x^2$

$x^2 = 256$
x = 16

Breadth = 16 cm
Length = 32 cm

Perimeter = 2(l + b)
= 2(32 + 16)= 2 × 48= 96 cm

Worksheet-2

1. Correct Answer: D) 6 cm

Area of triangle = 39 cm^2, Height = 13 cm. Find base.

Area of triangle = ½ × base × height

39 = ½ × b × 13

39 = 13b / 2

78 = 13b

b = 6 cm

2. Correct Answer: A) 36 m²

Area of parallelogram with base 8 m and altitude 4.5 m

Area = base × height
= 8 × 4.5= 36 m²

3. Correct Answer: C) 72

Side of parallelogram = 12 cm, altitude = 6 cm

Area = base × height
= 12 × 6 = 72 sq.cm

4. Correct Answer: C) 16 cm²

In ΔABC, BC = 8 cm, altitude from A to BC = 4 cm

Area of triangle = ½ × base × height
= ½ × 8 × 4= 16 cm²

5. Correct Answer: B) 10 cm

Area of parallelogram = 80 sq.cm, height = 8 cm. Find base.

Area = base × height

80 = b × 8

b = 10 cm

6. Correct Answer: C) x(x − 1)

The area of the parallelogram is _____

From the figure:
Base = x
Height = x − 1

Area of parallelogram = base × height
= x(x − 1)

7. Correct Answer: A) 20 cm

Area of triangle = 80 cm², Height = 8 cm. Find base.

Area = ½ × base × height

80 = ½ × b × 8

80 = 4b

b = 20 cm

8. Correct Answer: 65 cm²

Area of shaded portion in the figure

The figure is a rectangle divided diagonally.
Length = 13 cm
Height = 10 cm

Area of rectangle = 13 × 10 = 130 cm²

The diagonal divides the rectangle into two equal triangles.

Area of shaded portion = 130 ÷ 2
= 65 cm²

9. Correct Answer: A) 6 sq.cm

The diagonals of a rhombus are 4 cm and 3 cm. Find its area.

Area of rhombus = ½ × d_1 × d_2

= ½ × 4 × 3= ½ × 12= 6 sq.cm

10. Correct Answer: B) 24 sq.cm

The area of the figure is _____

From the figure:
The diagonals intersect at right angles.
Each half of one diagonal = 3 cm
Each half of the other diagonal = 4 cm

So full diagonals are:
d_1 = 3 + 3 = 6 cm
d_2 = 4 + 4 = 8 cm

Area = ½ × d_1 × d_2
= ½ × 6 × 8= ½ × 48= 24 sq.cm

Competitive Corner

SINGLE CORRECT OPTOIN TYPE

1. Correct Answer: A) 60 cm

Length : Breadth = 3 : 2

Let length = 3x and breadth = 2x

Area = length × breadth
= 3x × 2x= 6x²

Given area = 216 cm²

6x² = 216
x² = 36
x = 6

Length = 3x = 18 cm
Breadth = 2x = 12 cm

Perimeter = 2(l + b)
= 2(18 + 12)= 2 × 30= **60 cm**

2. Correct Answer: A) 90 m

Let breadth = x
Length = 3x

Perimeter = 2(l + b)
= 2(3x + x)= 2(4x)= 8x

Given perimeter = 240 m

8x = 240
x = 30

Length = 3x = 90 m

3. Correct Answer: B) 60 m

Area of square = 225 m²

Side = √225 = 15 m

Perimeter = 4 × side
= 4 × 15= **60 m**

4. Correct Answer: C) l = 35 cm, A = 1050 cm²

Perimeter = 130 cm
Breadth = 30 cm

Perimeter = 2(l + b)
130 = 2(l + 30)
65 = l + 30
l = 35 cm

Area = l × b
= 35 × 30= **1050 cm²**

5. Correct Answer: B) 10 cm

Area of square = 50 cm²

Side² = 50
Side = √50 = 5√2 cm

Diagonal of square = side × √2
= 5√2 × √2= 5 × 2= **10 cm**

6. Correct Answer: B) 44 m

Area of trapezium = 480 m²
Distance between parallel sides (height) = 15 m
One parallel side = 20 m
Let other parallel side = x

Area formula:

$$\frac{1}{2}(a+b)h$$

$$480 = \frac{1}{2}(20+x)(15) = \frac{15}{2}(20+x)$$

Multiply both sides by 2:

$$960 = 15(20+x)$$
$$960 = 300 + 15x$$
$$660 = 15x$$
$$x = 44$$

7. Correct Answer: A) 30 cm

Area of rhombus = 240 cm²
One diagonal = 16 cm
Let other diagonal = x

Area formula:

$$\frac{1}{2}d_1d_2$$
$$240 = \frac{1}{2}(16)(x)$$
$$240 = 8x$$
$$x = 30$$

8. Correct Answer: B) 41 cm²

Diagonals = 10 cm and 8.2 cm

Area = $\frac{1}{2}d_1d_2 = \frac{1}{2}(10)(8.2) = 5 \times 8.2$
$= 41$

9. Correct Answer: B) 26 units

Triangle sides = 5, 12, 13

This is a right triangle.

Area = $\frac{1}{2} \times 5 \times 12 = 30$
Rectangle length = 10 units

Area of rectangle = 30

$$10 \times breadth = 30$$
$$breadth = 3$$

Perimeter = $2(10 + 3) = 26$

10. Correct Answer: A

Original square side = 12 m

When corners are cut equally, the side of octagon becomes:

$$\frac{12}{1+\sqrt{2}}$$

Rationalizing: $= \frac{12(1-\sqrt{2})}{(1+\sqrt{2})(1-\sqrt{2})}$

$$= \frac{12(1-\sqrt{2})}{1-2} = 12(\sqrt{2}-1)$$

This matches option: $\frac{\sqrt{2}}{\sqrt{2}+1}$

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Correct Answers: A and C

Perimeter = 68 cm
Breadth = 14 cm

Perimeter = 2(l + b)

$$68 = 2(l + 14)$$

$$34 = l + 14$$
$$l = 20 \text{ cm}$$

Correct options:

- 20 cm
- ½(40 cm) = 20 cm

12. Correct Answers: B and C

Area of square = 625 cm²

Side = √625 = 25 cm

Perimeter = 4 × 25 = 100 cm

Correct options:

- B) 100 cm
- C) 50 × 2 cm = 100 cm

13. Correct Answer: A and D

Area of square = 40 × 40 = 1600 m²

Area of parallelogram = base × height

$$1600 = base \times 20$$
$$base = 80 \text{ m}$$

Correct options:

- A) 4 × 20 m² ✅
- D) 80 m ✅

14. Correct Answer: C,D

Let base = x
Height = 2x

Area of triangle:

$$\frac{1}{2} \times x \times 2x = x^2$$

Given area = 400

$$x^2 = 400$$
$$x = 20$$

Height = 2x = 40 cm

Correct options:

- A) 50 cm ❌
- B) 20 cm ❌
- C) 40 cm ✅
- D) 2√400 cm ✅

15. Correct Answer: A,C

Area of rectangle = 20 × 15 = 300 cm²

Area of triangle: $\frac{1}{2} \times 30 \times h$

$$15h = 300$$
$$h = 20 \text{ cm}$$

Correct options:

- A) 20 cm ✅

ASSERTION AND REASON TYPE

16. Correct Answer: (A)

Assertion (A): The perimeter of a rectangle is the sum of all four sides.
✓ True

Reason (R): The formula for perimeter of a rectangle is 2 × (length + breadth).
✓ True

Since

$$2(l + b) = l + b + l + b$$

which is the sum of all four sides,

✓ Reason correctly explains the Assertion.

17. Correct Answer: (D)

Assertion (A): The area of a triangle is always greater than its perimeter.
✗ False

(Example: A triangle with base 4 cm and height 3 cm
Area = ½ × 4 × 3 = 6 cm²
Perimeter can be greater than 6.)

Reason (R): Area of a triangle is given by ½ × base × height.
✓ True

Since Assertion is false but Reason is true:

18. Correct Answer: (A)

Assertion (A): Area is a two-dimensional quantity.
✓ True

Reason (R): Area is measured in square units.
✓ True

Area is two-dimensional because it measures length × breadth, which results in square units.
Thus, the Reason correctly explains the Assertion.

19. Correct Answer: (D)

Assertion (A): The perimeter of a circle is called its area.
✗ False
(The perimeter of a circle is called its *circumference*.)

Reason (R): The area of a circle is πr^2.
✓ True

Assertion is false, but Reason is true.

20. Correct Answer: (A)

Assertion (A): If all sides of a square are doubled, the perimeter also doubles.
✓ True

Reason (R): Perimeter of a square is 4 times the side.
✓ True

If side becomes 2s,
Perimeter = 4 × 2s = 8s,
which is double of 4s.

Thus, Reason correctly explains the Assertion.

MATRIX MATCHING TYPE

21. Correct Answer: A–3, B–1, C–2, D–4

Column A → Column B

A. Area of rectangle → 3. length × breadth
B. Perimeter of square → 1. 4 × side
C. Area of triangle → 2. ½ × b × h
D. Circumference of circle → 4. 2πr

22. Correct Answer: A–2, B–1, C–4, D–3

Column A → Column B

A. Unit of area → 2. square cm or square m
B. Unit of perimeter → 1. cm or m
C. Shape with maximum area → 4. For same boundary length
D. Shape with minimum perimeter → 3. Circle

COMPREHENSION TYPE

Passage – I

Given:
Base : Height = 5 : 2
Area of parallelogram = 360 m²

Let base = 5x
Height = 2x

Area = base × height

$$360 = 5x \times 2x$$
$$360 = 10x^2$$
$$x^2 = 36$$
$$x = 6$$

So,

Base = 5x = 5 × 6 = **30 m**
Height = 2x = 2 × 6 = **12 m**

23. Correct Answer: C) 30 m

The base of the parallelogram is 30m

24. Correct Answer: B) 180 m²

The area of △ABC is

Diagonal divides parallelogram into two equal triangles.

$$\frac{360}{2} = 180 \text{ m}^2$$

25. Correct Answer: A) 12 m

The height of the parallelogram is 12 m

Passage-2

26. Correct Answer: B) 45

Length = 9
Breadth = 5

Area = ℓ × b
= 9 × 5= **45**

27. Correct Answer: B) 40

Length = 12
Breadth = 8

Perimeter = 2(ℓ + b)
= 2(12 + 8)= 2 × 20= **40**

28. Correct Answer: B) 12 cm

Area = ℓ × b
72 = ℓ × 6

ℓ = 72 ÷ 6= **12 cm**

INTEGER TYPE

29. Correct Answer: $\frac{d^2}{2}$

Diagonal of square = d

We know,
Diagonal = side × √2

$$\text{side} = \frac{d}{\sqrt{2}}$$

Area = side²

$$= \left(\frac{d}{\sqrt{2}}\right)^2 = \frac{d^2}{2}$$

30. Correct Answer: 24

Base = 8 cm
Height = 6 cm

Area of triangle =

$$\frac{1}{2} \times b \times h \frac{1}{2} \times 8 \times 6 = 24$$

31. Correct Answer: $\frac{\sqrt{3}x^2}{3}$

Height of equilateral triangle = x

Formula:

$$\text{Height} = \frac{\sqrt{3}}{2} \times \text{side}$$

$$x = \frac{\sqrt{3}}{2} s$$

$$s = \frac{2x}{\sqrt{3}}$$

Area = $\frac{\sqrt{3}}{4}s^2 = \frac{\sqrt{3}}{4}\left(\frac{2x}{\sqrt{3}}\right)^2 = \frac{\sqrt{3}}{4} \times \frac{4x^2}{3} = \frac{\sqrt{3}x^2}{3}$

32. Correct Answer: 120 sq. cm

Diagonals = 10 cm and 24 cm

Area of rhombus =

$$\frac{1}{2} d_1 d_2 = \frac{1}{2} \times 10 \times 24 = 120$$

Previous Year Questions

1. Correct Answer: 116 cm (Option D)

Find the perimeter of the figure

Given:
AB = BC = CD
AD = 42 cm

So,

$$AB = BC = CD = \frac{42}{3} = 14 \text{ cm}$$

- Outer semicircle diameter = AD = 42 cm
  Radius $R = 21$cm
- Inner semicircle diameter = BC = 14 cm
  Radius $r = 7$cm

Perimeter includes:

- Outer semicircle arc = $\pi R = 21\pi$
- Inner semicircle arc = $\pi r = 7\pi$
- Straight parts = AB + CD = 14 + 14 = 28 cm

Total Perimeter $= 21\pi + 7\pi + 28$
$= 28\pi + 28$

Using $\pi = \frac{22}{7}$:

$$= 28\left(\frac{22}{7}\right) + 28 = 88 + 28 = 116 \text{ cm}$$

2. Correct Answer: ₹ 6600 (Option A)

Cost of fencing the circular field

Area = 6.16 hectares

1 hectare $= 10{,}000$ m²
Area $= 6.16 \times 10{,}000 = 61{,}600$ m²

Since area of circle $= \pi r^2$

$$\pi r^2 = 61{,}600$$

Using $\pi = \frac{22}{7}$:

$$r^2 = \frac{61{,}600 \times 7}{22} = 19{,}600$$

$$r = 140 \text{ m}$$

Circumference $= 2\pi r$

$$= 2 \times \frac{22}{7} \times 140 = 880 \text{ m}$$

Cost of fencing = 880 × 7.50 = ₹6600

3. Correct Option: B. (i) 40 days (ii) 6237

(i) Cow grazing problem

Diameter of field = 112 m
Radius = 56 m

Rope length = 56 m → Cow can graze the entire field (since rope = radius).

Area of field

$$= \pi r^2 = \frac{22}{7} \times 56 \times 56 = 9856 \text{ m}^2$$

Cow grazes 250 m² per day.

$$\text{Time} = \frac{9856}{250} \approx 39.4 \approx 40 \text{ days}$$

(ii) Area of semicircle

Perimeter of semicircle = 324 cm

Perimeter formula:

$$\pi r + 2r = 324$$

Using $\pi = \frac{22}{7}$:

$$\frac{22}{7}r + 2r = 324$$
$$\frac{36}{7}r = 324$$
$$r = \frac{324 \times 7}{36} = 63 \text{ cm}$$

Area of semicircle:

$$= \frac{1}{2}\pi r^2$$
$$= \frac{1}{2} \times \frac{22}{7} \times 63 \times 63 = 6237 \text{ cm}^2$$

4. Correct Answer: D. ₹ 3750

Cost of covering the floor

Length of room = 3 m
Width of room = 1.5 m

Convert into cm:

$$3 \text{ m} = 300 \text{ cm}$$
$$1.5 \text{ m} = 150 \text{ cm}$$

Area of floor

$$= 300 \times 150 = 45{,}000 \text{ cm}^2$$

Area of one tile

$$= 12 \times 5 = 60 \text{ cm}^2$$

Number of tiles required

$$= \frac{45{,}000}{60} = 750 \text{ tiles}$$

Cost per tile = ₹ 5

Total cost= $750 \times 5 = ₹3750$

5. Correct Answer: C. 61.25 cm²

Step 1: Area of RectangleArea = $\mathbf{17} \times \mathbf{7} = \mathbf{119}$ cm²
Step 2: Area of Triangle CEB

AB = 17 cm
AE = 5 cm

$$EB = 17 - 5 = 12 \text{ cm}$$

Triangle CEB is right-angled Area = $\frac{1}{2} \times 12 \times 7 = 42$ cm²
Step 3: Area of Semicircle

AD = 7 cm (same as BC)

Radius = $\frac{7}{2} = 3.5$cm

$$\text{Area of semicircle} = \frac{1}{2}\pi r^2$$

Using $\pi = \frac{22}{7}$:= $\frac{1}{2} \times \frac{22}{7} \times 3.5 \times 3.5$
$= 19.25$ cm²
Step 4: Unshaded Area

$= \text{Rectangle} - \text{Triangle} - \text{Semicircle}$
$= 119 - 42 - 19.25$
$= 57.75$ cm²

But this option is not given.

Now check shaded part carefully:

Unshaded region = Triangle + Semicircle

$$= 42 + 19.25 = 61.25 \text{ cm}^2$$

6. Correct Answer: C. 851

Length of rectangular sheet = **222 cm**
Breadth = **138 cm**
Side of each square card = **6 cm**

Step 1: Number of squares along length

$$\frac{222}{6} = 37$$

Step 2: Number of squares along breadth

$$\frac{138}{6} = 23$$

Step 3: Total number of square cards

$$37 \times 23 = 851$$

7. Correct Answer: D $\boxed{\mathbf{406 \text{ cm}^2}}$

Step 1: Area of Rectangle

$$\text{Area} = 24 \times 14 = 336 \text{ cm}^2$$

Step 2: Area of Two Semicircles

Two semicircles = one full circle of radius 7 cm

$$\text{Area} = \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154 \text{ cm}^2$$

Step 3: Area of Central Triangle inside Rectangle

Triangle FDC:

$$\text{Area} = \frac{1}{2} \times 24 \times 14 = 168 \text{ cm}^2$$

So, the remaining shaded part inside rectangle:

$$336 - 168 = 168 \text{ cm}^2$$

Step 4: Area of Right Triangle $\boldsymbol{DCE}$

$$\text{Area} = \frac{1}{2} \times 24 \times 7 = 84 \text{ cm}^2$$

Step 5: Total Shaded Area

$$168 + 154 + 84 = 406 \text{ cm}^2$$

8. Correct Answer: D

Diameter of circular garden = **21 cm**

$$\text{Inner radius } r = \frac{21}{2} = 10.5 \text{ cm}$$

Width of running track = **3.5 cm**

Outer radius $R = 10.5 + 3.5 = 14$ cm

Area of running track

(Area of outer circle − Area of inner circle)

$= \pi(R^2 - r^2)$

$= \frac{22}{7}(14^2 - 10.5^2) = \frac{22}{7}(196 - 110.25)$

$= \frac{22}{7}(85.75) = 22 \times 12.25 = 269.5$ cm^2

Final Answer: $\boxed{269.50 \text{ cm}^2}$

9. Correct Answer: A

Step 1: Area of Rectangle

$$\text{Area} = 35 \times 25 = 875 \text{ cm}^2$$

Step 2: Area of Circle

$\text{Area} = \pi r^2$

$= \frac{22}{7} \times 14 \times 14$

$= \frac{22}{7} \times 196 = 22 \times 28 = 616$ cm^2

Step 3: Area of Shaded Region

(Shaded part = Rectangle − Circle)

$$875 - 616 = 259 \text{ cm}^2$$

Final Answer: $\boxed{259 \text{ cm}^2}$

10. Algebraic Expressions

Solutions

Worksheet – 1

1. **Correct Answer: (B) $2x - 8$kg**

   Weight of the Table

   Given:Weight of chair = x kg
   Twice the weight of chair = $2x$ kg
   Table is 8 kg less than twice the chair's weight.

   Weight of table $= 2x - 8$ kg

2. **Correct Answer: (A) −10**

   Value of the Expression

   Given:$p = -2,\ q = -1,\ r = 3$

   Expression:$3p^2q + 5pq^2 + 2pqr$
   $3(-2)^2(-1) + 5(-2)(-1)^2 + 2(-2)(-1)(3)$
   $= 3(4)(-1) + 5(-2)(1) + 12$
   $= -12 - 10 + 12 = -10$

3. **Correct Answer: (A)**

   Which of the following is/are false?

   (A) In $9xy$, coefficient of x is $9y$, not $-9y \rightarrow$ False
   (B) In $-7abc$, coefficient of a is $-7bc \rightarrow$ True
   (C) In $-xyz$, coefficient of xyz is $-1 \rightarrow$ True
   (D) In $-abc$, coefficient of b is $-ac \rightarrow$ True

4. **Correct Answer: (A)**

   Like terms in $6x^2y + 5xy^2 - 8xy^2 - 7yx^2$

   Note: $yx^2 = x^2y$

   Like terms are:$6x^2y$ and $-7x^2y$

5. **Correct Answer: (B) 5x**

   Simplify: $7x + 3x - 5x$

   $(7 + 3 - 5)x = 5x$

6. **Correct Answer: (C) $19x^2 - 4x - 6$**

   Identify the correct result of:$15x^2 - 7x + 4x^2 + 3x - 6$

   $(15x^2 + 4x^2) + (-7x + 3x) - 6$
   $= 19x^2 - 4x - 6$

7. **Correct Answer: (A) 0**

   Find the value of $[16 - (-2n)]$when $n = -8$

   $-2n = -2(-8) = 16$
   $16 - 16 = 0$

8. **Correct Answer: (B) $4m + 4n + 5$**

   Simplify: $6m + 4n - 2m + 5$

   Combine like terms:$(6m - 2m) + 4n + 5 = 4m + 4n + 5$

9. **Correct Answer: (C) $12 - 4x$**

   Which expression represents the subtraction of $4x$from 12?

   "Subtraction of $4x$from 12" means $12 - 4x$.

10. **Correct Answer: (B) like**

    If two monomials do not have the same numerical factor but have the same literal factors, they are called like monomials.

Worksheet – 2

1. **Correct Answer: (C) 9x + 4**

   Four more than nine times of a number x"

   Nine times of x = 9x

   Four more than that = 9x + 4

2. **Correct Answer: (B) n – 7**

   Jeff scored n points. Bill scored seven points less than Jeff

   Seven less than n = n – 7

3. **Correct Answer: (D) 3C – 5**

   Phil ate five chocolates less than thrice the number eaten by Steve

   Steve ate C chocolates

   Thrice of C = 3C

   Five less than that = 3C – 5

4. **Correct Answer: (B) 3**

Solve the equation: 36 + 4x = 48

Subtract 36 from both sides:
4x = 12

Divide by 4:
x = 3

5. **Correct Answer: (C) 13**

Solve the equation: 2b − 11 = 15

Add 11 to both sides:
2b = 26

Divide by 2:
b = 13

6. **Correct Answer: (D) 2 is a constant, while p, l, and w are variables**

Given: p = 2l + 2w

The number 2 is a fixed number → constant

p, l, and w can take different values → variables

7. **Correct Answer: (B) 13**

Find the value of $(p+q)^2 - pq$ if $p = 3$ and $q = 1$

Substitute values:

$$(3+1)^2 - (3)(1) = 4^2 - 3 = 16 - 3 = 13$$

8. **Correct Answer: (C) 3x − 5**

Tom's age is five years less than thrice Ben's age

Ben's age = x

Thrice of Ben's age = $3x$

Five years less than that = $3x - 5$

9. **Correct Answer: (B) Raymond has Rs. 10 more than Sam**

Sam has Rs. x and Raymond has Rs. (x + 10)

Raymond's amount − Sam's amount

$$= (x + 10) - x = 10$$

So, Raymond has Rs. 10 more than Sam.

10. **Correct Answer: (D) 13**

Find the value of $2a^2 + 3ab - b^2$ if $a = 2$ and $b = 1$

Substitute the values:

$$2(2)^2 + 3(2)(1) - (1)^2 = 8 + 6 - 1 = 13$$

Worksheet - 3

1. **Correct Answer: (A) −40xyz²**

Find the product of $5x, -2y$, and $4z^2$.

Multiply the numerical coefficients:

$$5 \times (-2) \times 4 = -40$$

Multiply the variables:

$$x \cdot y \cdot z^2 = xyz^2$$

$$\text{Product} = -40xyz^2$$

2. **Correct Answer: (A) −8x²y + 12xy³**

Find the product of $-4xy$ and $(2x - 3y^2)$.

Multiply each term of the binomial by $-4xy$:

$$-4xy \cdot 2x = -8x^2y$$
$$-4xy \cdot (-3y^2) = +12xy^3$$
$$\text{Product} = -8x^2y + 12xy^3$$

3. **Correct Answer: (B) $x^4 - 7x^3 + 4x^2$**

$(x^6 - 7x^5 + 4x^4) \div x^2$

Divide each term by x^2:

$$x^6 \div x^2 = x^4, -7x^5 \div x^2 = -7x^3, 4x^4 \div x^2 = 4x^2$$
$$= x^4 - 7x^3 + 4x^2$$

4. **Correct Answer: (A) $10x^4 - 8x^3 + 3x$**

$$(10x^7 - 8x^6 + 3x^4) \div x^3$$

$$10x^7 \div x^3 = 10x^4, -8x^6 \div x^3 = -8x^3, 3x^4 \div x^3 = 3x$$
$$= 10x^4 - 8x^3 + 3x$$

5. **Correct Answer: (A) $-x$**

$abx^2 \div (-abx)$

$$\frac{abx^2}{-abx} = -x^{2-1} = -x$$

6. **Correct Answer: (B) $-4p^4q^4r^4$**

$16p^6q^7r^8 \div (-4p^2q^3r^4)$

$16 \div (-4) = -4$

$p^{6-2} = p^4, q^{7-3} = q^4, r^{8-4} = r^4$

$= -4p^4q^4r^4$

7. **Correct Answer: (B) −3x + 2y + 4**

$$\left(-\frac{5}{2}x^2 + \frac{5}{3}xy + \frac{10}{3}x\right) \div \frac{5}{6}x$$

Dividing by $\frac{5}{6}x$ is same as multiplying by $\frac{6}{5x}$

$$= \left(-\frac{5}{2}x^2\right) \div \left(\frac{6}{5x}\right) + \left(\frac{5}{3}xy\right) \div \left(\frac{6}{5x}\right) + \left(\frac{10}{3}x\right) \div \left(\frac{6}{5x}\right)$$
$$= -3x + 2y + 4$$

8. Correct Answer: (A) x – 5

$(x^2 - 11x + 30) \div (x - 6)$

Factor the numerator: $x^2 - 11x + 30 = (x - 5)(x - 6)$

Cancel $(x-6)$**:**

$$= x - 5$$

9. Correct Answer: (C) 3x – 7

$(3x^2 - x - 14) \div (x + 2)$

Factor the numerator: $3x^2 - x - 14 = (3x - 7)(x + 2)$

Cancel $(x+2)$**:**

$= 3x - 7$

10. Correct Answer: (D) $12x^2y^2z^4$

LCM of $3x^2y^2z^2$ and $4xy^2z^4$

LCM of coefficients: LCM(3,4) = 12

Take highest powers of variables:

x^2, y^2, z^4

LCM = $12x^2y^2z^4$

11. Correct Answer: (C) 11ht

HCF of $22h^2t$ and $33ht^2$

HCF of coefficients: HCF(22,33) = 11

Lowest powers of variables:

h^1, t^1

HCF = $11ht$

COMPETITIVE CORNER

SINGLE ANSWER CORRECT TYPE

1. Correct Answer: (A) 5a – x

$(15a^2 + 17ax - 4x^2) \div (3a + 4x)$

Rearrange and factor numerator:

$15a^2 + 17ax - 4x^2 = (5a - x)(3a + 4x)$

Cancel $(3a+4x)$:

$$= 5a - x$$

2. Correct Answer: (A) $11p^2 - 8p - 3$

Simplify: $9p^2 - 3p + 7 + 2p^2 - 5p - 10$

Combine like terms:

$(9p^2 + 2p^2) + (-3p - 5p) + (7 - 10)$
$= 11p^2 - 8p - 3$

3. Correct Answer: (C) $2a^2 + 5ab + b^2$

Subtract $3a^2 - 4ab + b^2$ from $5a^2 + ab + 2b^2$

$(5a^2 + ab + 2b^2) - (3a^2 - 4ab + b^2)$
$= 2a^2 + 5ab + b^2$

4. Correct Answer: (A) x^3

The HCF of x^4 and $8x^3$

Lowest power of x is x^3

5. Correct Answer: (D) q

The HCF of pq and qrs

Common variable = q

6. Correct Answer: (B) 2xy

The HCF of $4x^2y$, $6xy$, $8x^2y^2$

HCF of coefficients: HCF(4,6,8) = 2

Lowest powers: x^1, y^1

$$\text{HCF} = 2xy$$

7. Correct Answer: (C) 2a + 2

$(12a^2 + 10a - 2) \div (6a - 1)$

Factor numerator:

$12a^2 + 10a - 2 = 2(6a^2 + 5a - 1)$
$6a^2 + 5a - 1 = (6a - 1)(a + 1)$

Cancel $(6a-1)$:

$$= 2(a + 1) = 2a + 2$$

8. Correct Answer: (D) 12, 18, 30

Three numbers are in the ratio 2 : 3 : 5 and their HCF is 6

Let the numbers be $2k, 3k, 5k$.

Given HCF = 6 ⇒ $k = 6$

So the numbers are:

$$2 \times 6 = 12, 3 \times 6 = 18, 5 \times 6 = 30$$

9. Correct Answer: (B) x^2yz, xy^2z, xyz^2

Three monomials are in the ratio x : y : z and their HCF is xyz

If HCF = xyz, then each monomial must contain at least x, y, and z.

Also, the monomials should be proportional to $x : y : z$.

Option B gives:

$$x^2yz,\ xy^2z,\ xyz^2$$

HCF of these monomials:

$$x^1y^1z^1 = xyz$$

Ratio:

$$x : y : z$$

10. Correct Answer: (D)

Three monomials are in the ratio $xy:yz:zx$ and their HCF is xyz

If HCF = xyz, then each monomial must contain x, y, z.

Multiplying each ratio term by xyz:

$$xy \cdot xyz = x^2y^2z$$
$$yz \cdot xyz = xy^2z^2$$
$$zx \cdot xyz = x^2yz^2$$

So the monomials are:

$$x^2y^2z,\ x^2yz^2,\ xy^2z^2$$

11. Correct Answer: (C) abcd

LCM of abc and cd

Take each variable with the highest power appearing:

$$a^1,\ b^1,\ c^1,\ d^1$$
$$\text{LCM} = abcd$$

12. Correct Answer: (D)

Divide $\left(\frac{1}{4}a^2x - \frac{1}{10}abx - \frac{3}{8}acx\right) \div \frac{3}{8}ax$

Dividing by $\frac{3}{8}ax$ means multiplying by $\frac{8}{3ax}$:

$$\frac{1}{4}a^2x \cdot \frac{8}{3ax} = \frac{2}{3}a$$
$$-\frac{1}{10}abx \cdot \frac{8}{3ax} = -\frac{4}{15}b$$
$$-\frac{3}{8}acx \cdot \frac{8}{3ax} = -c$$
$$= \frac{2}{3}a - \frac{4}{15}b - c$$

13. Correct Answer: (C) xyz

LCM of xy, yz and xz

LCM is obtained by taking all variables with the highest power appearing:

$$x^1,\ y^1,\ z^1$$
$$\text{LCM} = xyz$$

14. Correct Answer: (B) $360x^6y^8$

LCM of $120x^2y^3$ and $36x^6y^8$

LCM of coefficients:

$$120 = 2^3 \cdot 3 \cdot 5, 36 = 2^2 \cdot 3^2$$
$$\text{LCM} = 2^3 \cdot 3^2 \cdot 5 = 360$$

Highest powers of variables:

$$x^6,\ y^8$$
$$\text{LCM} = 360x^6y^8$$

15. Correct Answer: (A) $a^2b^2c^2$

LCM of a^2bc, ab^2c, abc^2

Take highest power of each variable:

$$a^2,\ b^2,\ c^2$$
$$\text{LCM} = a^2b^2c^2$$

ONE OR MORE THAN ONE CORRECT ANSWER TYPE

16. Correct Answers: A, B, C

Consider the expression: $-6x^2y + 4xy - 9$

Statements:

A. The constant term is −9

B. The coefficient of x^2y is −6

C. The coefficient of xy is 4

D. The expression has four terms

Explanation: The expression has three terms:

$$-6x^2y,\ 4xy,\ -9$$

17. Correct Answers: A, B, D

Identify all pairs of like terms

Like terms must have exactly the same variables with the same powers.

A. $5x^2y$ and $-3x^2y$

B. $4ab$ and $4ba$ (order doesn't matter)

C. $7x^2$ and $7x$ (powers of x differ)

D. $9pq^2$ and $-pq^2$

18. Correct Answers: A, B, C

Simplify: $3x^2 - 5x + 7 + 4x^2 + x - 2$

Combine like terms:

$$(3x^2 + 4x^2) + (-5x + x) + (7 - 2)$$
$$= 7x^2 - 4x + 5$$

Now check each statement:

A. The simplified expression is $7x^2 - 4x + 5$

B. Like terms were combined

C. The constant term is 5

D. The coefficient of x is −6 (it is −4)

19. Correct Answers: A, C

If $x = -2$ and $y = 3$

Evaluate:$x^2 + 2xy + y^2$
$(-2)^2 + 2(-2)(3) + (3)^2 = 4 - 12 + 9 = 1$
Also,$x^2 + 2xy + y^2 = (x + y)^2$
Now check each statement:

A. The value is 1

B. The value is 9

C. The expression represents $(x+y)^2$

D. The value is 6

20. Correct Answers: A, B, C, D

A. HCF uses the least power of common variables

This is the rule for finding HCF of monomials.

B. LCM uses the greatest power of all variables
This is the rule for finding LCM of monomials.

C. *HCF of $6x^2y$ and $9xy^2$ is $3xy$*
HCF of coefficients: HCF(6, 9) = 3
Least powers: x^1, y^1
So HCF = $3xy$

D. *LCM of x and y is xy*
Different variables → LCM is their product.

ASSERTION AND REASON TYPE

21. Correct Answer: (A)

$2x$ and $3x$ are like terms (same variable x, same power 1).

Like terms can be added by adding their coefficients.

Therefore, $2x + 3x = (2 + 3)x = 5x$.

Both A and R are true, and R explains why A is true.

22. Correct Answer: (C) (A is true, but R is false)

$x^2 + 2x + 1$ has three terms, so it is a trinomial → A is true.

A trinomial has three terms, not two → R is false.

23. Correct Answer: (D) (A is false, but R is true)

In $4xy - 3y + 7$, the terms are $4xy$, $-3y$, and 7.

The constant term is 7, not −3 (since −3y contains the variable y).

Therefore, Assertion (A) is false.

The reason is a correct definition.

24. Correct Answer: (A)(Both A and R are true, and R is the correct explanation of A)

In $-5abc$, the numerical factor is −5.

Both the assertion and the reason are true, and the reason explains the assertion.

25. Correct Answer: (D) (A is false, but R is true)

Algebraic expressions can be combined (added) only when terms are like, not unlike.

Therefore, Assertion (A) is false.

The reason correctly defines unlike terms, so Reason (R) is true.

MATRIX MATCHING TYPE

26. Correct Matching:A–2, B–1, C–4, D–3

A. Monomial → 2. Single term expression

B. Binomial → 1. Expression with two terms

C. Trinomial → 4. Expression with three terms

D. Polynomial → 3. Expression with more than three terms

27. 1. $3x + 5$ when $x = 4$

$$3(4) + 5 = 12 + 5 = 17$$

→ C (17)

$a^2 + 2ab + b^2$ when $a = 1, b = 3$

$$1 + 6 + 9 = 16$$

→ D (16)

$m^2 - n^2$ when $m = 6, n = 2$

$$36 - 4 = 32$$

→ A (32)

$2p + 5q$when $p = 2, q = 4$

$$4 + 20 = 24$$

→ B (24)

Final Correct Matching :1–C, 2–D, 3–A, 4–B

COMPREHENSION TYPE

Passage – 1

28. Correct Answer: (C) 5

The coefficient of a variable is the number multiplying it.

In $5a$, the coefficient of a is 5.

29. Correct Answer: (C) Trinomial

The expression has three terms:

$$5a,\ 3b,\ 4m$$

An expression with three terms is called a trinomial.

30. Correct Answer: (B) ₱97

$5a + 3b + 4m = 5(10) + 3(5) + 4(8)$
$= 50 + 15 + 32 = 97$

Passage -2

31. Correct Answer: (C) 4x + 3y

Which algebraic expression represents the cost of 4 pencils and 3 erasers?

Cost of 1 pencil = ₱x → cost of 4 pencils = 4x

Cost of 1 eraser = ₱y → cost of 3 erasers = 3y

Total cost = 4x + 3y

32. Correct Answer: (A) 6x + 3y

After buying 2 more pencils, which expression represents the new total cost (before discount)?

Earlier pencils = 4, more pencils = 2 → total pencils = 6

Cost of pencils = 6x

Erasers remain = 3 → cost = 3y

New total cost = 6x + 3y

33. Correct Answer: (B) 6x + 3y – 4

If ₱4 discount is given on the new total, which expression shows the final amount paid?

New total before discount = 6x + 3y
Discount = ₱4

Final amount paid = 6x + 3y – 4

INTEGER TYPE

34. Correct Answer: $\boxed{6y^2}$

Subtract $-5y^2$ from y^2

$$y^2 - (-5y^2) = 6y^2$$

35. Correct Answer: $\boxed{-1}$

Find the value of $3(2x - 1) - 4(x + 2)$when $x = 5$

$$3(10 - 1) - 4(7) = 27 - 28 = -1$$

36. Correct Answer: $\boxed{-x}$

If
$A = -7x - 3x - 5x = -15x$
$B = 9x + 3x + 2x = 14x$

$A + B = -15x + 14x = -x$

37. Correct Answer: $\boxed{0.10645}$

If $x = 2, y = 1.15, z = 0.01$, find

$xz(x^2 + y^2)$
$= 2(0.01)(4 + 1.3225) = 0.02 \times 5.3225$
$= 0.10645$

38. Correct Answer: $\boxed{14mn - 1}$

Add: $-7mn + 5,12mn + 2,9mn - 8$

$(-7 + 12 + 9)mn + (5 + 2 - 8)$
$= 14mn - 1$

39. Correct Answer: $\boxed{7x + 5}$ Add:

$14x + 10y - 12xy - 13$
$18 - 7x - 10y + 8xy$
$4xy$

$(14x - 7x) + (10y - 10y) + (-12xy + 8xy + 4xy) + (-13 + 18)$
$= 7x + 5$

40. Correct Answer: $\boxed{a^2b^4}$

LCM = $12a^2b^4c$

HCF = a^2b^2
One monomial = $12a^2b^2c$

Using

Product of monomials = LCM × HCF

$$\text{Other monomial} = \frac{12a^2b^4c \times a^2b^2}{12a^2b^2c}$$
$$= a^2b^4$$

PREVIOUS YEAR QUESTIONS

1. Correct Answer: B

Step 1: Find the sum

$(x^2y^2 + y^2z + 4) + (-y^2z + 3y^2 + z^2)$
$= x^2y^2 + 3y^2 + z^2 + 4$

Step 2: Subtract the given expression

$(x^2y^2 + 3y^2 + z^2 + 4) - (8y^2 + 4x^2y^2 - 2z^2)$

Step 3: Combine like terms

$= -3x^2y^2 - 5y^2 + 3z^2 + 4$

2. Correct Answer: A

A binomial is an algebraic expression with exactly two unlike terms.

$5a^2 + 3b \rightarrow$ 2 terms (Binomial)

$2a + 3b \rightarrow$ 2 terms (Binomial)

$8a^2 \rightarrow$ 1 term (Monomial)

$9a^2 + 5a + 3b \rightarrow$ 3 terms (Trinomial)

$7ab + 3a^2 \rightarrow$ 2 terms (Binomial)

Total binomials = 3

3. Correct Answer:B

Let the minimum marks be x.

According to the question:

Maximum marks = 5 ×minimum marks +3

Given maximum marks = 24

$$24 = 5x + 3$$

4. Correct Answer:A

Let the required expression be E.

$(x^3 + 4x^2 + 3x - 2) - E = x^3 - 2x^2 + 4$

So,

$E = (x^3 + 4x^2 + 3x - 2) - (x^3 - 2x^2 + 4)$
$E = x^3 + 4x^2 + 3x - 2 - x^3 + 2x^2 - 4$
$E = 6x^2 + 3x - 6$

5. Correct Answer: A

Marbles left

$= (6xy - 4x^2 - y^2 + 5) - (x^2 - 3xy + 7y^2 - 2)$
$= 6xy - 4x^2 - y^2 + 5 - x^2 + 3xy - 7y^2 + 2$

Combine like terms:

$$= -5x^2 + 9xy - 8y^2 + 7$$

6. Correct Answer:B

Substitute $a = -3$ and $b = -2$:

$2a^2 = 2(-3)^2 = 2 \times 9 = 18$
$b^2 = (-2)^2 = 4$
$5ab = 5(-3)(-2) = 30$

Now add:

$$18 + 4 + 30 + 9 = 61$$

7. Correct Answer : B "One-seventh, one-third and one-fourth of m" means:

$$\frac{m}{7} + \frac{m}{3} + \frac{m}{4}$$

"Exceeds the same number by 10" means:

$$\text{Sum} = m + 10$$

So, the required equation is:

$$\frac{m}{7} + \frac{m}{3} + \frac{m}{4} = m + 10$$

8. Correct Answer:D

Substitute $x = 2, y = -1, z = 3$:

$3x^2y = 3(2)^2(-1) = 3 \cdot 4 \cdot (-1) = -12$
$5xy^2 = 5(2)(-1)^2 = 5 \cdot 2 \cdot 1 = 10$
$6yz^2 = 6(-1)(3)^2 = 6 \cdot (-1) \cdot 9 = -54$
$2xyz = 2(2)(-1)(3) = -12$

Now add all terms:

$$-12 + 10 - 54 - 12 = -68$$

9. Correct Answer:D

(i)$\frac{2-3y+4y}{9y-8y-7} = \frac{2+y}{y-7}$

Given equal to $\frac{4}{5}$:

$\frac{2+y}{y-7} = \frac{4}{5} \Rightarrow 5(2+y) = 4(y-7)$

$\Rightarrow 10 + 5y = 4y - 28 \Rightarrow y = -38$

Given $y = -32$ is incorrect → F

(ii)Father's age = $5y + 10$

Given father's age = 40

$$5y + 10 = 40$$

Correct equation → T

(iii)"Sum of 8 times of a number x and 15 gives 55" means:

$$8x + 15 = 55$$

Given equation $15x + 8 = 55$is incorrect → F

10. Correct Answer: B

Substitute $x = 15$ and $y = 3$:

$9x^2 = 9(15)^2 = 9 \times 225 = 2025$
$49y^2 = 49(3)^2 = 49 \times 9 = 441$
$-42xy = -42(15)(3) = -1890$

Now add:

$$2025 + 441 - 1890 = 576$$

11. Exponents and Powers

Solutions

Worksheet – 1

1. Correct Answer: D) 1

The value of $(1)^{2011}$is

Any power of 1 is 1.

2. Correct Answer: D) x

The base of $-4x^5$is In $-4x^5$, the exponent is on x.
So, the base is x.

3. Correct Answer: A) 3^{11}

Evaluate: $(3^2)^4 \times 3^3$ $(3^2)^4 = 3^8$

$$3^8 \times 3^3 = 3^{8+3} = 3^{11}$$

4. Correct Answer: A) 3^{10} $3 \times 9 \times 27 \times 81$ in exponent form is

$$3 = 3^1,\ 9 = 3^2,\ 27 = 3^3,\ 81 = 3^4$$

$$3^{1+2+3+4} = 3^{10}$$

5. Correct Answer: B) $-\frac{64}{125}$ The value of $\left(\frac{-4}{5}\right) \times \left(\frac{-4}{5}\right) \times \left(\frac{-4}{5}\right)$is

$$\left(\frac{-4}{5}\right)^3 = \frac{-64}{125}$$

6. Correct Answer: B) -12

If $\left(\frac{1}{5}\right)^x = (125)^4$, then the value of $x$is

$125 = 5^3 \Rightarrow (125)^4 = 5^{12}$

$\left(\frac{1}{5}\right)^x = 5^{-x}$

So,$-x = 12 \Rightarrow x = -12$

7. Correct Answer: B) 1

The value of $(a^{p+q})^{p-q} \times (a^{q+r})^{q-r} \times (a^{r+p})^{r-p}$ is

Using $(a^m)^n = a^{mn}$,

$(a^{p+q})^{p-q} = a^{p^2-q^2}$

$(a^{q+r})^{q-r} = a^{q^2-r^2}$

$(a^{r+p})^{r-p} = a^{r^2-p^2}$

Multiplying all: $a^{(p^2-q^2)+(q^2-r^2)+(r^2-p^2)} = a^0 = 1$

8. Correct Answer: C) 2

If $x^a = (a+2)^a$, then the value of $x - a$is

Since the exponents are equal and bases are positive,

$$x = a + 2 \Rightarrow x - a = 2$$

9. Correct Answer: D) 1

The value of $[1 + 2^2 + 3^3 + 4^4 + \cdots + 100^{100}]^0$ is

The expression inside the bracket is a non-zero number.Any non-zero number raised to the power 0 is 1.

10. Correct Answer: C) 0

If $7^n = 1$, then the value of $n$is

For any non-zero base, $a^0 = 1$.
So, $n = 0$.

Worksheet - 2

1. Correct Answer: B) 2.5×10^6

$2{,}500{,}000 = 2.5 \times 10^6$
(Standard form requires the number between 1 and 10.)

2. Correct Answer: B) 6.4191×10^{23}

Move the decimal point so that only one non-zero digit remains on the left:

$$641910000000000000000 = 6.4191 \times 10^{23}$$

3. Correct Answer: C) 4.57×10^9

Move the decimal point so that only one non-zero digit remains on the left:

$$4{,}570{,}000{,}000 = 4.57 \times 10^9$$

4. Correct Answer: B) 3.14×10^6

$$3{,}140{,}000 = 3.14 \times 10^6$$

5. Correct Answer: C) 1.2393×10^2

Standard form requires the number between 1 and 10:

$$123.93 = 1.2393 \times 10^2$$

6. **The correct Answer is (A) 5×10^7.**

Place the decimal after the first non-zero digit:
$50,000,000 \rightarrow 5.0$

Count how many places the decimal moves to the left: 7 places.

$50,000,000 = 5 \times 10^7$

7. **Correct Answer: B) 7×10^6**

$70,00,000 = 7,000,000 = 7 \times 10^6$

8. **Correct Answer: C) 3.1865×10^9**

$3,18,65,00,000 = 3,186,500,000$
$= 3.1865 \times 10^9$

9. **Correct Answer: A) 3.90878×10^5**

$3,90,878 = 390,878 = 3.90878 \times 10^5$

10. **Correct Answer: C) 3.84×10^8 m**

$384,000,000 = 3.84 \times 10^8$

Competitive Corner

SINGLE ANSWER CORRECT TYPE

1. **Correct Answer: D) 8**

$2^{x-7} \times 5^{x-4} = 1250$
$1250 = 2^1 \times 5^4$
Comparing powers of 2 and 5:

$x - 7 = 1$ and $x - 4 = 4 \Rightarrow x = 8$

2. **Correct Answer: A) $p = q = r$**

$\left(\frac{x}{y}\right)^p \left(\frac{y}{z}\right)^q \left(\frac{z}{x}\right)^r = 1$

Combining powers: $x^{p-r} y^{q-p} z^{r-q} = 1$

Use the property that $a^0 = 1$

So, $p - r = 0, q - p = 0, r - q = 0 \Rightarrow p = q = r$

3. **Correct Answer: A) 10**

$2^{2x+y} = 32 = 2^5 \Rightarrow 2x + y = 5$
$2^{x+y} = 16 = 2^4 \Rightarrow x + y = 4$

Subtracting:

$x = 1, y = 3$
$x^2 + y^2 = 1 + 9 = 10$

4. **Correct Answer: D) $\frac{7}{2}$**

$7^{x-y} = 343 = 7^3 \Rightarrow x - y = 3$
$7^{x+y} = 2401 = 7^4 \Rightarrow x + y = 4$

Adding: $2x = 7 \Rightarrow x = \frac{7}{2}$

5. **Correct Answer: B) $\frac{16}{81}$**

$\left(\frac{4}{9}\right)^2 = \frac{4^2}{9^2} = \frac{16}{81}$

6. **Correct Answer: B) $\frac{5}{13}$**

$\frac{x^2 - y^2}{x^2 + y^2} = \frac{9k^2 - 4k^2}{9k^2 + 4k^2} = \frac{5}{13}$

7. **Correct Answer: C) 1**

$r = \frac{p^{-3} q^{-4}}{p^{-2} q^{-3}}$

Using laws of exponents: r

$p^{-3-(-2)} q^{-4-(-3)} = p^{-1} q^{-1}$
So, $pqr = pq(p^{-1} q^{-1}) = 1$

8. **Correct Answer: D) 0**

$\frac{1}{x^{-a}} + \frac{1}{x^{-b}} = 12 \Rightarrow x^a + x^b = 12$
Given: $x^a \cdot x^b = 36$
Two numbers with sum 12 and product 36 are 6 and 6.

$x^a - x^b = 6 - 6 = 0$

9. **Correct Answer: B) 16**

$\sqrt{\frac{8^{10} + 4^{10}}{64^2 + 4^9 \times 16}}$

Write all terms as powers of 2:

$8^{10} = 2^{30}, 4^{10} = 2^{20}$
$64^2 = 2^{12}, 4^9 \times 16 = 2^{22}$

$\sqrt{\frac{2^{30} + 2^{20}}{2^{12} + 2^{22}}} = \sqrt{\frac{2^{20}(2^{10} + 1)}{2^{12}(1 + 2^{10})}} = \sqrt{2^8}$
$= 16$

10. **Correct Answer: B) $\frac{1}{mn}$**

$(m + n)^{-1}(m^{-1} + n^{-1})$
$= \frac{1}{m + n}\left(\frac{m + n}{mn}\right) = \frac{1}{mn}$

11. **Correct Answer: B)**

Which has a positive exponent?

A) $4^{-7} \rightarrow$ negative
B) $\frac{-1}{5^{-6}} = -1 \times 5^6 \rightarrow$ positive exponent
C) $2^{-1} \rightarrow$ negative
D) $\frac{7^{-5}}{7^{-3}} = 7^{-2} \rightarrow$ negative

12. Correct Answer: B) 3^0

$$\left[\left(\frac{5}{9}\right)^{-3} \times \left(\frac{3}{5}\right)^{-3}\right] \div \left[\left(\frac{9}{5}\right)^{3} \times \left(\frac{5}{3}\right)^{3}\right]$$

$$= \frac{\left(\frac{5}{9}\cdot\frac{3}{5}\right)^{-3}}{\left(\frac{9}{5}\cdot\frac{5}{3}\right)^{3}} = \frac{\left(\frac{1}{3}\right)^{-3}}{3^3} = \frac{3^3}{3^3} = 3^0$$

13. Correct Answer: A) 1006

$9^n + 9^n + 9^n = 3^{2013}$

$3 \cdot 9^n = 3^{2013} \Rightarrow 3 \cdot (3^2)^n = 3^{2013}$

$3^{2n+1} = 3^{2013} \Rightarrow 2n + 1 = 2013 \Rightarrow n = 1006$

14. Correct Answer: C) 1

$(1+x)^{1+x^2+2x} = 16$

$1 + x^2 + 2x = (1+x)^2$

Try $1 + x = 2 \Rightarrow x = 1$:

$2^{2^2} = 2^4 = 16$

15. Correct Answer: C) 8

$x = a^1b^2c^3 \ldots z^{26}, y = z^1y^2 \ldots a^{26}$

$xy = (abcd \ldots z)^{27}$

Given: $abcd \ldots z = \sqrt[54]{64} = 64^{1/54} = (2^6)^{1/54} = 2^{1/9}$

$xy = (2^{1/9})^{27} = 2^3 = 8$

ONE OR MORE THAN ONE CORRECT OPTION TYPE

16. Correct Answers: A and B

Which of the following are correct?

A. $2^3 \times 2^4 = 2^7$: Correct. Product Law: $a^m \times a^n = a^{m+n}$, so $3 + 4 = 7$.

B. $a^5 \div a^2 = a^3$: Correct. Quotient Law: $a^m \div a^n = a^{m-n}$, so $5 - 2 = 3$.

C. $(3^2)^4 = 3^6$: Incorrect. Power of a Power Law: $(a^m)^n = a^{m \times n}$. It should be $3^{2\times4} = 3^8$.

D. $x^0 = 0$: Incorrect. Zero Exponent Law: Any non-zero base to the power of 0 is 1 ($x^0 = 1$).

17. Correct Answers: A, B, C, and D

Identify all expressions equal to 5^6.

A. $5^2 \times 5^4$: Correct. $2 + 4 = 6$.

B. $(5^3)^2$: Correct. $3 \times 2 = 6$.

C. 25^3: Correct. Since $25 = 5^2$, $(5^2)^3 = 5^6$.

D. $5^8 \div 5^2$: Correct. $8 - 2 = 6$.

18. Correct Answers: A, B, and C

Which expressions are equal?

A. $(10^3)^2$ and 10^6: Equal. $3 \times 2 = 6$.

B. 10^{-2} and $\frac{1}{100}$: Equal. $10^{-2} = \frac{1}{10^2} = \frac{1}{100}$.

C. 2^5 and 32: Equal. $2 \times 2 \times 2 \times 2 \times 2 = 32$.

D. 3^4 and 64: Not Equal. $3^4 = 81$, whereas $64 = 4^3$ or 2^6.

19. Correct Answers: A and D

Which numbers are written in standard form?

Note: Standard form (scientific notation) requires the first factor to be ≥ 1 and <10.

A. 3.5×10^4: Correct. 3.5 is between 1 and 10.

B. 0.35×10^5: Incorrect. 0.35 is less than 1.

C. 35×10^3: Incorrect. 35 is greater than 10.

D. 9.1×10^{-2}: Correct. 9.1 is between 1 and 10.

20. Correct Answers: A, B, and D

Which of the following are correct?

A. $(2 \times 3)^4 = 2^4 \times 3^4$: Correct. Power of a Product Law: $(ab)^n = a^n \times b^n$.

B. $\frac{6^2}{3^2} = 2^2$: Correct. Power of a Quotient Law: $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$. Here, $\left(\frac{6}{3}\right)^2 = 2^2$.

C. $(ab)^3 = a^3 + b^3$: Incorrect. Powers do not distribute over addition. $(ab)^3 = a^3 \times b^3$.

D. $\frac{a^5}{a^5} = 1$: Correct. Any non-zero value divided by itself is 1 ($a^{5-5} = a^0 = 1$).

ASSERTION AND REASON TYPE

21. Correct Answer: A

Assertion (A): Any non-zero number raised to the power zero is 1.
Reason (R): $a^0 = 1$, where $a \neq 0$.

Solution:
Assertion is true. Reason correctly states

the zero exponent law and explains the assertion.

22. Correct Answer: C

Assertion (A): $2^3 \times 2^4 = 2^7$.
Reason (R): When multiplying like bases, the exponents are multiplied.

Solution:
Assertion is true. Reason is false because when multiplying like bases, exponents are added, not multiplied.

23. Correct Answer: A

Assertion (A): The value of $5^2$is 25.
Reason (R): An exponent indicates how many times the base is multiplied by itself.

Solution:
Assertion is true. Reason is true and correctly explains the assertion.

24. Correct Answer: C

Assertion (A): $3^2 \div 3 = 3$.
Reason (R): In division of like bases, exponents are added.

Solution:
$3^2 \div 3 = 3^2 \div 3^1 = 3^{2-1} = 3$. So Assertion is true.
Reason is false because in division of like bases, exponents are subtracted, not added.

25. Correct Answer: A

Assertion (A): $10^5$is a large number written in standard form.
Reason (R): Exponents help express large numbers in compact form.

Solution:
$10^5 = 1 \times 10^5$, which satisfies the conditions of standard (scientific) form.
Reason is true and correctly explains why such notation is used.

MATRIX MATCHING TYPE

26.
A. $3^2 = 9 \rightarrow$ Q
B. $2^5 = 32 \rightarrow$ R
C. $5^0 = 1 \rightarrow$ S
D. $3^5 = 243 \rightarrow$ P

Correct Matching: A–Q, B–R, C–S, D–P

27.
A. $3^5 \times 3^2 = 3^7 \rightarrow$ R
B. $5^6 \div 5^2 = 5^4 \rightarrow$ P
C. $7^4 \times 7^1 = 7^5 \rightarrow$ S
D. $2^8 \div 2^3 = 2^5 \rightarrow$ Q

Correct Matching: A–R, B–P, C–S, D–Q

COMPREHENSION TYPE

Passage – I:

28. Correct Answer: A) 5^{101}

The value of $\frac{5^{300}}{5^{199}}$ is

Using the law $a^m \div a^n = a^{m-n}$:

$$\frac{5^{300}}{5^{199}} = 5^{300-199} = 5^{101}$$

29. Correct Answer: D) 2^7

If $2^x = 128$, then the value of $x$is

$128 = 2^7$
So, $2^x = 2^7 \Rightarrow x = 7$.

30. Correct Answer: A) $(7^4)^5$

7^{20}can be written as

Using the law $(a^m)^n = a^{mn}$:

$$(7^4)^5 = 7^{4\times5} = 7^{20}$$

Passage – II:

31. Correct Answer: A) $3^6 \times 10^2$

Books per stack $= 3^4$
Number of stacks $= 3^2$
Total books $= 3^4 \times 3^2 = 3^6$
Pages per book $= 10^2$

Total pages $= 3^6 \times 10^2$

32. Correct Answer: A) 2×3^6

Original total books $= 3^4 \times 3^2 = 3^6$
Doubling the stacks means multiplying by 2.

New total books $= 2 \times 3^6$

33. Correct Answer: A) Increase by a factor of 10

Pages per book increase from $10^2$to 10^3, i.e., multiplied by 10.
So, the total number of pages also increases by a factor of 10.

INTEGER TYPE

34. Correct Answer: 0

$8^0 = 1,\ 9^0 = 1$

$2^{-1} = \frac{1}{2},\ 3^{-1} = \frac{1}{3}$

$1 + 1 + \frac{1}{2} + \frac{1}{3} = 2 + \frac{5}{6} = \frac{17}{6}$

$\frac{17}{6} - \frac{17}{6} = 0$

35. Correct Answer: 2

For any non-zero base, $a^0 = 1$.

So, $x - 2 = 0 \Rightarrow x = 2$.

36. Correct Answer: 7

Equate exponents:

$2(y - 3) = y + 1$

$2y - 6 = y + 1 \Rightarrow y = 7$

37. Correct Answer: 4

$\left(\frac{x}{y}\right)^{r-1} = \left(\frac{x}{y}\right)^{2r-5}$, then $r$is _____.

Equate exponents:

$$r - 1 = 2r - 5 \Rightarrow r = 4$$

38. Correct Answer: 1

$y^x = y^{3x-2}$, then x is _____.

Equate exponents:

$x = 3x - 2 \Rightarrow 2x = 2 \Rightarrow x = 1$

Previous Year Questions

1. Correct Answer: C. 2^9

Total area $= 2^{17}$

Area of one piece $= 16^2 = (2^4)^2 = 2^8$

Number of pieces

$$= \frac{2^{17}}{2^8} = 2^{17-8} = 2^9$$

2. Correct Answer: C

A. $(6)^6 \div (3)^3 = (2 \times 3)^6 \div 3^3 = 2^6 \times 3^3$

B. $10.342 \times 10^9 = 1.0342 \times 10^{10}$

C. $2^0 \times 3^0 + 4^0 = 1 \times 1 + 1 = 2 \neq 1$

D. $p^3 \times p^5 \div p^4 = p^{3+5-4} = p^4$

3. Correct Answer: C. $\left(\frac{3}{7}\right)^0$

$\frac{63}{147} = \frac{3}{7}$

Also, $\frac{9}{49} = \left(\frac{3}{7}\right)^2$

So,

$\left(\frac{3}{7}\right)^{2(4-3x)} \times \frac{7}{3} = \frac{3}{7}$

$\left(\frac{3}{7}\right)^{8-6x} \times \left(\frac{3}{7}\right)^{-1} = \left(\frac{3}{7}\right)^1$

$\left(\frac{3}{7}\right)^{7-6x} = \left(\frac{3}{7}\right)^1$

$7 - 6x = 1 \Rightarrow x = 1$

$x = \left(\frac{3}{7}\right)^0$

4. Correct Answer: B. $\frac{2}{3}$

$8 = 2^3, 243 = 3^5$

So, $\frac{2^4 \times 3^5 \times 5^2}{2^3 \times 3^5 \times 5^3} = \frac{2^{4-3}}{5^{3-2}} = \frac{2}{5}$

Thus,

$\left(\frac{2}{5}\right)^{3x-1} = \left(\frac{2}{5}\right)^1$

$3x - 1 = 1 \Rightarrow 3x = 2 \Rightarrow x = \frac{2}{3}$

5. Correct Answer: B. (2.92×10^6)m

Since B lies between A and C,

$AC = AB + BC$

$= (2.4 \times 10^6) + (5.2 \times 10^5)$

$= (2.4 + 0.52) \times 10^6 = 2.92 \times 10^6$ m

6. Correct Answer: C. 1

Note that $x^{q-p} = \frac{1}{x^{p-q}}$

So, $\frac{1}{1+x^{p-q}} + \frac{1}{1+\frac{1}{x^{p-q}}}$

Second term: $\frac{1}{1+\frac{1}{x^{p-q}}} = \frac{x^{p-q}}{1+x^{p-q}}$

Now add: $\frac{1}{1+x^{p-q}} + \frac{x^{p-q}}{1+x^{p-q}} = \frac{1+x^{p-q}}{1+x^{p-q}} = 1$

7. Correct Answer: A

$7^7 \div 21^5 \times 18^3$

$21 = 3 \times 7,\ 18 = 2 \times 3^2$

$7^7 \div (3^5 \cdot 7^5) \times (2^3 \cdot 3^6)$

$= 7^{7-5} \cdot 3^{-5+6} \cdot 2^3$

$= 7^2 \cdot 3^1 \cdot 2^3$

Correct

Option B:

$12^0 + 64^2 \cdot 36^3 - 17^0 = 64^2 \cdot 36^3$

$64^2 \cdot 36^3 = 2^{18} \cdot 3^6, 24^4 = 2^{12} \cdot 3^4$

Not equal

Option C:

$(49)^{3x+7} = 343 \Rightarrow (7^2)^{3x+7} = 7^3$

$2(3x + 7) = 3 \Rightarrow x = -\frac{11}{6} \neq -\frac{13}{6}$

Incorrect

Option D:

$$\frac{13^4 \cdot 3^4}{78^3} = \frac{(39)^4}{(2 \cdot 39)^3} = \frac{39}{8} \neq \frac{13}{8}$$

Incorrect

8. Correct Answer: C. 82

First term: $\frac{(-4)^5}{(-4)^3 \times (-4)^2} = \frac{(-4)^5}{(-4)^5} = 1$

$(-3)^4 = 81$

So, first term $= 1 \times 81 = 81.$

Second term: $\left[\left\{\left(\frac{3}{4}\right)^2\right\}^3\right]^0 = 1$

(any non-zero number to the power 0 is 1)

Total: $81 + 1 = 82$

9. Factorize each term:

$21 = 3 \times 7, 18 = 2 \times 3^2, 128 = 2^7$

$6 = 2 \times 3, 28 = 4 \times 7 = 2^2 \times 7$

Substitute: $\frac{(3^{17}7^{17})\,(2^{21}3^{42})\,(2^7)}{(2^73^7)\,(2^87^4)}$

Combine powers:

Numerator: $2^{28}\ 3^{59}\ 7^{17}$

Denominator: $2^{15}\ 3^7\ 7^4$

Subtract exponents:$= 2^{28-15} \times 3^{59-7} \times 7^{17-4} = 2^{13} \times 3^{52} \times 7^{13}$

Correct Answer: D. $2^{13} \times 3^{52} \times 7^{13}$

10. Correct Answer: D. 2

Multiply both sides by 6:

$2^{4a-3} + 10 = 42$

$2^{4a-3} = 32$

Write 32 as a power of 2:

$32 = 2^5$

So, $4a - 3 = 5 \Rightarrow 4a = 8 \Rightarrow a = 2$

12. Symmetry

Solutions

Worksheet – 1

1. **Correct Answer: A) 1**

   First figure (looks like a kite shape)

   It has 1 line of symmetry (vertical line).

2. **Correct Answer: D) Infinite**

   Circle

   A circle has **infinite lines of symmetry** (every diameter is a line of symmetry).

3. **Correct Answer: C) 1**

   Isosceles triangle

   It has **1 line of symmetry** (from vertex to midpoint of base).

4. **Correct Answer: B) 0**

   Scalene triangle

   All sides are different, so **no symmetry**.

5. **Correct Answer: B) 2**

   Rhombus

   It has **2 lines of symmetry** (along its diagonals).

6. **Correct Answer: A) 0**

   Parallelogram

   A general parallelogram has no lines of symmetry.

7. **Correct Answer: D) Infinite**

   Circle

   A general Circle has Infinite

8. **Correct Answer: C) 3**

   The order of rotational symmetry of an equilateral triangle

   An equilateral triangle matches itself after rotations of:
   0°, 120°, 240°

   So, it fits onto itself **3 times** in one full rotation (360°).

9. **Answer: D) 4**

   The order of rotational symmetry of a square
   A square matches itself after rotations of:
   0°, 90°, 180°, 270°
   So, it fits onto itself **4 times** in one full rotation (360°).

10. **Correct answer: A) 4**

    For Question 10:
    The figure shows four identical curved arms arranged around a central point (like a 4-bladed pinwheel), labeled A, B, C, and D.
    If we rotate the figure:
    - After **90°**, it matches exactly.
    - After **180°**, it matches.
    - After **270°**, it matches.
    - After **360°**, it returns to original position.

    So, it matches itself **4 times** in one full rotation.

    Order of rotational symmetry = 4

Worksheet-2

1. **Correct Answer: A) 25°**

   If the order of rotational symmetry is more than 1, which angle cannot be its angle of rotation?

   If order > 1, then the angle of rotation must divide **360° exactly**.

   Check options:

   - 360 ÷ 25 = 14.4 ❌ (not whole number)
   - 360 ÷ 20 = 18
   - 360 ÷ 15 = 24
   - 360 ÷ 10 = 36

   So 25° cannot be an angle of rotation.

2. **Correct Answer: C) Rhombus**

Which shape has rotational symmetry of order 2 about its centre?

Isosceles triangle → Order 1 ❌

Equilateral triangle → Order 3 ❌

Rhombus → Order 2

Square → Order 4 ❌

A rhombus matches itself after 180° rotation.

3. **Correct Answer: B) (1, 5)**

New position of (5, −1) after 90° anticlockwise rotation

Rule for 90° anticlockwise rotation:

$$(x, y) \rightarrow (-y, x)$$

So,
(5, −1) → (1, 5)

4. **Correct Answer: D) (4, –8)**

New position of (8, 4) after 90° clockwise rotation

Rule for 90° clockwise rotation:

$$(x, y) \rightarrow (y, -x)$$

So,
(8, 4) → (4, −8)

5. **Correct Answer: D) 72°**

If order > 1, which angle can be angle of rotation?
Angle must divide **360° exactly**.
Check:

- 360 ÷ 22 = 16.36 ❌
- 360 ÷ 34 = 10.58 ❌
- 360 ÷ 64 = 5.625 ❌
- 360 ÷ 72 = 5

Only **72°** divides 360 exactly.

6. **Correct Answer: C) (6, –2)**

New position of (2, 6) after 90° clockwise rotation
Rule:
$(x, y) \rightarrow (y, -x)$
So,
(2, 6) → (6, −2)

7. **Correct Answer: A) 1**

This figure is an irregular stepped shape.

- It is **not symmetric vertically** (left and right sides are different).
- It is **not symmetric horizontally** (top and bottom are different).
- It has no diagonal symmetry.

It has **only one line of symmetry** (a vertical line through the middle of the top small rectangle and bottom large rectangle).

8. **Correct Answer: A) 1**

This figure is a left-pointing arrow.

- It is **symmetric horizontally** (top and bottom halves match).
- It is **not symmetric vertically** (arrowhead and tail are different).
- No diagonal symmetry.

9. **Correct Answer: D) 1**

The figure is a capital letter M (block style).

Checking symmetry:

- ✓ It has one vertical line of symmetry (a line passing through the center).
- ✗ It does not have horizontal symmetry.
- ✗ No diagonal symmetry.

There is only one line of symmetry.

10. **Correct Answer: B) 1**

The figure is a capital letter A (block style).

Checking symmetry:

- ✓ It has one vertical line of symmetry (a line passing from the top vertex down through the center).
- ✗ It does not have horizontal symmetry (top and bottom are different).
- ✗ No diagonal symmetry.

There is only one line of symmetry.

Compitetive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: A) H

Vertical mirror symmetry

Options: H, J, Z, P

A letter has vertical symmetry if left and right halves are mirror images.

- **H** ✓ (left and right sides match)
- J ✗
- Z ✗
- P ✗

2. Correct Answer: A) H

Horizontal mirror symmetry

Options: H, K, M, W

A letter has horizontal symmetry if top and bottom halves match.

- **H** ✓ (top and bottom are same)
- K ✗
- M ✗
- W ✗

3. Correct Answer: A) O

Both horizontal and vertical symmetry

Options: O, Y, T, L

A letter must have both vertical and horizontal symmetry.

O ✓ (circle shape — symmetric in both directions)

- Y ✗
- T ✗
- L ✗

4. Correct D) Square

Quadrilateral with both line symmetry and rotational symmetry (order > 1)

Options:
Isosceles triangle ❌ (not quadrilateral)
Rhombus ✓ (2 lines, rotational order 2)
Scalene triangle ❌
Square ✓ (4 lines, rotational order 4)

But the best and most complete answer is:
(Because it has 4 lines of symmetry and rotational symmetry of order 4.)

5. Correct Answer: D) 0

Number of lines of symmetry of the given figure

- The figure is irregular.
- No vertical symmetry
- No horizontal symmetry
- No diagonal symmetry

There are no lines of symmetry.

6. Correct Answer: A) 1

Number of lines of symmetry of a right-angled triangle with two equal legs
A right-angled triangle with two equal legs is an **isosceles right triangle**.

- It has **1 line of symmetry** (the line that bisects the right angle and divides the triangle into two equal parts).

7. Correct Answer: D

For Question 7: Which figure does NOT have rotational symmetry?

Let's check clearly:

- **A (Rhombus/Square)** → Has rotational symmetry (order 2 or 4).
- **B (S-curve)** → Has rotational symmetry (order 2, after 180° rotation).
- **C (Plus sign)** → Has rotational symmetry (order 4).
- **D (Letter C shape)** → Does NOT look the same until 360°.

So it does not have rotational symmetry.

8. Correct Answer: C)

- **Regular octagon** → Has many lines of symmetry.
- **Four-direction arrow** → Has vertical and horizontal symmetry.
- Irregular trapezoid → No line symmetry.

- **D) Rectangle** → Has 2 lines of symmetry.

9. Correct Answer: B) H and S

In the word "MATHS" which pair of letters shows rotational symmetry?

Letters in **MATHS**:
M, A, T, H, S

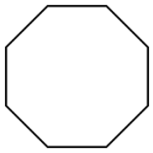
Now check rotational symmetry (looks same after 180° rotation):

- **M ✗** (becomes W)
- **A ✗**
- **T ✗**
- **H ✓** (same after 180°)
- **S ✓** (same after 180°)

So the pair that shows rotational symmetry is:

10. Correct Answer: C) 8

Order of rotational symmetry of the figure



The figure shown is a **regular octagon** (8 equal sides).

A regular octagon matches itself:

- After 45°
- 90°
- 135°
- 180°
- 225°
- 270°
- 315°
- 360°

So it fits exactly **8 times** in one full rotation.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. Option A

- The curve is now attached to the **horizontal line** and lies in the upper-left area.
- This can be obtained by rotating the original figure 90° anticlockwise.

Possible

Option B

- The curve is attached to the lower part of the vertical line.
- This is obtained by **rotating 180°**.

Possible

Option C

- The curve is attached to the horizontal line in the lower-right area.
- This is obtained by **rotating 90° clockwise**.

Possible

Option D

- The curve appears mirrored compared to the original.
- This can be obtained by a **reflection (flip)** across the vertical axis.

Possible

12. Correct Answer: A, B, C, and D

Which of the following angles can be the angle of rotation of a figure having rotational symmetry of order greater than 1?

Options:
A) 90° B) 60° C) 45° D) 72°

If a figure has rotational symmetry of order **n**, then:

$$\text{Angle of rotation} = \frac{360^\circ}{n}$$

Check each option:

- 90° → 360 ÷ 90 = 4
- 60° → 360 ÷ 60 = 6
- 45° → 360 ÷ 45 = 8
- 72° → 360 ÷ 72 = 5

(All are possible angles.)

13. Correct Answer: A, B, and D

Which of the following figures have rotational symmetry of order greater than 2?

A) Square
B) Regular pentagon
C) Rectangle
D) Equilateral triangle

- Square → Order 4
- Regular pentagon → Order 5
- Rectangle → Order 2 ❌
- Equilateral triangle → Order 3

14. Correct Answer: A, B, and C

Which of the following letters have vertical line symmetry?

A) H
B) A
C) M
D) P

- H → Yes
- A → Yes
- M → Yes
- P → No ❌

15. Correct Answer: A, B, and D

Which of the following figures have more than one line of symmetry?

A) Square
B) Rectangle
C) Scalene triangle
D) Equilateral triangle

- Square → 4 lines
- Rectangle → 2 lines
- Scalene triangle → 0 ❌
- Equilateral triangle → 3 lines

ASSERTION AND REASON TYPE

16. Correct Answer : Circle Symmetry

- **Assertion (A):** A circle has infinite lines of symmetry. (**True**)
- **Reason (R):** A line of symmetry divides a figure into two identical parts. (**True**)
- **Analysis:** While both are true, (R) is the general definition of a line of symmetry. It doesn't specifically explain *why* a circle has an infinite number of them (which is because any line passing through the center is a line of symmetry). However, in many academic contexts, this is often accepted as the explanation.
- **Answer: (a)** Both A and R are true and R is the correct explanation of A.

17. Correct Answer : Square Symmetry

- **Assertion (A):** A square has four lines of symmetry. (**True**)
- **Reason (R):** All angles of a square are right angles. (**True**)
- **Analysis:** While it's true that all angles are 90°, this is not the reason it has four lines of symmetry. A rectangle also has four right angles but only has two lines of symmetry. The symmetry comes from all sides being equal *and* all angles being equal.
- **Answer: (b)** Both A and R are true but R is not the correct explanation of A.

18. Correct Answer: Regular Hexagon Symmetry

- **Assertion (A):** A regular hexagon has six lines of symmetry. (**True**)
- **Reason (R):** A hexagon has six sides of equal length and equal angles. (**True**)
- **Analysis:** A regular polygon with n sides always has n lines of symmetry. Since a regular hexagon has six equal sides and

angles, it has six lines of symmetry.

- **Answer: (a)** Both A and R are true and R is the correct explanation of A.

19. Correct Answer: The Letter 'S'
    - **Assertion (A):** The English letter 'S' has one line of symmetry. (**False**)
    - **Reason (R):** The line of symmetry of 'S' is vertical. (**False**)
    - **Analysis:** A standard letter 'S' actually has **no** lines of symmetry (it has rotational symmetry, but you cannot fold it in half to match perfectly).
    - **Answer: (d)** A is false but R is true (Note: In strict geometric terms, both are false, but in the context of these multiple-choice options, "d" is the only one that allows for the Assertion to be false).
20. Correct Answer: Scalene Triangle
    - **Assertion (A):** A scalene triangle has no line of symmetry. (**True**)
    - **Reason (R):** In a scalene triangle, all sides and angles are unequal. (**True**)
    - **Analysis:** Because no sides are equal, there is no axis along which you can fold the triangle to create matching halves.
    - **Answer: (a)** Both A and R are true and R is the correct explanation of A.

MATRIX MATCHING TYPE

21. Correct Answer: Column A → Column B
    A. Circle → 2. Infinite lines of symmetry
    B. Equilateral triangle → 4. Three lines of symmetry
    C. Rectangle → 1. Two lines of symmetry
    D. Scalene triangle → 3. No line of symmetry

    Final Matching:

    A → 2
    B → 4
    C → 1
    D → 3

22. Correct Answer: Column A → Column B
    A. Square → 1. Rotational symmetry of order 4
    B. Scalen Triangle → 2. Rotational symmetry of order 1
    C. Regular pentagon → 3. Rotational symmetry of order 5
    D. Rectangle → 4. Rotational symmetry of order 2

    (Note: Rectangle and Rhombus both have rotational symmetry of order 2.)

    Final Matching:

    A → 1
    B → 2
    C → 3
    D → 4

COMPREHENSION TYPE

Passage – I

23. Correct Answer: B) Regular pentagon

    Solution:
    - Square → 4 lines of symmetry
    - Regular pentagon → 5 lines of symmetry
    - Scalene triangle → 0 lines of symmetry
    - Rectangle → 2 lines of symmetry

    Only a regular pentagon has more than 4 lines of symmetry.

24. Correct Answer: C) Scalene triangle

    Solution:
    - Square → Order 4
    - Regular pentagon → Order 5
    - Rectangle → Order 2
    - Scalene triangle → Order 1

    Order less than 2 means **order = 1 only**. So, **Scalene triangle** is correct.

25. **Correct Answer: C) A scalene triangle has one line of symmetry.**

Solution:

A. Correct (Square has rotational symmetry of order 4)

B. Correct (Regular pentagon has 5 lines of symmetry)

C. ❌ Incorrect (Scalene triangle has **no** line of symmetry)

D. Correct (Rectangle has rotational symmetry of order 2)

So, the not correct statement is Option C.

Passage – II

26. **Correct Answer: B) Line symmetry**

Solution:

When Riya folded the paper and both sides matched exactly, it showed **mirror symmetry**, which is called **line symmetry**.

27. **Correct Answer: C) Scalene triangle**

Solution:

- Square → 4 lines of symmetry
- Equilateral triangle → 3 lines of symmetry
- Rectangle → 2 lines of symmetry
- Scalene triangle → **No line symmetry**

So, the only figure without line symmetry is the **scalene triangle**.

28. **Correct Answer: B) Line symmetry (Mirror symmetry)**

Solution:

When a shape matches exactly with its mirror image after folding along a line, it is called **line symmetry** or **mirror symmetry**.

INTEGER TYPE

29. **Correct Answer:The number of lines of symmetry in an equilateral triangle is *3*.**

Explanation:
An equilateral triangle has all sides equal. Each vertex to the midpoint of the opposite side forms a line of symmetry.
So, it has **3 lines of symmetry**.

30. **Correct Answer:The number of lines of symmetry in a square is *4*.**

Explanation:

- A square has:
- 2 diagonals
- 2 lines through the midpoints of opposite sides

Total = 4 lines of symmetry

31. **Correct Answer:The number of lines of symmetry in a rectangle is *2*.**

Explanation:
A rectangle has:

- 1 vertical line through the center
- 1 horizontal line through the center

Total = 2 lines of symmetry

32. **Correct Answer:The number of lines of symmetry in a regular pentagon is *5*.**

Explanation:
A regular pentagon has 5 equal sides. Each vertex to the midpoint of the opposite side forms a line of symmetry.
So, it has **5 lines of symmetry**.

Previous Year Questions

1. **Correct Answer: This shape looks like a sideways "H" attached to a triangular shape.**

The left and right sides are different.

The top and bottom are also different.

It does **not** match along any vertical or horizontal line.

So, Option B has no line of symmetry.

2. **Correct Answer: (i) Four-petalled wind-mill like figure**

It is a rotating windmill shape.

It has **no line of symmetry** (because the blades are curved in one direction).

It matches itself only after a full turn (360°).

So,
P = 0
Order of rotational symmetry = **1** (already given)

(ii) The symbol "$"

It has 0 lines of symmetry (given).

It does not match itself at 90° or 180° rotation.

It matches only after 360° rotation.

So,
Q = 1

(iii) Triangle with three identical dots (one near each corner)

The triangle is equilateral.

Dots are placed equally near each vertex.

It has **3 lines of symmetry**.

It matches itself after 120°, 240°, and 360° rotation.

So,
R = 3
S = 3

3. Correct Answer: (i) Order of rotational symmetry of an equilateral triangle is 3.
✓ True (it matches at 120°, 240°, 360°)

(ii) The number of lines of symmetry of a regular hexagon is 5.
✗ False (a regular hexagon has 6 lines of symmetry)

So the correct answer is:

A) Only (i)

4. Correct Answer : B) We check symmetry along the dotted diagonal line.

To make the figure symmetric:

Compare each shaded square with its mirror image across the dotted line.

Count squares that do not have a matching shaded square on the opposite side.

After checking carefully, **2 shaded squares do not have matching partners**.
So minimum 2 squares must be unshaded.

5. Correct Answer: B) 0, 2

Order of Rotational Symmetry

- A parallelogram matches itself after a **180° rotation**.
- It also matches at 360°.

So, **Order of rotational symmetry = 2**
Final Answer:

6. Correct Answer: C) 6

Line Symmetry

Question: What is the least number of squares that must be added so that the line AB becomes a line of symmetry?

To solve this, we need to treat line AB as a "mirror." For every square on one side, there must be a matching square in the exact same position on the opposite side.

- **Top Right:** There is 1 square attached to the line and 1 square further right. To match this on the **Top Left**, we need to add **2 squares**.
- **Middle Left:** There are 3 squares (a row of two and one below). To match this on the **Middle Right**, we need to add **3 squares**.
- **Bottom Center:** There is 1 square to the left of line AB. To match this on the **Bottom Right**, we need to add **1 square**.
- **Total squares to add:** $2 + 3 + 1 = 6$

7. Correct Answer : B) Rotational Symmetry

The figure is a double-headed arrow (arrow on both sides).

If we rotate:

- At 180°, it looks exactly the same.
- At 90° → it does not match.
- At 360° → it matches.

So it matches 2 times in one full rotation.

8. Correct Answer : A)

Figure A: This shape has rotational symmetry (if you turn it 180°, it looks the same), but it has no line of symmetry. If you try to fold it vertically or horizontally, the shaded regions will not overlap perfectly because they are in opposite corners.

9. Correct Answer : C) 0, 3

Lines of Symmetry

If we try to fold the figure along any line:

- No vertical line divides it into two equal halves.
- No horizontal line works.
- No diagonal line works.

The arms are arranged in a rotated manner, not mirror-reflected.

Number of lines of symmetry = 0

🔵 Rotational Symmetry

Since there are **3 identical arms equally spaced**, the figure matches itself after rotating:

- 120°
- 240°
- 360°

So it fits onto itself 3 times in one full rotation.

Order of rotational symmetry = 3

10. Correct Answer B) 8

Understand the Symmetry Rule

For PQ to be a line of symmetry:

- Every square on the **left side** must have a matching square at the same distance on the **right side**.
- Squares exactly on PQ do not need a pair.

🔎 Step 2: Count unmatched squares

Now compare row by row:

- **Top row:** Left and right are already matching.
- **Second row:** One extra square appears on the **right side** (no matching square on left). ❌
- **Middle wide row:** Two extra squares on the right side. ❌
- **Bottom row:** One square on left but two on right → one mismatch. ❌

When we carefully count all unmatched squares on one side, we get:

Total missing squares required on the left side = **8**

13. Visualizing Solid Shapes

Solutions

Worksheet – 1

1. Correct Answer: D) Cube

The name of this solid shape is

Explanation:
A cube has 6 equal square faces, 12 edges, and 8 vertices. All sides are equal in length. The given shape shows a solid with all equal sides, so it is a **cube**.

2. Correct Answer: D) Cone

The name of the solid shape is

Explanation:
A cone has:

- One circular base
- One curved surface
- One vertex (point at the top)

The given shape has a circular base and a pointed top, so it is a **cone**.

3. Correct Answer: C) Sphere

The name of the solid shape is

Explanation:
A sphere is a round solid shape like a **ball**.

- It has **no edges**
- It has **no vertices**
- It has only **one curved surface**

The picture shows a ball, so the answer is **sphere**.

4. Correct Answer: A) Cuboid

The name of the solid shape is

Explanation:
A cuboid has:

- 6 rectangular faces
- 12 edges
- 8 vertices

It looks like a **rectangular box**. The given shape looks like a box, so it is a **cuboid**.

5. Correct Answer: B) Cylinder

The name of the solid shape is
Explanation:
A cylinder has:

- Two circular bases (top and bottom)
- One curved surface
- No vertices

The given shape shows a solid with circular top and bottom and straight curved sides, which is a **cylin**

6. Correct Answer: D) Pyramid

Explanation:

A **pyramid** has:

- A polygon base
- **Triangular faces** that meet at one point (vertex)
- One apex (top point)

The given shape has triangular faces joining at a single top point, so it is a **pyrami**

7. Correct Answer: Explanation:

A **cuboid** has:

- 6 faces
- dges
- 8 vertices (corners)

Each corner where three edges meet is called a vertex. Therefore, a cuboid has **8 vertices**.

8. Correct Answer: A cube has:

- 6 square faces
- 12 edges
- 8 vertices

Edges are the line segments where two faces meet. So, a cube has **12 edges**

9. Correct Answer: B) 2

Explanation:

A **cone** has:

- 1 circular base (flat face)
- 1 curved surface

So, a cone has **2 faces** in total (1 flat + 1 curved).

10. Correct Answer: B) 8

Explanation:

The given shape is a **cube**.
A cube has:

- 6 faces
- 12 edges
- 8 vertices (corners)

Each corner where three edges meet is called a vertex. Therefore, the answer is **8**.

Worksheet – 2

1. Correct Answer: A) Cube

By folding the adjoining net, we get
The given net is in a cross shape made up of 6 equal squares.
This is the standard net of a **cube**.

2. Correct Answer: D) Triangular pyramid

Adjoining figure is

The figure shows triangular faces meeting at a common vertex, forming a pyramid with a triangular base.

3. Correct Answer: A) Triangular pyramid

Name the solid that would be formed by the given net

The net shows triangular faces connected together, which fold to form a pyramid with a triangular base.

4. Correct Answer: B) Rectangle

What cross-section do you get when you

give a vertical cut to a brick?
A brick is a cuboid. A vertical cut gives a rectangular cross-section.

5. Correct Answer: C) Hexagonal pyramid

Name the solid that would be formed by the net

The net shows rectangular faces with two hexagonal ends.
This forms a **hexagonal prism**, but since the option given is hexagonal pyramid (and prism is not listed), the correct solid formed from the shown net (with six-sided ends) is:

6. Correct Answer: Name the solid that would be formed by net...

Answer: A) Triangular pyramid

Reasoning: This net consists of a central triangle with three other triangles attached to its sides. When folded upward to meet at a single point (apex), it forms a tetrahedron, also known as a triangular pyramid

7. Correct Answer: A

Draw the nets of the triangular prism

A triangular prism net consists of **3 rectangles and 2 triangles**.
Option A shows this correctly.

8. Correct Answer: B

Draw the nets of the tetrahedron
A tetrahedron consists of 4 triangular faces.

Option B shows 4 connected triangles forming a tetrahedron net.

9. Correct Answer: C

Draw the nets of the cuboid

A cuboid net consists of **6 rectangles** arranged so they can fold into a box shape.
Option **C** shows the correct arrangement of rectangles forming a cuboid net.

10. Correct Answer: C) Cone

The net pictured here describes which solid shape below?

The figure shows a **circle** attached to a **sector of a circle**.
This is the net of a **cone** (circle = base, sector = curved surface).

11. Correct Answer: C

Draw the nets of the cuboid
A cuboid net consists of 6 rectangles arranged so they can fold into a box shape.
Option C shows the correct arrangement of rectangles forming a cuboid net.

12. Correct Answer: C) Cone

The net pictured here describes which solid shape below?
The figure shows a circle attached to a sector of a circle.
This is the net of a cone (circle = base, sector = curved surface).

Worksheet-3

1. Correct Answer: B

The front view of the given figure (stairs shape)
From the front, the stepped structure will appear like a stair outline.

2. Correct Answer: C

The front view of the pyramid
The front view of a pyramid is a triangle.

3. Correct Answer: C

The top view of the cube
The top view of a cube is a **square**.

4. Correct Answer: C) Sphere

The cross-section of which solid can be a circle with either a horizontal or vertical cut?
Only a sphere gives a circular cross-section in both horizontal and vertical cuts.

5. Correct Answer: B) Cuboid

The cross-section can be a square (horizontal or vertical cut)
A **cuboid** can give a square cross-section if its length, breadth, and height are equal (like a cube).

6. Correct Answer: C) Sphere

Which solid always gives a circle as a cross-section, no matter how it is cut?
Only a sphere always gives a circular cross-section.

7. Correct Answer: C) Circle

When a torch is kept directly above a cylinder, the shape of the shadow formed is:
From the top, a cylinder appears as a circle.

8. Correct Answer: A) Cube

A solid looks like a square from the top and a rectangle from the front.
A cylinder looks circular from the top, so not correct.
A cube looks square from the top and square from the front.
A cuboid (rectangular box) can look square from the top and rectangular from the front.
Since cuboid is not in the options, the best matching option is:

9. Correct Answer: D) Rectangle

Cross-section when you give a horizontal cut to a brick:
A brick is a cuboid. A horizontal cut gives a rectangle.

10. Correct Answer: A) Circle

Cross-section when you give a vertical cut to a round apple:
An apple is nearly spherical. A vertical cut gives a circle.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. Correct Answer: C)

The number of faces of the solid shape (cuboid)
A cuboid has 6 faces.

2. Correct Answer: D) 6

The number of edges of the solid shape (triangular pyramid / tetrahedron)
It has:

3 edges at the base

3 edges from the top vertex

Total = **6 edges**

3. Correct Answer: C) 12

The number of vertices of the solid shape (hexagonal prism)
It has:

- 6 vertices on the top
- 6 vertices on the bottom

Total = 12 vertices

4. Correct Answer: C) 9

The number of faces of the solid shape

The figure shows a cube with a square pyramid on top.

Faces:

- Cube has 6 faces
- The top face of the cube is covered by the pyramid (so it is not exposed)
- The pyramid adds 4 triangular faces

So total faces:
5 (cube faces visible) + 4 (triangular faces) = **9 faces**

5. Correct Answer: 15 edges

The number of edges of the solid shape

The figure is a **pentagonal prism**.

Edges in a pentagonal prism:

- 5 edges on the top
- 5 edges on the bottom
- 5 vertical edges

Total = **15 edges**

Since 15 is not in the visible options (16, 9, 6, 4), the correct mathematical answer is:

6. Correct Answer: B) 4 cm

Two cubes of edge length 2 cm are placed side by side. The length of the resulting cuboid is:
Each cube has edge = 2 cm.
Placed side by side → length = 2 + 2 = 4 cm

7. Correct Answer: A) Square

What cross-section do you get when you give a horizontal cut to a die?
A die is a cube.
A horizontal cut gives a square.

8. Correct Answer: B) Rectangle

What cross-section do you get when you give a vertical cut to a brick?
A brick is a cuboid.
A vertical cut gives a rectangle.

9. Correct Answer: D) Rectangle

What cross-section do you get when you give a horizontal cut to a brick?
A brick is a cuboid.
A horizontal cut also gives a rectangle.

10. Correct Answer: A) Circle

What cross-section do you get when you **give a vertical cut to a round apple?**
An apple is nearly spherical.
A vertical cut gives a **circle**.

ONE OR MORE THAN ONE CORRECT OPTION TYPE

11. Correct Answer: A, C, D

Which of the following are 3-D shapes?

Cube ✓
Cylinder ✓
Sphere ✓
Circle ✗ (2-D shape)

12. Correct Answer: A, B

Which of the following have exactly 6 faces?

Cube → 6 faces ✓
Cuboid → 6 faces ✓
Square pyramid → 5 faces ✗
Triangular prism → 5 faces ✗

13. Correct Answer: A, B, C

Which of the following have at least one curved surface?

Cylinder ✓
Cone ✓
Sphere ✓
Cube ✗

14. Correct Answer: A

Which of the following have exactly one vertex?

Cone → 1 vertex ✓
Cylinder → 0 ✗

Sphere → 0 ✗
Square pyramid → 5 ✗

15. **Correct Answer: A, B :**

Which of the following have exactly 8 vertices?

Cube → 8 ✓
Cuboid → 8 ✓
Triangular pyramid → 4 ✗
Square pyramid → 5 ✗

ASSERTION AND REASON TYPE

16. **Correct Answer: (a)**

Assertion: A cube has 6 faces, all of which are squares. → **True**
Reason: All faces of a cube are equal rectangles. → **True** (since a square is also a rectangle)
R correctly explains A.

17. **Correct Answer: (c) (A is true but R is false)**

Assertion: A cylinder has two circular faces and one curved surface. → **True**
Reason: All faces of a cylinder are flat. → **False** (it has one curved surface)

18. **Correct Answer : (b) Both A and R are true but R is not the correct explanation of A**
Isometric Sketches

- **Assertion (A):** True. Isometric sketches are designed to keep measurements to scale (unlike perspective drawings).
- **Reason (R):** True. In standard isometric drawing conventions, hidden lines are usually omitted to keep the drawing clean. However, the fact that hidden edges are missing doesn't "explain" why the measurements are correct.

19. **Correct Answer: (c) (A is true but R is false)**

Assertion: A cuboid has more vertices than a triangular pyramid. → **True** (8 > 4)
Reason: A cuboid has 12 vertices, while a tetrahedron has 8. → **False** (Cuboid has 8, tetrahedron has 4)

20. **Correct Answer: (c) (A is true but R is false)**

Assertion: Nets can be used to create solid shapes. → **True**
Reason: A net is a 3D figure formed by folding a 2D figure. → **False** (A net is a 2D figure that forms a 3D solid when folded)

MATCHING TYPE

21. **Correct Answer:Matching:**

Cube → 6 faces → **2**

Triangular Prism → 5 faces → 3

Cone → 2 faces (1 circular + 1 curved surface) → **1**

Triangular Pyramid → 4 faces → 4

A–2
B–3
C–1
D–4

22. **Correct Answer : Column A (Solid Shape)**

- → Column B (Properties)
- Cylinder → 2 circular faces, 1 curved surface → 2
- Cone → 1 circular face, 1 curved surface → 3
- Cube → All faces are equal squares → 4
- Cuboid → 6 faces, 12 edges, 8 vertices → 1

A–2
B–3
C–4
D–1

COMPREHENSION TYPE

Passage 1: Faces, Edges and Vertices

23. **Correct Answer: C) 12**

The number of edges in a cube is:
A cube has 12 edges.

24. **Correct Answer: B) 1**

The number of vertices in a cone is:
A cone has 1 vertex.

25. **Correct Answer: B) Edge**

The line segment where two faces meet is called:

It is called an **edge**.

Passage 2: Cross-sections and Shadows

26. **Correct Answer: B) Circle**

A horizontal cut to a cylinder gives a:
A horizontal cut gives a circle.

27. **Correct Answer: C) Rectangle**

A vertical cut through the centre of a cylinder gives a:

A vertical cut through the centre gives a **rectangle**.

28. **Correct Answer: C) 2D**

The shadow of a solid object is always:
A shadow is a 2D shape.

INETEGR TYEPE

29. **The number of faces of a square pyramid is 5.**
(1 square base + 4 triangular faces)

30. **The curved face of a cylinder can be opened up as a rectangle.**

31. **The angle between two adjacent edges of a cube and cuboid is 90°.**

32. **A sphere has how many faces? 0 face**

Previous Year Question

1. **The given solid figure is a pentagonal prism because:**
   - The top face is a **pentagon**.
   - The bottom face is also a **pentagon** (shown with dotted lines).
   - There are **5 rectangular lateral faces** joining corresponding sides of the two pentagons.

   Now, the net of a pentagonal prism must contain:
   - 2 pentagons
   - **5 rectangles** arranged in a strip (for the lateral faces)

   Option B → Contains **5 rectangles** and **2 pentagons** arranged correctly

2. **Correct Answer: Second solid**

   It is a hexagonal prism.

   For an n-sided prism:
   - Edges = $3n$
   - Faces = $n + 2$
   - Vertices = $2n$
   - Given:

   Faces = 8

   So,

   $$n + 2 = 8$$
   $$n = 6$$

   Hence,

   $$Q = 3n = 3 \times 6 = 18$$

3. **Correct Answer: A. 4 Third solid**

   Given:

   Edges = 10

   So, $2n = 10$

   $$n = 5$$

   Therefore, $R = n + 1 = 6$

   Now calculate: $Q - 2P - R$

   $$= 18 - 2(4) - 6$$
   $$= 18 - 8 - 6$$

   $= 4$

4. **Correct Answer: From the given 3D figure:**
   - The bottom row has 4 cubes.
   - On the leftmost cube, there are two more cubes stacked (making height = 3).
   - On the right side, there is one extra cube stacked (making height = 2).
   - The middle cubes remain height = 1.

   So from the front, the heights appear as:

   $$3, 1, 1, 2$$

   Now compare with options:
   - A → Matches 3 on left, 1–1 in middle, and 2 on right

5. Correct Anuswe A : is incorrect because a cone has only 2 faces, not 3

Explanation:

- A **cone** has **2 faces** (1 flat circular base + 1 curved surface).
  It does **not** have 3 faces.
- A **sphere** has **1 curved surface** (often counted as one face in elementary level).
- A **cube** has **6 faces ✓**
- A **cylinder** has **3 faces** (2 flat circular faces + 1 curved surface) ✓

Therefore:

6. Correct Answer: B — 4 cubes

First, count cubes in each figure.

Fig. (A)

- Bottom layer: 3 cubes
- One cube on top

Total in Fig. (A) = **4 cubes**

7. Correct Answer :

Fig. (B)

- Bottom row: 3 cubes
- Middle right: 2 cubes
- One cube above
- One more cube at back/right
- Total in Fig. (B) = **8 cubes**

Cubes to be added:

$$8 - 4 = 4$$

8. Correct Answer: A — Faces = 5, Edges = 8

Faces:

- 1 square base
- 4 triangular faces

Total faces = **5**

Edges:

- 4 edges of base
- 4 side edges

Total edges = **8**

9. Correcrt Answer: D) Rectangle

A horizontal cut to a smart phone (which is shaped like a rectangular prism) gives a rectangle.

10. Correct Answer: A) Circle

A horizontal cut to a ball (which is a sphere) gives a circle.

11. Correct Answer: B. 11

faces